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REAL-TIME PLASMA POSITION ESTIMATION IN ISTTOK: A MARTE2-BASED IMPLEMENTATION WITH SIMULINK MODELING

Dissertation within the scope of the Master's Degree in Mechanical Engineering, with a specialization in Energy and Environment, supervised by Professors Bernardo Brotas de Carvalho, Ph.D., and Pedro Alexandre Dias Martins, Ph.D., and presented to the Department of Mechanical Engineering of the University of Coimbra.

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Real-Time Plasma Position Estimation in ISTTOK: A MARTe2-Based Implementation with Simulink Modeling

A dissertation submitted in partial fulfilment of the requirements for the degree of Master in Mechanical Engineering in the specialty of Energy and Environment

Estimação em Tempo Real da Posição do Plasma no ISTTOK: Implementação em MARTe2 com Modelagem em Simulink

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"*There can be no success without sacrifice*."

John C. Maxwell, 2016.

Aos meus pais.

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Com a conclusão desta etapa marcante de um percurso acadêmico repleto de conquistas e de superações, não poderia deixar de expressar meus sinceros agradecimentos a todos que contribuíram para meu sucesso nesta jornada.

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ABSTRACT

This work presents the development and validation of a real-time plasma position estimation architecture for the ISTTOK tokamak, based on the integration between the Simulink modeling environment and the MARTe2 framework. The methodology was structured into three main stages: initial configuration and validation of the MARTe2 framework, integration of Simulink models through the "SimulinkWrapperGAM", and development of an estimation approach for the plasma current centroid position using magnetic and electric diagnostics. The magnetic reconstruction was based on Mirnov coil measurements, while the electric reconstruction was obtained from Langmuir probe signals, with the corresponding estimators combined through a sensor fusion strategy based on extended Kalman filters. The proposed approach was evaluated using experimental ISTTOK discharge data and compared with reference reconstructions previously used in successful machine operation. The results showed that the developed estimators reproduce the main temporal trends of the plasma position, although systematic deviations remain due to model simplifications and the influence of passive structures in the machine. The fused estimate presented a more continuous and balanced behavior than the individual estimators. However, the real-time performance evaluation showed that the current implementation does not yet satisfy the nominal 100 μs requirement needed for 10 kHz ISTTOK operation. Therefore, this work validates the feasibility of the proposed approach as a prototyping and development tool, while also identifying the optimizations required for future closed-loop control application.

Keywords: Nuclear fusion, Plasma position estimation, MARTe2, Simulink, Sensor fusion.

RESUMO

Este trabalho apresenta o desenvolvimento e a validação de uma arquitetura de estimação em tempo real da posição do plasma no tokamak ISTTOK, baseada na integração entre o ambiente de modelação Simulink e a framework MARTe2. A metodologia foi estruturada em três etapas principais: configuração e validação inicial da framework MARTe2, integração de modelos Simulink através da "SimulinkWrapperGAM", e desenvolvimento de uma abordagem de estimação da posição do centroide de corrente do plasma utilizando diagnósticos magnéticos e elétricos. A reconstrução magnética foi baseada em medições das bobinas de Mirnov, enquanto a reconstrução elétrica foi obtida a partir dos sinais das sondas de Langmuir, sendo os respetivos estimadores combinados através de uma estratégia de fusão sensorial baseada em filtros de Kalman estendidos. A abordagem proposta foi avaliada utilizando dados experimentais de descargas do ISTTOK e comparada com reconstruções de referência previamente utilizadas com sucesso na operação da máquina. Os resultados demonstraram que os estimadores desenvolvidos reproduzem as principais tendências temporais da posição do plasma, embora persistam desvios sistemáticos devido a simplificações do modelo e à influência das estruturas passivas da máquina. A estimativa fundida apresentou um comportamento mais contínuo e equilibrado do que os estimadores individuais. No entanto, a avaliação do desempenho em tempo real mostrou que a implementação atual ainda não satisfaz o requisito nominal de 100 µs necessário para a operação do ISTTOK a 10 kHz. Assim, este trabalho valida a viabilidade da abordagem proposta como ferramenta de prototipagem e desenvolvimento, ao mesmo tempo que identifica as otimizações necessárias para uma futura aplicação em controlo em malha fechada.

Palavras-chave: Fusão nuclear, Estimação da posição do plasma, MARTe2, Simulink, Fusão sensorial.

DECLARATIONS

Declaration of integrity

I declare that I have acted with integrity in the preparation of this academic study and confirm that I have not resorted to plagiarism or any form of misuse or falsification of information or results in any of the stages leading to its preparation.

I further declare that I know and have respected the Code of Ethical Conduct of the University of Coimbra.

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I acknowledge the use of ChatGPT-4o (Open AI, https://chat.openai.com) to organize concepts, improve text clarity, review the final draft, and to support the development and optimization of programming tasks.

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CONTENTS

Agradecimentos i
Abstract ii
Resumo iii
Declarations iv
Contents v
List of Figures vii
List of Tables ix
List of Symbols and Acronyms/Abbreviations x
List of Symbols x
Acronyms/Abbreviations xiii
1. Introduction 1
1.1. Research Background and Motivation 1
1.2. Objectives and Methodology Overview 7
1.3. Structure of the Dissertation 9
2. Literature Review & State-of-the-Art 11
2.1. Tokamaks and Plasma 11
2.2. ISTTOK and Plasma Control Challenges 16
2.2.1. ISTTOK Configuration and Parameters 16
2.2.2. ISTTOK Control & Data Acquisition System 18
2.2.3. ISTTOK Plasma Control Challenges 24
2.3. Plasma Control Models & Techniques 26
2.3.1. Plasma Equilibrium and Linearized Modeling 26
2.3.2. Control-Oriented Plasma Modeling Codes 27
2.3.3. Simplified Modeling Approaches for ISTTOK 28
2.3.4. Modern Strategies for Plasma Control 29
2.4. MARTe and Simulink: Tools for Plasma Control 29
2.4.1. MARTe Overview 29
2.4.2. Generic Application Modules 31
2.4.3. DataSources & Brokers 32
2.4.4. Real-Time Threads 33
2.4.5. Simulink and MARTe for Control Applications 34
3. Research Methodology 36
3.1. MARTe2 Engine Setup and Standalone Validation 36
3.1.1. Compilation and Configuration of MARTe2 36
3.1.2. Executing a Real-Time Application with MARTe2 37
3.1.3. Implementation of the Water Tank Problem 39
3.2. Simulink Integration with MARTe2 44
3.2.1. C/C++ Code Generation from Simulink Models 44
3.2.2. Integration with MARTe2 via ‘SimulinkWrapperGAM’ 45
3.2.3. Validation and Behavior Analysis 46
3.3. Plasma Current Centroid Position Estimation in ISTTOK 47
3.3.1. Acquisition, Processing, and Calibration of Plasma Diagnostics for Position Estimation 47

3.3.2. Sensor Fusion for Plasma Position Estimation Using Kalman Filters 55
4. Results & Discussions 63
4.1. Validation of the Estimation Approach 63
4.2. Sensor Fusion and Robustness Analysis 74
4.3. Real-Time Performance Evaluation 77
5. Conclusion 82
References 86
Annex A: MATLAB Script to Setup Simulink Models for MARTe2 Integration 91
Appendix A: MARTe2 Compilation Guide 93
Appendix B: Parameters for Magnetic Diagnostics Signal Processing 95
Appendix C: Magnetic and Electric Diagnostics Initial Covariance Matrices 97
Appendix D: Detailed Plasma Position Estimation Metrics 99

LIST OF FIGURES

Figure 1: Illustration of coils wrapped the torus and the generated toroidal magnetic field $B\phi$ (extracted from [7, p. 14]) 3

Figure 2: Illustration of the magnetic field configuration in a tokamak, showing **(a)** the plasma torus shape created by toroidal magnetic field $B\phi$, and the poloidal magnetic field Bp induced by toroidal current $I\phi$, and **(b)** the helical trajectory of the field lines around the torus stemmed from the combination of both magnetic fields (adapted from [7, p. 14]) 3

Figure 3: Source of vertical instability in different plasma configurations, showing **(a)** a vertically elongated plasma (adapted from [9, p. 4]), and **(b)** a circular plasma (adapted from [11, p. 9]) 5

Figure 4: Illustration of poloidal magnetic fields and the resulting forces under idealized conditions 15

Figure 5: Frontal view of ISTTOK, highlighting its main components (adapted from [40, p. 2]) 17

Figure 6: Simplistic illustration of ISTTOK tokamak 18

Figure 7: Overview of the main ISTTOK's diagnostics 19

Figure 8: Illustration of ISTTOK's poloidal cross-section containing the set of Mirnov coils (adapted from [40, p. 3]) 21

Figure 9: Showcase of ISTTOK sections with problems in the arrangement of the poloidal field coils 23

Figure 10: Core MARTe2 components for real-time applications 31

Figure 11: Overview of the internal logic of Generic Application Modules 32

Figure 12: Overview of the communication logic managed by DataSources and supervised by Brokers 33

Figure 13: Overview of the structure of the simplest functional MARTe real-time application 34

Figure 14: Log for the first 5 seconds runtime and subsequent MARTe termination of 'RTApp-1-1.cfg' 39

Figure 15: Water tank model used for feedback control analysis 40

Figure 16: Water tank feedback control simulation models, showing **(a)** the block-based representation of MARTe configuration, and **(b)** the Simulink water tank model used for feedback control analysis 43

Figure 17: Comparison of the water tank feedback control system responses obtained from **(a)** the MARTe2 simulation and **(b)** the Simulink simulation 44

Figure 18: Simulink water tank model for MARTe2 integration 45

Figure 19: Water tank feedback control system response **(a)** using the Simulink water model integrated into MARTe2 with the 'SimulinkWrapperGAM' and **(b)** using the 'WaterTankGAM' 46

Figure 20: Comparison of cycle duration histograms for different MARTe2 implementations, namely **(a)** the Simulink water tank model integrated into MARTe2 through the 'SimulinkWrapperGAM'and **(b)** the 'WaterTankGAM' implementation..... 47

Figure 21: Time evolution of magnetic field measurements from 12 Mirnov coils for shot 53071, obtained through **(a)** MDSplus (new ATCA board) and **(b)** the ISTTOK Shared Data Access System (old ATCA board) 50

Figure 22: Comparison of plasma current obtained from magnetic reconstruction and Rogowski coil measurements for shot 53071 51

Figure 23: Plasma position reconstruction from magnetic measurements for shot 53071, showing **(a)** the radial and **(b)** the vertical components..... 52

Figure 24: Comparison between **(a)** the radial and **(b)** the vertical plasma positions reconstructed from magnetic and electric measurements for shot 53071 54

Figure 25: Estimated plasma **(a)** radial and **(b)** vertical positions as obtained by magnetic coils, electric probes, and fused state for shot 53071..... 61

Figure 26: Plasma radial position for shot 53071 estimated by magnetic coils, electric probes, fused state, and reference reconstructions..... 66

Figure 27: Plasma vertical position for shot 53071 estimated by magnetic coils, electric probes, fused state, and reference reconstructions..... 66

Figure 28: Plasma radial position for shot 48555 estimated by magnetic coils, electric probes, fused state, and reference reconstructions..... 68

Figure 29: Plasma vertical position for shot 48555 estimated by magnetic coils, electric probes, fused state, and reference reconstructions..... 69

Figure 30: Plasma radial position estimation metrics for magnetic coils, electric probes, and fused state, using the reference reconstruction as benchmark..... 71

Figure 31: Plasma vertical position estimation metrics for magnetic coils, electric probes, and fused state, using the reference reconstruction as benchmark..... 71

Figure 32: Estimation-cycle duration histograms for shots 48555 and 53071 evaluated around the nominal 100 µs real-time cycle duration..... 79

Figure 33: Estimation-cycle duration histograms for shots 48555 and 53071 evaluated in the 160-200 µs interval 80

LIST OF TABLES

Table 1: Research Objectives 8
Table 2: Collection of different tokamak characteristics..... 13
Table 3: MARTe Startup Arguments 38
Table 4: Water tank model and control parameters used in the simulation 43
Table 5: Selected shots for weighted averaging process for determination of Langmuir coefficients 53
Table 6: Resulting global Langmuir coefficients with mantissas rounded to five decimal places 54
Table 7: Discharge selection for assessing the developed estimation approach 64
Table 8: Comparison between calculated mean plasma positions during 272-293 ms flat top phase for shot 53071 67
Table 9: Comparison between calculated mean plasma positions during 105-130 ms flat top phase for shot 48555 70
Table 10: Relevant Parameters for the Mirnov Coil Signal Conversion Algorithm 95
Table 11: Detailed plasma position estimation metrics computed over the selected flat top intervals for shots presented in Table 7 99

LIST OF SYMBOLS AND ACRONYMS/ABBREVIATIONS

List of Symbols

General plasma and tokamak quantities

I_P – Plasma current

R_0 – Major radius

a – Minor radius

Q – Fusion gain factor

R_p – Plasma radial position

Z_p – Plasma vertical position

Φ_P – Poloidal magnetic flux

I_ϕ – Toroidal current

B_ϕ – Toroidal magnetic field

B_p – Poloidal magnetic field

ψ – Stream function for the poloidal magnetic field

$p(\psi)$ – Free function associated with plasma pressure

$F(\psi)$ – Free function related to the net poloidal current flowing in the plasma

$\boldsymbol{J}$ – Current density vector

q – Particle charge

μ_0 – Plasma poloidal flux

Electromagnetic quantities

$\boldsymbol{F}$ – Force vector

F_R – Radial/horizontal force

F_V – Vertical force

$\boldsymbol{E}$ – Electric field vector

$\boldsymbol{B}$ – Magnetic field vector

B_R – Radial/horizontal magnetic field

B_V – Vertical magnetic field

$\boldsymbol{v}$ – Velocity vector

ε – Induced voltage

ϕ – Angle between the magnetic field lines and the area vector $\boldsymbol{S}$

Mirnov coil quantities

$\boldsymbol{S}$ – Mirnov probe cross-sectional area vector

S – Mirnov probe cross-sectional area

S_{eff} – Mirnov probe cross-section effective area

B_{Mirnov_i} – Magnetic field intensity measured by the i-th Mirnov coil

R_{Mirnov_i} – Relative radial position of the i-th Mirnov coil

Z_{Mirnov_i} – Relative vertical position of the i-th Mirnov coil

pol – Polarity of a Mirnov coil

K – Gain of a Mirnov coil

ADC_i – Digital signal acquired by the i-th Mirnov coil

r_{Mirnov} – Radial distance from the vacuum-vessel center to each Mirnov coil

Langmuir probe quantities

$V_{f_{\text{top}}}$ – Floating potential measured at the top Langmuir probe

$V_{f_{\text{outer}}}$ – Floating potential measured at the outer Langmuir probe

$V_{f_{\text{bottom}}}$ – Floating potential measured at the bottom Langmuir probe

$V_{f_{\text{inner}}}$ – Floating potential measured at the inner Langmuir probe

Water tank model quantities

h – Water level

q_i – Inlet flow

q_o – Outlet flow

A – Area of the water tank cross-section

V – Action signal (understood as a voltage applied to the pump)

b – Constant related to the proportional relationship between the input voltage and the inlet flow rate generated by the pump

c – Constant related to the physical characteristics of the outlet

A_o – Orifice area

v – Velocity

Kalman filter and estimation quantities

x_k – State vector

P – State covariance matrix

P_0 – Initial state covariance matrix

Q – Process noise covariance matrix

R – Measurement noise covariance matrix

y_k – Diagnostic measurements vector

$h(\cdot)$ – Measurement function

∇h_k – Measurement function Jacobian

K_k – Kalman gain

$A_{\text{Mirnov}_{k|k-1}}$ – Constraint matrix

c_{Mirnov_k} – Constraint vector

α – Fusion coefficient

Real-time implementation quantities

f_{RT} – Real-time system clock frequency

d_{RT} – Real-time decimation factor

Acronyms/Abbreviations

AC – Alternating Current

CEMMPRE – Centro de Engenharia Mecânica, Materiais e Processos

CODAS – Control and Data Acquisition System

DS – DataSource

ECCD – Electron Cyclotron Current Drive

FCT – Fundação para a Ciência e a Tecnologia

GAM – Generic Application Module

IPFN – Instituto de Plasmas e Fusão Nuclear

ISS – Internal State Machine

ISTTOK – Instituto Superior Técnico TOKamak

ITER – International Thermonuclear Experimental Reactor

LQG – Linear-Quadratic-Gaussian

LQR – Linear-Quadratic Regulator

MAE – Mean Absolute Error

MARTe/MARTe2 – Multi-threaded Application Real-Time executor

MaxAE – Maximum Absolute Error

MIMO – Multiple-Input Multiple-Output

PCC – Pearson Correlation Coefficient

PCS – Plasma Control System

PFC – Poloidal Field Coil

PID – Proportional-Integral-Derivative

RMSE – Root Mean Squared Error

RTA – Real-Time Application

RT – Real-time

RTT – Real-Time Thread

SDAS – Shared Data Access System

SISO – Single-Input Single-Output

TFC – Toroidal Field Coil

VE – Vertically Elongated

VV – Vacuum Vessel

WTM – Water Tank Model

1. INTRODUCTION

This dissertation was written as part of the Master's in Mechanical Engineering, specializing in Energy and Environment, at the University of Coimbra, between February 2025 and October 2026. In this sense, this research focuses on the application of real-time estimation techniques to Nuclear Fusion systems, particularly addressing the real-time estimation of the plasma position in the ISTTOK tokamak. This problem is directly related to plasma stability and constitutes a fundamental component for advanced control strategies in magnetic confinement devices. The work explores an estimation approach based on diagnostic signals, supported by the integration between the MARTe2 real-time framework and the Simulink[1] environment for modeling and validation purposes. This introductory chapter is divided into three sections: the first section provides the contextual foundation of the research to highlight its motivation and relevance; the following section outlines the objectives and the research methodology followed throughout the dissertation; and the final section closes this chapter by presenting the structure of the remaining chapters.

1.1. Research Background and Motivation

In the global effort to mitigate climate change, low-carbon electricity generation technologies are crucial. The development of fusion power has gained significant interest in recent years due to its promising potential of becoming a source of safe and clean energy [1], [2], since it produces zero greenhouse gas emissions, and does not create long-term radioactive waste [3], [4], [5]. In this context, nuclear fusion holds unique relevance due to the enormous amount of energy released by the reactions, particularly in the case of deuterium and tritium nuclei reactions, which are not only likely to be the first harnessed [6] but are also proposed for power generation [4]. For instance, fusing just 1 kg of deuterium-tritium fuel would release approximately 10^8 kWh of energy, which is enough to fulfill the requirements of a 1 GW electrical power station for a day [7]. Moreover, compared to a non-sustainable source of energy, fusion energy released from a single gram of this fuel is equivalent to the energy of approximately 7.4 tons of oil [4].

The fusion process occurs when particles possess sufficient kinetic energy to overcome the electrostatic repulsive forces between them, to the point that the short-range nuclear forces of attraction are greater in magnitude [6]. The most appealing method of supplying

[1] MATLAB Simulink documentation, https://mathworks.com/products/simulink.html.

the energy required for the deuterium-tritium reaction to take place is to heat the fuel to increase thermal velocities of the nuclei until fusion occurs. When the reaction is achieved in this way, it is known as thermonuclear fusion [7].

Thermonuclear fusion is characterized by extremely high temperatures of about 100 million degrees Celsius. This crucial aspect rules out the possibility of sustaining controlled reactions in reactors that rely solely on material walls for confinement [7]. Moreover, at such high temperatures, the fuel transitions from the gaseous state into a state known as plasma, which can be defined as "a quasineutral gas of charged and neutral particles which exhibits collective behavior" [8]. Consequently, the confinement of plasma with magnetic fields becomes feasible, hence fusion reactors based on this principle were developed through various approaches [4], [6].

Tokamaks represent the leading concept for controlled thermonuclear fusion [5], [8], [9], [10], [11] and were first conceived at the Kurchatov Institute in the 1950s to address the challenge of attaining fusion power [12]. Tokamaks are devices capable of maintaining and confining plasma within the reactor's vacuum vessel (VV) with the ultimate goal of investigating enhanced methods of confining and heating plasma in a way that sustained fusion reactions take place [3]. These toroidal (i.e. doughnut-shaped) devices use electromagnetic forces generated by external fields to confine hot plasma to a toroidal region kept away from the VV inner walls. The dominant magnetic field is the toroidal field; however, it also requires a poloidal field in order to allow confinement of the plasma [7], since plasma particles flowing in the toroidal direction experience vertical drift. This drift is counteracted by the combined action of both magnetic fields [11].

The toroidal field is produced by currents in coils wrapped around the reactor (see Figure 1), while the poloidal is mainly generated by current flowing in the toroidal direction, which is generated by the plasma itself [11]. Hence, the combination of both magnetic fields originates magnetic field lines that follow a helical trajectory around the torus [3], [4], [5], as shown in Figure 2.

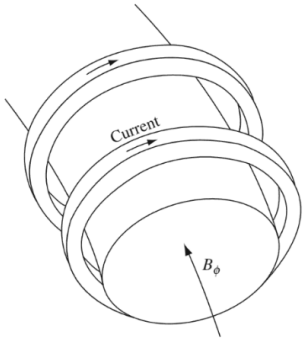


Figure 1: Illustration of coils wrapped the torus and the generated toroidal magnetic field B_ϕ (extracted from [7, p. 14])

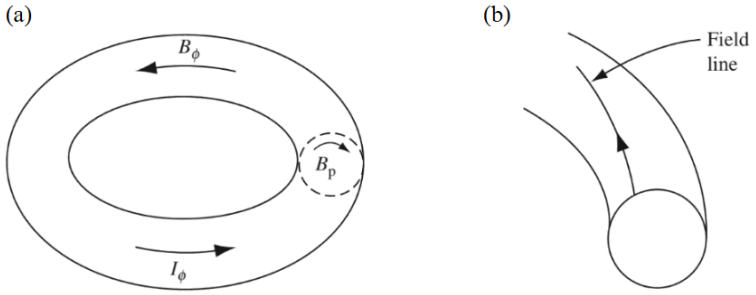


Figure 2: Illustration of the magnetic field configuration in a tokamak, showing **(a)** the plasma torus shape created by toroidal magnetic field B_ϕ, and the poloidal magnetic field B_p induced by toroidal current I_ϕ, and **(b)** the helical trajectory of the field lines around the torus stemmed from the combination of both magnetic fields (adapted from [7, p. 14])

At the heart of tokamak development lies the ability to accurately monitor and regulate plasma shape and position, which relies fundamentally on reliable diagnostic measurements and state estimation. In this context, modern tokamak operations envisage an increase in performance by achieving plasma that is forced to be vertically elongated (VE) [9], [13], [14]. The VE cross-sections lead to improvements in plasma parameters, such as an increase in energy confinement time, albeit at the expense of introducing vertical displacement instability [9], [15]. Furthermore, enhanced performance in terms of increased magnetohydrodynamic stability and reduced transport also arises from this approach [16], which justifies the preference for vertical elongation. As a result, this led to the evolution of older tokamaks into D-shaped tokamaks with VE cross-sections [8], [12].

Since plasma with high elongation is vertically unstable due to its large vertical instability growth rate [2], [4], [17], [18], accurate real-time (RT) knowledge of the plasma position becomes essential to properly address this instability and enable effective feedback control strategies. The vertical instability stems from the fact that the magnetic field in

tokamaks is not perfectly vertical, especially in the case of VE plasma. In this sense, the placement of the poloidal field coils (PFC, plural: PFCs) outside the vessel contributes to the lack of verticality, which in turn generates a radial component. According to the Lorentz force law, this radial component of the magnetic field induces vertical displacement of the plasma [11].

The vertical instability is more pronounced in elongated plasmas, not only because of the specific configuration of the magnetic field responsible for maintaining the plasma elongation in the poloidal cross-section, but also due to the way the plasma current is distributed. The elongated shape leads to a current profile that is inherently more susceptible to the vertical instability. Conversely, due to the imperfect vertical magnetic field used for plasma positioning, plasma with circular cross-section also becomes unstable when perturbed. For instance, in both plasma configurations, if the plasma shifts upward from its equilibrium position, the net force is also directed upward, further increasing both the vertical displacement and the upward force [4]. The derivative of the force along the vertical axis is a source of the plasma instability [19], which reinforces the need for accurate RT tracking of the plasma position.

Figure 3(a) illustrates the source of vertical instability in an elongated plasma, where the radial magnetic field component (B_R) is responsible for the unstable behavior. Figure 3(b) shows the same phenomenon in a circular plasma, highlighting that the vertical magnetic field (B_V) generated by the vertical PFC is not purely vertical due to the influence of both plasma current (I_P) and coil placement, which in turn results in the appearance of B_R. It is also worth mentioning that the gray areas in both figures represent the plasma current distribution. As long as the current distribution and the magnetic field are both simultaneously symmetric about the midplane ($z = 0$), the net force will be zero, and the plasma remains in equilibrium [9].

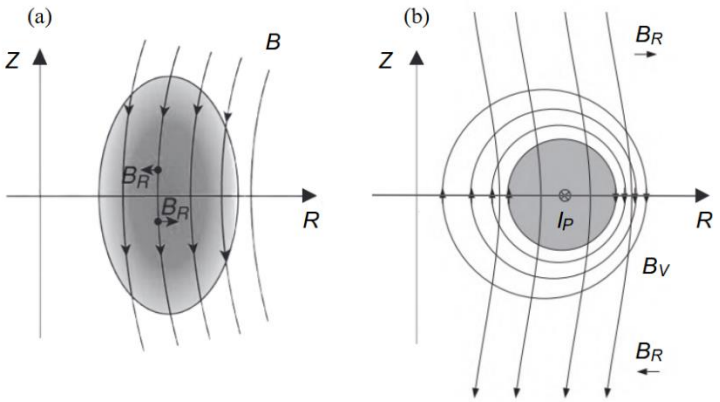


Figure 3: Source of vertical instability in different plasma configurations, showing **(a)** a vertically elongated plasma (adapted from [9, p. 4]), and **(b)** a circular plasma (adapted from [11, p. 9])

The vertical stabilization system addresses the issue of instability by driving current in a set of PFCs to produce a radial/horizontal magnetic field [4], [13] in response to changes in the plasma vertical position [19], which must be accurately determined in RT. Indeed, magnetic control systems are a standard feature of tokamaks with VE plasma and will be present in the International Thermonuclear Experimental Reactor (ITER) [16], [20], which is regarded as "the next-step toward the realization of electricity-producing fusion power plants" [17] and is the largest fusion experiment up to date [21]. These systems play key roles, including stabilizing vertical instability, as well as controlling the plasma current and centroid position [16]. This highlights that, while magnetic control remains a core engineering challenge in tokamaks [11], its effectiveness fundamentally depends on the accurate estimation of the plasma position.

The complete termination of the plasma current can arise from unrecoverable disruptions in the plasma (major disruptions), which are deemed extremely undesirable due to the potential damaging effects they pose for the reactor [9], [15]. Although modern D-shaped tokamaks have magnetic control systems designed to stabilize vertical displacement of plasma, the reliable determination of plasma position remains a critical and challenging task. Therefore, further research is crucial for the success of future fusion power plants, as failures in such systems could cause major disruptions [12]. It is worth noting that, if the plasma position is not accurately tracked and maintained, a major disruption in the plasma occurs, leading to a sudden loss of confinement [7], [8] and thermal energy due the uncontrolled growth of the large-scale instability in the plasma [9].

The fast plasma dynamics are particularly relevant and impose a hard constraint on plasma control cycle, [22], emphasizing the need for position control on an extremely fast

time scale [9], [17]. For example, on the ISTTOK tokamak, which represents the testbed for this research, control cycles of 100 μs were implemented [11], [18], [23]. Moreover, future tokamaks designed for fusion power will operate in pulses, each lasting up to several minutes [3], [9], highlighting the importance of the position control system in maintaining stability during such operational scenarios [24].

The successful operation of a future thermonuclear fusion reactor will be dependent on the RT knowledge of several plasma parameters, with the plasma position figuring as a key parameter [11], [18]. Indeed, RT control of plasma parameters is a critical aspect of advanced operation of tokamaks, as it enables the attainment of high confinement while allowing for the timely detection and prevention of major disruptions [10]. Furthermore, plasma control in tokamak engineering plays a pivotal role in achieving the growing performance demands of the industry [14].

Achieving the required performance strictly depends on the flexibility and reliability of the RT systems that govern the operation of the plant during the experiment [25]. Therefore, hard RT control systems are essential in the operation of such devices for ensuring the completion of control cycles within the strict deadlines dictated by plasma behavior, while guaranteeing that no sample is lost [26]. Failure to meet these deadlines can lead to the catastrophic consequences that plasma instability can cause [18], [27].

The 'Multi-threaded Application Real-Time executor' (MARTe[2]) is a control C++ modular multi-platform framework designed for hard RT projects [18], [25], [27] and is used in several fusion devices [11], [25], [28], [29], [30], [31]. It enables high portability, facilitating code reuse across other applications, and ensures maintainability while minimizing the impact on performance in terms of jitter and latency [13], [25], [27]. In this context, it provides the infrastructure required for RT data processing and state estimation in fusion experiments.

MARTe2, the evolution of the original MARTe framework and a standard in the context of magnetic confinement fusion RT control system deployment [26], was tailored to enhance these features by implementing a strict quality assurance procedure to guarantee that external contributions adhere to the framework's intended quality [32]. Indeed, this allows safe integration contributions from a heterogeneous community that includes industrial suppliers as well as academic institutions [31], [32].

[2] In this dissertation, MARTe and MARTe2 are used interchangeably when referring to the framework.

Despite the framework's powerful features, with one of its most important being its flexibility [30], this quality comes with the downside of extreme configuration verbosity and complexity. Indeed, this represents the most critical issue in MARTe usability, requiring significant expertise to construct and maintain the configuration [31], [33]. Moreover, these configurations of the MARTe2 engine entails writing hundreds or up to thousands of lines of code, rendering the manual preparation an extremely lengthy and error prone process [21], [31], [33]. Thus, an intuitive interface with the framework, easily manageable and understandable by non-specialists, offers the benefit of quickly implementing advanced control algorithms in the context of plasma control systems (PCS) [13], [31].

1.2. Objectives and Methodology Overview

At the time of writing, the majority of ISTTOK's embedded control systems still operate under the former version of the MARTe framework [11], [23], [34], [35], which hinders the evolution of the PCS, as algorithms are not certified by the same software quality assurance processes enforced over the trailblazing updated version [11]. In this view, this constraint hinders tracking of external developments and creates obstacles for contributions from the fusion research community.

The MARTe2 upgrade boasts the advantage of overcoming the difficulty related to the distribution and maintenance of software without the continuous support of the original developers [11]. Moreover, given that ISTTOK serves as a testbed for many young researchers pursuing doctoral research [11], the deployment of a user-friendly control design environment capable of seamless integration with the RT PCS shortens the associated lengthy learning curve [22], [31].

This is of paramount importance for fusion research as it allows for swifter advancement of control algorithms in PCS, which inherently depend on the accurate RT estimation of plasma parameters. In this sense, this is ultimately crucial for the development of future successful tokamaks [11], [17]. Therefore, the overarching goal of this dissertation is to develop and validate a robust RT plasma position estimation framework to support plasma position control for the ISTTOK tokamak, leveraging the MARTe2 framework and its integration with the MATLAB Simulink block diagram environment for model-based design.

This goal can be achieved by adhering to a well-structured methodology, following a step-by-step procedure based on the objectives defined in Table 1.

Table 1: Research Objectives

Steps	Underlying Objectives
A1	Introduce the MARTe framework and test simpler configuration files.
A2	Validate the compiled MARTe engine for RT control applications using the 'Water Tank Problem'.
B1	Deploy Simulink models and introspect executable code via the 'SimulinkWrapperGAM'.
B2	Test the water tank model created in Simulink using the 'SimulinkWrapperGAM' and benchmark it against the GAM developed in the previous step for the same purpose using MARTe2.
C1	Reconstruct plasma parameters using magnetic measurements and integrate them into the control system for RT processing.
C2	Develop and fine-tune the algorithm for plasma position estimation.

In this sense, this research is divided into three main parts, the first being an introductory approach to developing control algorithms with MARTe. This prerequisite is of the utmost importance due to the complexity inherent not only to the framework itself but also to working with embedded control systems, which have recently garnered attention as a platform for model-driven design [25].

Evidently, this first part entails the compilation of the MARTe2 engine and its offline operation in order to test simpler configuration files, which dictate real-time application (RTA) structures in a Simulink-like (i.e. block-based) fashion [26], [29], [31]. Moreover, the 'Water Tank Problem' – a classic control engineering problem extensively described in [36] – is formulated within MARTe by means of a designated Generic Application Module (GAM) [13]. Due to its simplicity and non-linear dynamics, this step is meant as validation of the compiled MARTe engine for RT control purposes. The non-linearity associated with this problem is appealing in the context of fusion, given that the complex plasma dynamics are extremely non-linear [3], [37], [38] and even uncertain in the control perspective [39].

The second phase focuses on the deployment of Simulink models through introspection of executable code using MARTe [22]. One of the main advantages of leveraging Simulink for this purpose is that C or C++ code defining the control algorithm can be automatically generated owing to the Embedded Coder[3] Simulink add-on. In this

[3] Embedded Coder documentation, https://mathworks.com/help/ecoder/index.html.

sense, a Simulink water tank model (WTM) is developed to be exported and integrated into MARTe in order to validate the same solution tested in the previous part exploiting the capability of loading Simulink models' embedded code into MARTe through the 'SimulinkWrapperGAM' [21], [22]. This step concludes the validation of model-based control designs within MARTe, bridging the gap between simulation and deployment.

Finally, the third part focuses on addressing the overarching problem of plasma position regulation in the ISTTOK tokamak, with particular emphasis on the estimation of the plasma current centroid – the geometrical center for the plasma current distribution [11]. For this purpose, previously acquired experimental data is used to reconstruct the plasma current and centroid position, based on measurements from Mirnov coils (magnetic diagnostics) [11], [34], [35] fused with Langmuir probe signals (electric diagnostics), since these quantities are not directly measured [2], [3], [4], [40], [41]. Moreover, this data enables the comparison of the proposed estimation approach with previously developed solutions, by evaluating the reconstructed plasma centroid position against reference estimations obtained from past ISTTOK discharges. In parallel, the reconstructed plasma current is compared with direct measurements from the Rogowski coil, providing a means to validate the underlying magnetic reconstruction model. Consequently, this comparison supports the assessment of the proposed method, particularly in the context of combining extended Kalman filters (KF, plural: KFs) for sensor fusion [42]. This is of extreme significance for ISTTOK mainly due to its alternating current (AC) pulsed operation, which urges control of the plasma position [43].

The conclusion of all these steps will essentially:

- validate the claim that the integration between MARTe and Simulink lessens the work and difficulty of implementing RT estimation and data processing applications;
- create a user-friendly environment for swifter prototyping of advanced estimation and monitoring algorithms for future researchers interested in ISTTOK;
- represent a step towards the upgrade of ISTTOK's RT systems architecture to the latest version of the Multi-threaded Application Real-Time executor.

1.3. Structure of the Dissertation

Besides this first introductory chapter, this master's dissertation is divided into another four chapters. The following chapter, entitled "Literature Review & State-of-the-Art", dives into the core concepts related to the topics covered in this research, namely tokamaks, plasma

behavior and control, RT PCS, the MARTe framework, modeling approaches, and the specific applications in ISTTOK, providing an in-depth review of the existing literature and the latest advancements in these fields.

Following that, the research methodology and procedure are described in further detail in the “Research Methodology” chapter, outlining the course of action taken to achieve the defined objectives. This chapter presents a comprehensive overview of the experimental setup, the Simulink integration with MARTe2, and the development, testing, as well as the fine-tuning of the proposed estimation framework.

Naturally, the “Results & Discussions” chapter presents the outcomes of applying the proposed estimation approach to historical ISTTOK data, highlighting the main findings obtained. In particular, the reconstructed plasma position is analyzed and compared with experimentally validated estimations from past discharges in order to assess the accuracy and consistency of the proposed method. Finally, the last chapter, “Conclusion”, summarizes the main conclusions extracted from this dissertation in the context of modeling-based approaches to plasma position estimation and outlines directions for future research.

2. LITERATURE REVIEW & STATE-OF-THE-ART

This chapter provides an overview of the current body of knowledge and the theoretical foundations of the key topics relevant to this research. In this view, it reviews the existing literature on tokamaks devices, plasma behavior, RT monitoring and control systems in the context of nuclear fusion, and the MARTe framework. Additionally, a more detailed description of the ISTTOK tokamak is presented, since it represents the testbed for the research conducted in this dissertation. Control techniques implemented in fusion research are covered, including the ones stemmed from emerging technologies. To conclude, the Simulink platform for design of control algorithms is also covered, due to its relevance not only in the domain of control engineering, but also for the conduction of this research.

2.1. Tokamaks and Plasma

Tokamaks (from the Russian acronym 'токамáк', meaning "toroidal chamber with an axial magnetic field" [8]) are the most advanced and widely researched devices for magnetic confinement fusion [3]. These experimental reactors are designed to confine and contain the extremely hot plasma required for nuclear fusion reactions to occur [7], [8], [25] within its VV by means of powerful induced magnetic fields [5], [7], [40]. In comparison to other existing concepts such as stellarators, reversed-field pinch reactors, and inertial confinement reactors, tokamaks boast the highest potential for hosting nuclear fusion reactions capable of reaching ignition [10] (i.e. self-sustained nuclear fusion [3]), consequently, allowing the achievement of energy production through thermonuclear fusion [11], [18], [40]. In this sense, the ratio between the thermonuclear power produced and the power supplied, better known as Q, represents a "measure of the success in approaching reactor conditions" [7], and, when ignition is achieved, $Q \rightarrow \infty$.

Tokamaks have spread throughout the world since their inception, with a wide variety of configurations and sizes now in existence [3]. For instance, in 2022, 50 of the 226 tokamaks built up to that date were still operational [12], and plans for upcoming tokamaks are ongoing, with some currently under construction. One notable example is the ITER tokamak[4], a one-of-a-kind experimental reactor that is significantly larger and more powerful than its predecessors [4], [39], representing the pinnacle of the tokamak

[4] As of October 2026, ITER is still under construction, with research operations scheduled to begin in 2034, according to official sources (https://www.iter.org/project/assembly-overview).

technology. In this context, ITER is an ambitious megaproject developed by a multi-country consortium, aiming to produce and control plasma that is far more energetic than that of any existing tokamaks, with the ultimate goal of demonstrating the technological feasibility of fusion power [4], [39]. Indeed, for the achievement of fusion power, it is necessary to confine a 30 keV (approximately 348 million degrees Celsius[5]) plasma with magnetic fields for at least a second [8], hence ITER's initial operation will be focused on control research since its performance is critically reliant on the required control techniques for sustaining such plasmas [4].

Moreover, although ITER is not centered around the production of electricity, the cutting-edge device is foreseen as the first device capable of hosting a self-sustained fusion reaction [2], [3], [8], and the project aims to achieve a tenfold energy gain ($Q \geq 10$) by generating 500 MW of fusion power from an input of 50 MW [4], [20]. On the other hand, ITER boasts crucial features of envisioned fusion power reactors, with the long duration pulsed operation figuring as a requirement for the success of these reactors [4]. In this perspective, successfully achieving this objective mandates active plasma control and shaping for hundreds of seconds [20], a time scale way longer than the characteristic time scale of many plasma instabilities [9].

Furthermore, magnetic control and, consequently, accurate plasma position estimation are essential to avoid not only contact with internal components, especially in the case of high-energy plasmas such as those in ITER, as well as prevent disruptive plasma terminations resulting from the loss of stability [4], [20], [44]. Therefore, plasma position control on a fast time scale represents an essential feature of tokamaks [17], and, since ITER's operation "will depend critically on control", the project plays a crucial part in the current investigation on plasma control [4].

Concurrently, other smaller tokamaks, some of which are installed in universities [3], also play a vital role in plasma control research. Among these, the ISTTOK tokamak, which will be covered in further detail in the following subsection, stands out for its scientific program that includes operation and control of plasma AC regimes [34], [40], [43]. In this sense, given that the time scale requirements for plasma control are linked to its confinement time and passive plasma stabilization systems [18], each tokamak presents their own unique challenges related to their configuration and size. Table 2 presents a collection of the main tokamak parameters of different reactors, namely the major radius R_0, the minor radius a,

[5] 1 eV = 11,600 K [8]

and the aspect ratio. These parameters represent, respectively, the distance from the center of the torus to the plasma center, the radius of the plasma cross-section, and the ratio of the major radius to the minor radius [3], which is related to plasma energy confinement time scaling [4].

Table 2: Collection of different tokamak characteristics

Parameters	ITER	JET	TCV	ADITYA-U[6]	ISTTOK
Major Radius R_0 [m]	6.2	2.96	0.89	0.75	0.46
Minor Radius a [m]	2.0	1.25	0.25	0.25	0.085
Aspect Ratio	3.1	~2.37	3.56	3.0	~5.4
Maximum Plasma Current I_p [kA]	15000.0 (estimated)	~6000.0	1200.0	~175.0	~7.0
Plasma Duration [s]	~1000.0 (estimated)	~5.0	~4.0 (ECCD[7])	~0.300	~1.0 (AC)
Configuration	D-shaped	D-shaped	D-shaped	Circular plasma	Circular plasma

Despite ISTTOK's circular cross-section, which differs from the D-shaped configuration of tokamaks such as ITER [12], plasma position control remains a requirement for its operation and garners even greater interest, because plasma vertical stability control is characterized by a faster time scale than plasma shape control [45]. In this context, achieving precise plasma control requires that the control cycle be faster than a plasma parameter closely related to the size of the device: the particle confinement time [10], which for ISTTOK is approximately 300 μs [18]. Therefore, characteristic time scales are critical for tokamak control, as the fast time scales demand both rapid RT actuation and control calculations [9].

In devices that rely on magnetic confinement, RT control of plasma is crucial for achieving high confinement efficiency and for detecting and preventing major disruptions. The stabilization of the plasma position underscores the need for feedback control of both

[6] Upgraded version of ADITYA tokamak [47], a device with dimensions and configuration similar to ISTTOK [10]. The difference between the plasma duration in both devices despite their similarities is due to ISTTOK's advanced AC operation.
[7] This plasma duration was achieved in advanced tokamak scenario of Electron Cyclotron Current Drive (ECCD) operation.

the vertical and horizontal magnetic fields [10]. Since plasma is an ionized gas, this control is grounded in the interaction of charged particles with magnetic fields [3], governed by the Lorentz force, given by the following equation:

$$\boldsymbol{F} = q \cdot (\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}) \tag{1}$$

where $\boldsymbol{F}$ is the force vector experienced by a particle with a charge q moving with velocity vector $\boldsymbol{v}$ in electric and magnetic field vectors, denoted by $\boldsymbol{E}$ and $\boldsymbol{B}$, respectively.

This is accomplished in tokamaks by driving electrical currents in PFCs, which in turn generates magnetic fields that interact with the toroidal plasma current to produce forces to confine plasma [3], [4], [9], [40]. The vertical field is created to counteract the outward radial forces experienced by plasma, such as the tire force, originated from the plasma pressure, and the hoop force, which arises from the compression of the magnetic field on the inside part of the torus [4], and, therefore, generate an inward radial force to control the plasma radial position [11], [12]. The radial or horizontal field is necessary to oppose the plasma drift observed in the vertical direction, known as toroidal drift [11], by creating vertical forces responsible for the control of the vertical position and stabilization of the vertical instability [4], [12], [19].

Figure 4[8] illustrates, respectively, the magnetic fields generated by vertical and horizontal PFCs according to ISTTOK's configuration and design. It is important to note that this simplified representation neglects the influences of plasma current and passive structures on the resulting magnetic field [9], [16], [38], [43], which is why the magnetic field lines appear uniform. The magnetic field lines are shown in orange in both scenarios, with F_R and F_V representing the radial and vertical induced forces, respectively.

[8] Figure generated using Python code adapted from Ben Dudson, University of York, and other contributors (2016-2022). The original code is licensed under the GNU Lesser General Public License (LGPL), version 3 or later. Available at: https://github.com/freegs-plasma/freegs.

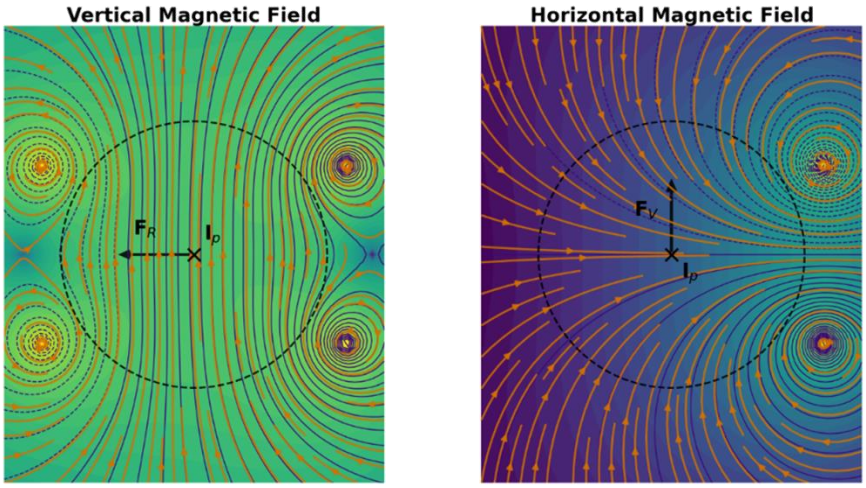


Figure 4: Illustration of poloidal magnetic fields and the resulting forces under idealized conditions

Improper placement of the PFCs can lead to loss of plasma control, which explains the current trend of placing the horizontal PFCs between the VV and toroidal field coils (TFC) to provide the best plasma vertical position controllability [12]. Despite ISTTOK's configuration being based on this principle, one of the main challenges in its plasma control arises from the non-axisymmetric and non-uniform arrangement of both vertical and horizontal PFCs [11] (see Figure 9).

Moreover, the eddy currents induced on the conducting walls of the VV reduces the ability of the magnetic fields generated by the PFCs to control plasma due to their shielding effect [9], hence the vertical control system envisioned for ITER is solely based on the operation of in-vessel coils [12], [16], [20]. In spite of the advantage of decoupling controllable parameters by separating the coil sets reserved for ohmic heating, the vertical field, and the radial field, this approach introduces the risk of damaging the in-vessel coils due to the fusion plasma heat load [9], [12].

Additionally, for older tokamaks with circular cross-sections, copper shells surrounding the VV have been employed as an effective controller to dampen the plasma motion. For modern D-shaped tokamaks, the shell acts as a direct acting controller, that is, it directly exploits plasma energy to reduce motion in VE plasma [12]. This auto-control feature is present in ISTTOK [11] and, while it offers the benefit of increasing the instability growth rate to the millisecond scale (~1 ms) [18], it also degrades signal quality due to significant contamination from eddy currents induced in the machine conducting structures [11]. Moreover, the shell properties, introduces a delay – known as skin time – of approximately 2.5 ms, representing the time required for magnetic fields to penetrate the

conducting walls [10], [11], further exacerbating the challenges in plasma control in ISTTOK.

2.2. ISTTOK and Plasma Control Challenges

2.2.1. ISTTOK Configuration and Parameters

The Instituto Superior Técnico TOKamak (ISTTOK) is a small-scale experimental tokamak located at the Instituto Superior Técnico in Lisbon, Portugal, and operated by the Instituto de Plasmas e Fusão Nuclear (IPFN) [11], [40]. The device is mainly dedicated to training novice researchers [11] and, due to its flexibility in terms of replacing experimental campaign-specific diagnostics, it serves as a testbed for various scientific programs, such as edge plasma turbulence studies and AC operation [11], [23], [34], [40], [43].

The large aspect ratio (with R_0 = 46 cm and a = 8.5 cm) circular cross-sectioned tokamak [10], [23], [40] boasts a toroidal VV composed of two half-tori made of a 0.15 mm thick INCONEL alloy 625, which is surrounded by a 1.5 cm thick copper shell [11], [18]. The shell is in turn surrounded by a set of 24 uniformly arranged [11] TFCs that generates a nominal toroidal field of 0.3–0.6 T and a maximum toroidal field of 2.8 T [40]. ISTTOK also presents a graphite limiter [40] designed to impede plasma contact with the stainless steel first wall (AISI 304L). Additionally, the device has an iron core transformer that generates 0.25 Vs of inductive flux [10], that, prior to the introduction of flexible power supplies for the AC operation, limited the duration of single-mode direct current plasma discharges to approximately 45 ms [10].

Figure 5 highlights ISTTOK's main components, including the currently disconnected central solenoid [40], which was previously responsible for generating plasma current. Plasma current generation is currently achieved through ohmic heating produced by the primary PFCs, with plasma currents reaching a maximum of approximately 7 kA [40] and nominal values of 4 kA in AC operation [18]. It is worth noting that the plasma in ISTTOK consists solely of hydrogen and fusion is not actually achievable in the device. This is primarily due to the fact that the temperatures are not nearly high enough to facilitate fusion reactions, as hotter plasmas in ISTTOK only reach around 150 eV (≈ 1.74 million Kelvin) [46].

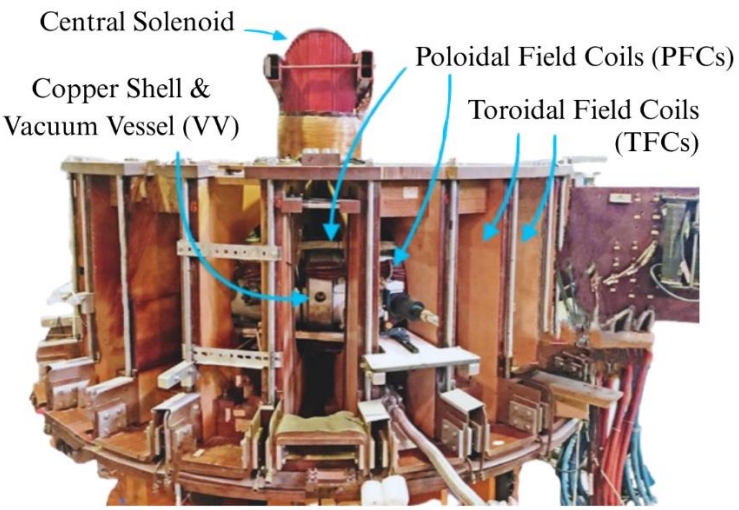


Figure 5: Frontal view of ISTTOK, highlighting its main components (adapted from [40, p. 2])

Although ISTTOK was commissioned using parts of the former Dutch tokamak TORTUR, several of its components were locally designed, such as the vacuum systems, the PFCs, the power supply for the TFCs, as well as its diagnostics, and control and data acquisition system (CODAS) [11]. In this context, since the start of its operation, ISTTOK's components – such as diagnostics, hardware acquisition system and control algorithms – have undergone continuous upgrades [43]. Among the most significant improvements is the upgrade of the flexible power supplies, which enables fast plasma AC reversal [18], [40], [43]. This enhancement has contributed to improve discharge stability and increase the number of AC discharge cycles, while maintaining stable plasma parameters throughout the discharge [23], [34].

Figure 6[9], which presents a simplified model of ISTTOK without its TFCs, illustrates the iron core transformer, the copper shell, the INCONEL alloy 625 VV, the toroidal plasma confined within the VV, and three sets of PFCs placed between the TFCs and the copper shell. These sets include the vertical field coils, horizontal field coils, and primary (or ohmic heating) coils, which are depicted in orange, blue, and white, respectively. It is worth noting that the horizontal field coils and the low field-side (the outboard side of the tokamak, farther from the central axis) vertical field coils in the model are intended to be understood as a single set of coils, as they are intertwined in the actual ISTTOK configuration.

[9] Figure generated using Python code adapted from Paramak, developed by Fusion Energy and Paramak contributors (2024). The original software is licensed under the MIT License. Available at: https://github.com/fusion-energy/paramak.

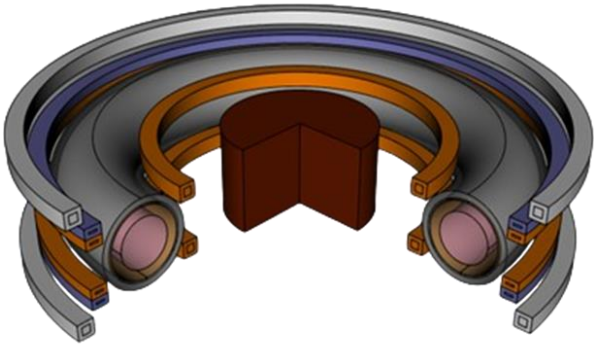


Figure 6: Simplistic illustration of ISTTOK tokamak

2.2.2. ISTTOK Control & Data Acquisition System

ISTTOK is "one of the few tokamaks with a regular AC plasma discharges scientific program" [23], [34], distinguishing it from similar devices like the ADITYA-U tokamak [47] due to its longer plasma duration (see Table 2). The development of this complex plasma current regime is rooted in the saturation of ISTTOK's iron core. Once saturated, it is impossible to continue driving the plasma current inductively in the same direction. Therefore, to sustain the discharge, an immediate reversal of the plasma AC is required, which justifies the use of AC operation [23].

"A modern CODAS must provide integrated management of supervision, data acquisition and RT control, and data centric view of the system combining raw, elaborated and configuration data" [28]. In this context, the advanced AC operation is achievable due to ISTTOK's intricate CODAS, which was designed following the Advanced Telecommunications Computing Architecture (ATCA) standard and operates with a 100 μs control cycle [18], [34]. Despite the constant upgrades to the PCS of ISTTOK, it remains outdated by MARTe standards, as it still relies on its earlier version [18], [40], [43].

2.2.2.1. ISTTOK Diagnostics

Diagnostic systems in tokamaks typically serve two main purposes: enabling proper control of the machine and acting as research tools themselves. Given the inherently noisy environment of tokamak operations [23], multiple diagnostic methods are employed in order to attain a comprehensive plasma equilibrium reconstruction from indirect measurements. Therefore, specialized techniques – based on plasma models, which will be explored in the following subsection – must be employed to reconstruct the desired quantities based on these measurements [40].

In ISTTOK's case, these systems are vital not only for maintaining the stability of plasma confinement but also advancing the understanding of plasma behavior. The diverse range of diagnostics present in ISTTOK enables acquisition of measurements related to various plasma parameters, facilitating position estimation and experimental campaign specific analysis. In this context, Figure 7 illustrates the main diagnostic systems present in ISTTOK, including those that have been used to estimate the plasma position, such as the visible light tomography, the heavy ion beam diagnostic, and, most importantly for this dissertation, the Langmuir probes and the Mirnov coils [18].

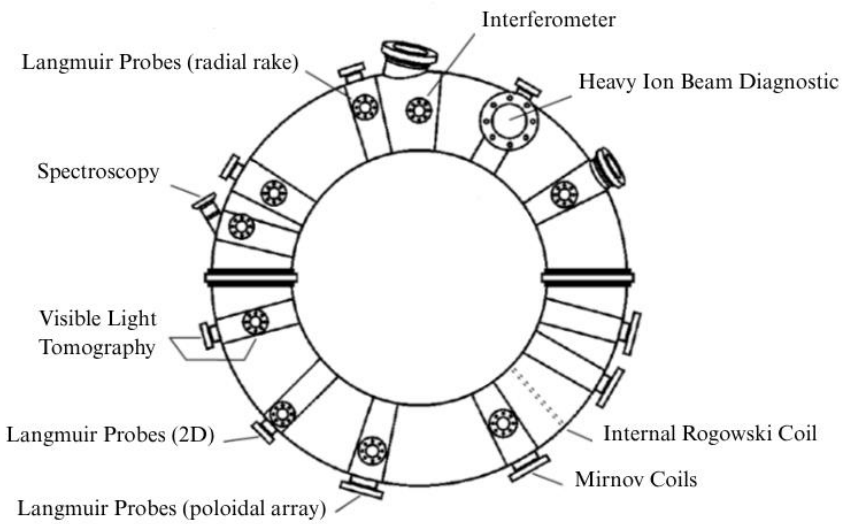


Figure 7: Overview of the main ISTTOK's diagnostics

Visible light tomography, described in further detail in [35], is a diagnostic system that enables the reconstruction of a 2-D plasma profile by measuring light emissions in the visible and ultra-violet spectrum. These emissions are related to key plasma parameters, allowing for the estimation of the plasma position [18]. The system operates by using RT algorithms to derive the plasma emissivity function from several line-integrated light measurements [35]. While the tomography system shows good agreement with other diagnostics in plasma displacement, its absolute value away from the axis of the VV may be incorrect. This limitation could be improved with the addition of more sensors, however, doing so would increase the computational time for calculation of the algorithms [35].

The heavy ion beam diagnostic is capable of performing simultaneous local measurements of a convolved plasma density and temperature in 12 distinct sample volumes, vertically distributed along the central region of the plasma (-6 mm < z < 6 mm) [18]. This measurement, known as generation factor, can be related to plasma pressure, which, in turn, is linked to plasma equilibrium, therefore, the generation factor profile is expected to provide

an accurate estimation of the plasma position. Furthermore, this diagnostic system performs a direct measurement across a nearly vertical plasma cord by injecting a Xe^{+} beam (primary beam) into the plasma, which then generates Xe^{2+} beams (secondary beams) as a result of ionization induced by the primary beam impact with plasma electrons [18]. Due to the nature of its operation, this diagnostic is expected to produce the most accurate and consistent measurements of the plasma vertical position in comparison to the other options present in ISTTOK [18]. However, despite this advantage, data processing remains a limiting factor, as the measurements are subject to noisy background that must be removed in each control cycle.

Langmuir probes can be used to estimate the plasma position by analyzing the difference in floating potential between two local measurements performed by opposing probes placed at the plasma periphery [18]. However, the main limitation of this diagnostic system for plasma current centroid position reconstruction is that it is only valid for small position variations ($\Delta z < 10$ mm) [18]. Additionally, the success rate for plasma AC inversion using measurements from Langmuir probes is only approximately 80%, while the success rate obtained with magnetic measurements from Mirnov coils is about 99.8%. This disparity is largely a result of their inability to measure negative plasma cycles [11], [43].

Mirnov coils are essential diagnostic tools for RT plasma position measurements used in tokamaks worldwide [18], as they provide valuable information about plasma instabilities within the system [14] and the necessary signals for reconstruction of the plasma current centroid position [11], [43]. In this sense, these magnetic probes measure the time derivative of the magnetic flux via Faraday's law of induction, allowing the determination of magnetic fields, which ultimately enables a model-based reconstruction of the plasma position and current estimates [11], [43].

Despite their critical role in plasma diagnostics, Mirnov coils have some limitations, including drifts on the integration of electronics [11], [18], [43], the negative impact of external magnetic fields, such as the ones produced by the external coils and by the eddy currents in the passive structures, on the accuracy of the plasma reconstruction [11], [18], [43], as well as their inoperability during plasma AC reversal [35]. ISTTOK is equipped with a set of 12 Mirnov coils, shown in Figure 8, arranged to measure the poloidal magnetic field across a poloidal cross-section [11], [40], [43]. These coils have been placed inside the VV at 93.5 mm from its axis and are equally spaced, with a separation of $\Delta\theta = 30^{o}$. Each coil has

an area of approximately 49 mm^2, 50 turns, and a length of 5 mm, and is encapsulated in graphite boxes [11].

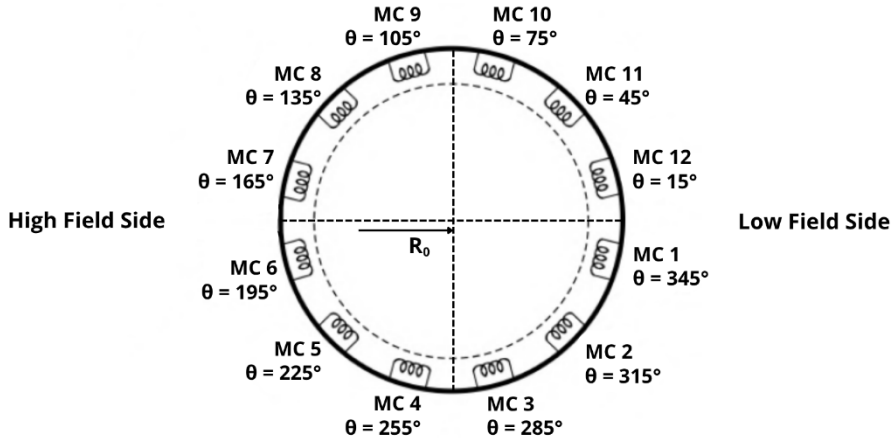


Figure 8: Illustration of ISTTOK's poloidal cross-section containing the set of Mirnov coils (adapted from [40, p. 3])

The operation of magnetic diagnostics is governed by Faraday's law of induction, which relates the induced voltage ε to the rate of change of magnetic flux passing through the probe cross-section for a coil of N turns, expressed as:

$$\varepsilon = -N \cdot \frac{d\Phi_P}{dt} \tag{2}$$

where Φ_P is the poloidal magnetic flux measured by the coils. Consequently, to obtain the magnetic field magnitude from the inductive probe signals, an integration step is required to first determine the poloidal magnetic flux [11]. In this sense, the magnetic field intensity can be derived from the following surface integral, where $\boldsymbol{B}$ represents the magnetic field vector and $\boldsymbol{S}$ denotes the area vector normal to the Mirnov probe cross-section:

$$\Phi_P = \iint_S \boldsymbol{B} \cdot d\boldsymbol{S} \tag{3}$$

However, given the small cross-section of the Mirnov coils featured in ISTTOK, the magnetic field can be assumed to be approximately constant across the probe volume [40]. Therefore, this relation simplifies to:

$$\Phi_P \approx \boldsymbol{B} \cdot \boldsymbol{S} = B \cdot S \cdot \cos\phi \tag{4}$$

where B is the magnetic field magnitude, S is the area of the Mirnov probe cross-section, and ϕ is the angle between the magnetic field lines and the area vector $\boldsymbol{S}$. Assuming an

effective area S_{eff}, which represents the area perpendicular to the magnetic field lines, the magnetic field intensity can be then expressed by the simple relation:

$$B = \frac{\Phi_P}{S_{\text{eff}}} \tag{5}$$

2.2.2.2. ISTTOK Actuators & Control Strategy

In tokamaks, actuators play a crucial role in manipulating the plasma position and shape, ensuring stability and preventing disruptive events. In ISTTOK, aside from the gas puffing system, the TFCs and PFCs figure as the primary actuators responsible for plasma discharges, as the plasma position is controlled by magnetic fields generated by these coils [18]. In this context, the PFCs are divided into primary, vertical, and radial fields coils, all connected to independently feedback-controlled power supplies [40], which were upgraded to extend the duration of AC discharges [34]. These coils are responsible for generating plasma current through ohmic heating and controlling the plasma current centroid radial and vertical positions, respectively [11], [40], [43]. On the other hand, the TFCs are powered by a locally designed flexible power supply that enables AC operation [11], [40].

Due to the strong space constraints and diagnostic ports installed on the device, the PFCs are not arranged uniformly and not toroidally axisymmetric, particularly the vertical and radial fields coils [11], [40]. Additionally, according to the Lorentz force law, the coils experience forces during each discharge, which lead to further movement of these coils if they are not properly fixed [20]. As a result, the positions of these coils have shifted significantly from their nominal values since the start of the operation of ISTTOK, and there is no knowledge of how the internal vertical field coils have moved after more than 53,000 discharges [11], [40]. Figure 9 illustrates sections where the lack of symmetry in the vertical and radial fields coils (dark red cables) and their non-uniform arrangement are most prominent. It can also be observed that this issue is less significant in the case of the primary coils (white cables).



Figure 9: Showcase of ISTTOK sections with problems in the arrangement of the poloidal field coils

The arrangement issues of the PFCs affect the behavior of the control system in AC operation, as they lead to distinct overall topologies of the poloidal magnetic field distribution in the scenarios of positive and negative plasma current [40]. Given these issues and the uncertainties in the geometry of ISTTOK, adapting a numerical plasma equilibrium model, such as the CREATE-L codes [48], to ISTTOK presents significant challenges.

Consequently, attempts to develop a functioning physics-based model for magnetic control in ISTTOK have been unsuccessful [40]. [11] details a novel data-driven approach for plasma position control in ISTTOK, where the reconstruction of the plasma current centroid position involves extracting the sole magnetic contribution of the plasma current to the Mirnov coils, while data-driven discrete state-space models determine the magnetic contribution from eddy currents and PFCs to the measured signals.

For a long time, control strategy in ISTTOK relied on Single-Input Single-Output (SISO) PID controllers to manage the radial and vertical positions, with preset gains empirically selected based on controller performance across various experimental discharges [11], [40]. Moreover, another locally employed method that follows this approach is the auto-PID feature, which dynamically adjusts the initial user-defined PID constants to improve tracking performance [34].

For traditional tokamaks, the simplest control strategy involves two separate SISO controllers [9]. In this sense, the empirical approach in ISTTOK was rooted not only in the absence of a theoretical decoupled model for the plasma parameters but also in the simplicity of implementing the SISO PID controllers in MARTe. With the introduction of the enhancement described in [11], the strategy shifted to a 2x2 Multiple-Input Multiple-Output (MIMO) Linear-Quadratic Regulator (LQR) controller, using the currents of the vertical and radial field PFCs as inputs, and the vertical and radial plasma current centroid position estimates as outputs.

In this context, in order to achieve optimal sensor-based feedback, KFs were employed to enable Linear-Quadratic-Gaussian (LQG) control. However, PID control remains essential, as the data-driven model requires an initial phase managed by the PID controller, while the KF reconstructs the state vector to determine the initial conditions, but the instantaneous switching process can cause rapid spikes in the control signal, a phenomenon known as the 'bumpless transfer problem' [49].

2.2.2.3. ISTTOK Control System Software & Architecture

The ISTTOK CODAS is based on a recently enhanced hardware architecture that follows the ATCA standard, as described in further detail in [23]. This upgrade aimed to improve plasma discharges in terms of their duration and number of AC cycles. Furthermore, the RT capabilities demonstrated to be successful, as 100 μs control cycles with jitter lower than 0.4 μs were achieved using the MARTe framework.

The hardware includes a carrier board equipped with a quad-core Intel Q8200 processor running at 2.33 GHz. It also includes two acquisition boards, each with 32 galvanically insulated differential inputs acquired by analog-to-digital converter (ADC) modules. These modules are connected to a Virtex-4 Field-programmable gate array (FPGA) with an 18-bit resolution at 2 MSamples/s. Communication between the carrier and acquisition boards is handled via the PCI-Express standard [34].

The use of already established solutions can prove effective for the development of novel CODAS, since it reduces the development time and the likelihood of errors [28]. In this sense, ISTTOK features the use of three frameworks common to fusion research: EPICS addresses system supervision and provides graphical user interfaces; MDSplus, which furnishes not only support for data acquisition and storage, but also an API for data management; and MARTe for RT control [28]. Upgrading MARTe to MARTe2, brings the benefit of better decoupling between synchronization and data flow management, enabling automatic tracking of signals handled in RT, and, therefore, "adhering to the data centric paradigm of MDSplus" [31].

2.2.3. ISTTOK Plasma Control Challenges

ISTTOK plasma discharges are characterized by the unique capability of operating in AC mode [11]. Despite the multitude of advantages this plasma current regime offers, it also introduces challenges, particularly during AC transitions. Indeed, the reliable success of the plasma AC inversions requires highly accurate plasma position control, both just before the

AC reversal and during the restart process [43]. This requirement is rooted in the delicate and inherently unstable residual plasma that persists during the time needed to complete the inversion [43].

One such challenge is the 'bumpless transfer problem', which introduces undesirable jumps in plant inputs due to the constant switching between automatic and manual control [49]. During this transition, which lasts approximately 1 ms, a preprogrammed open-loop configuration of the PFCs currents is employed. To minimize these spikes, the control signal at the switching instant must be kept as close as possible to the preprogrammed value for manual mode [40], which can be achieved if the controlled variable is maintained near the setpoint when switching controllers.

Additional challenges include poor signal quality resulting from the low magnetic field generated by the plasma current and the consequent high contamination from eddy currents in the passive structures [18]. Due to the impracticality of using models derived from equilibrium codes – which will be covered in the following subsection – and the difficulties with theoretical models, ISTTOK has had to rely on data-driven discrete state-space models recently developed to account for the magnetic noise caused by the eddy currents [40].

The main obstacle to properly adjusting algorithms for the necessary correction fields lies in the poor arrangement of the PFCs [35], leading to the development of this extremely complex and "black box" control strategy. Given that ISTTOK lacks complete axisymmetry, the model requires separate models for positive and negative plasma current [40]. Additionally, since the initial state conditions cannot be determined *a priori*, in accordance with the control strategy described in [11], the system requires use of a different approach before switching to LQG control, leading to the 'bumpless transfer problem' once again.

In spite of the better results of advanced diagnostics for low plasma currents [50], such as visible light tomography and heavy ion beam diagnostics, using the magnetic diagnostics remains a viable option for plasma equilibrium reconstruction, as they allow for faster control cycles that meet the plasma confinement time constraint of 300 μs [18].

However, due to contamination from eddy currents in the current filament reconstruction process, when compared to the visible light tomography diagnostic, magnetic diagnostics indicate that the plasma vertical position is shifted upwards for positive plasma currents and downwards for negative plasma currents [35]. This comparison is valid since the average plasma emissivity center, measured by the tomography diagnostic, and the average center of plasma current, measured by the magnetics diagnostic are typically close

to one another, since the latter is sensitive to the plasma current density distribution. Lastly, another major issue related to the Mirnov coils diagnostic is their inoperability during plasma AC reversals [35], which motivates the previously discussed controller switching strategy.

2.3. Plasma Control Models & Techniques

2.3.1. Plasma Equilibrium and Linearized Modeling

Direct observation of the plasma current centroid position is not possible, therefore, it needs to be reconstructed from indirect measurements, such as magnetic data [2], [3], [40], [41]. In this context, the position can be obtained via two-dimensional magnetic-equilibrium reconstruction to solve an inverse problem for determination of the plasma current distribution that respects the Grad-Shafranov equation force balance and best matches the obtained measurements [2].

"Equilibrium is achieved by balancing the magnetic force with the plasma pressure gradient force" [51]. Consequently, according to [51], the Grad-Shafranov equation, "a second-order nonlinear partial differential equation describing axisymmetric toroidal equilibria", based on the standard right-handed cylindrical coordinate system (R, z, ϕ) is given by:

$$\begin{gathered}\Delta^{*}\psi = -\mu_0 \cdot R^2 \cdot \frac{\partial p(\psi)}{\partial \psi} - F(\psi) \cdot \frac{\partial F(\psi)}{\partial \psi} \\ \Delta^{*}\psi \equiv R \cdot \frac{\partial}{\partial R}\left(\frac{1}{R} \cdot \frac{\partial \psi}{\partial R}\right) + \frac{\partial^2 \psi}{\partial z^2}\end{gathered} \tag{6}$$

Moreover, the plasma magnetic field and current distribution are, respectively, given by Equations (7) and (8).

$$\boldsymbol{B} = \frac{1}{R} \cdot \nabla\psi \times \boldsymbol{e_\phi} + \frac{F(\psi)}{R} \cdot \boldsymbol{e_\phi} \tag{7}$$

$$\mu_0 \cdot \boldsymbol{J} = \frac{1}{R} \cdot \frac{\partial F(\psi)}{\partial \psi} \cdot \nabla\psi \times \boldsymbol{e_\phi} - \frac{1}{R} \cdot \Delta^{*}\psi \cdot \boldsymbol{e_\phi} \tag{8}$$

ψ is the stream function for the poloidal magnetic field, which is related to the plasma poloidal flux. μ_0 is the vacuum magnetic permeability, $p(\psi)$ is a free function associated with plasma pressure, and $F(\psi)$ is a free function related to the net poloidal current flowing

in the plasma. Additionally, $\boldsymbol{J}$ is the current density vector and $\boldsymbol{e_\phi}$ is the unit vector in the ϕ coordinate.

The design of PCSs is based on tokamak plasma models [12], which are, in turn, modeled based on the Grad-Shafranov equation [2]. In this sense, given the non-linearity and the second-order nature of the equilibrium equation, "the modeling of a tokamak plasma demands complex mathematical calculation, in depth physical knowledge and computational power for numerical calculation" [24], which led to the development of numerous approaches, mostly focused on the design of intricate equilibrium codes.

Even though physical systems are inherently nonlinear, most modern control techniques are reliant on linear time-invariant discrete-time system models [52], which are the basis of controller design for PCSs [9], [24]. The linear approximation of nonlinear systems is validated by the Hartman-Grobman theorem, which states that stability (or instability) of the nonlinear system is retained in its linearization around a fixed point [16]. In this perspective, the models are derived from a linearization around an equilibrium region where the key modes of operation are captured, as described for the plasma current centroid position model in [40]. Given the absence of net forces on a plasma equilibrium in the vertical direction, the linearization of the vertical position dynamics around this desired point of operation is valid [4].

2.3.2. Control-Oriented Plasma Modeling Codes

Accurate equilibrium reconstruction codes have thus been developed to deploy linearized plasma models in RT control applications [5], such as the well-established CREATE-L [48] and RZIP [53] linear models. However, some more intricate codes are only suitable for simulation purposes [24]. These linear models derive the time evolution of the plasma current from electrical circuit equations [24]. In this context, the CREATE-L code provides a linearized plasma equilibrium response model specifically designed for plasma current, position, and shape control in tokamaks [11], [24]. This code enables the retrieval of a linearized state-space model for a given reference configuration under a certain equilibrium condition and has been used for vertical stabilization simulations on the JET tokamak through integration with MARTe [54].

The RZIP code defines a non-deformable linearized plasma model for calculating the output voltages of the power supplies, the currents in the control coils, and the plasma position and current [24]. The model not only facilitates the analysis of control strategy and

algorithm performance but also allows for the optimization of control parameters [55]. Additionally, the simplifications assumed in RZIP – namely, the constant distribution of plasma current due to plasma rigidity, the ability of the plasma center to vary in position, and the total plasma current capacity to change in magnitude – enable easy implementation, eliminate the need for calculation of complete plasma equilibrium, and enhance the clarity of quantities relevant to plasma control, all while maintaining overall good accuracy [24]. RZIP has been extensively used in TCV for the development of controllers primarily during their design phases [5], [16], [24], [55] and can be deployed as a MATLAB-based code[10] [1]. Although this approach has yet to be reproduced, RZIP can also be integrated into MARTe through the Simulink integration with the framework, described in the following section, using the 'SimulinkWrapperGAM' described in full detail in [22].

2.3.3. Simplified Modeling Approaches for ISTTOK

On the other hand, simpler models treat the plasma as a non-deformable grid of conducting filaments [24] or as a single conducting filament of circular shape, which is a reasonable approximation for control purposes in large aspect ratio tokamaks, such as ISTTOK [10]. The single-filament approach leads to a good balance between accuracy and speed [10], but it comes at the cost of producing significantly more optimistic growth rates compared to those obtained with a more realistic multifilament model [9].

Moreover, this approach could introduce the risk of saturating the transformer's iron core, however, the ISTTOK AC operation avoids saturation while maintaining linearity [10]. Given the ISTTOK short plasma discharges, typically lasting no longer than 1 second [18], [43], the simplified single-filament model remains a valid and practical choice, as the combined effect of the passive shell slowing the instabilities and the fast-acting active control system allows for continuous correction before the cumulative effect of any model underestimation leads to catastrophic growth within the discharge duration.

Due to the limited set of magnetic probes and measured quantities, a full equilibrium reconstruction of flux and current profiles is not achievable in ISTTOK [56], which is why it does not feature a Grad-Shafranov solver [11]. Currently, the plasma current centroid position can be estimated in RT from the reconstruction of data captured by a poloidal array of twelve Mirnov probes, using a twelve-filament model, as well as from a data-driven model that computes the plasma response to variations in the PFCs currents [40]. The previously

[10] For more information, refer to the RZIP GitHub repository: https://github.com/atisharma/rzip.git.

employed single-filament model did not account for the contributions of the active coils and the eddy currents induced on the passive structures to the magnetic probes' signals, which led to inaccuracies in centroid position, however this approach was still able to reliably produce AC discharges [43]. [10] demonstrate that the time evolution of the position is similar for both multi- and single-filament models, although greater differences are observed in the reconstructed vertical position.

As a result of this approach, implementing linear controllers, such as PID or LQR for optimal control, based on linear models of the flat top phase of a discharge becomes an adequate control strategy due to their robustness and good performance throughout the whole discharge [24]. Moreover, KFs can be employed to mitigate noise from the measurements and account for simplifications and uncertainties in the model [42], making the simplified single-filament model a viable option.

2.3.4. Modern Strategies for Plasma Control

New more advanced control techniques based on machine learning and artificial intelligence have gained significant interest in the fusion community due to their success in challenging applications across various domains [37]. Indeed, the latter has been identified as a "Priority Research Opportunity" for fusion control [2]. These tools are being explored for magnetic stabilization and nonlinear control of plasma position and shape [38].

Furthermore, an emerging approach based on experimental data involves applying digital twins with real and digital control systems [12]. Recently, a digital replica of ISTTOK was developed by an industrial partner using data and information from 2024. While these novel approaches, such as reinforcement learning, offer significant advantages, they also present drawbacks that still restrict their application in PCSs [41]. The major challenges associated with these solutions include the lack of interpretability of results, which explains the preference for simpler control methods [4], and the considerable computational power required to execute the algorithms at the desired control frequencies [2].

2.4. MARTe and Simulink: Tools for Plasma Control

2.4.1. MARTe Overview

Originally developed at JET [25], [29] and envisioned for the ITER magnetics diagnostics [26], [32], MARTe is a C++ modular framework designed for the development

and deployment of RT control system applications, primarily in fusion research [11], [13], [25], [29], [54]. It is the only framework developed by nuclear fusion experimentalists that has been adapted to other application areas, such as the RT control of wind turbine blades on marine platforms [25].

Indeed, MARTe2 shows potential as a valuable tool for the field of industrial automation as it represents a "solid and scalable framework for the development of RT control systems applications" [32]. One of its key strengths is that MARTe overcomes the challenge of non-portable code tied to specific RT systems, as it enables the same code to run across different operating systems [13], [17], [25]. Additionally, its major advantage lies in the clear separation between algorithms, hardware interaction, and system configuration, facilitating control system simulation and incremental commissioning from different teams working in parallel [13], [25], [32].

A new version of the framework, MARTe2, has been developed to incorporate contributions from various sources by adopting a software quality assurance strategy based on the MISRA C++:2008 and the ISO⁄IEC 14882:2003 (C++03) standards [11], [26]. Moreover, MARTe2 introduces "a new and more powerful system abstraction" in comparison to its predecessor, along with two novel components aimed at decoupling synchronization and data flow management [31]. The overall structure of the framework remains unchanged, with the added advantage that distribution of developed components across various systems is facilitated by quality procedures, eliminating the need for continuous support from original developers [11].

The framework is structured as a fully configurable internal state machine (ISS) that governs the system execution cycle, enabling synchronization of the RTA states to external events through the use of messages [13], [22]. Additionally, MARTe2 features a "Makefile-based compilation engine" and relies on text-based configuration files, which define the RTA structure using a "very simple, block-based, text file transfer protocol" [26]. The configuration language has a syntax similar to JSON [21] and is parsed at startup using the StandardParser class [13], [29], which can be invoked programmatically, via a runtime initialization routine, or even through a bash script. Furthermore, bash scripts are used to compile and build the MARTe environment, including DataSources (DS, plural: DSs) and GAMs [25], [26]. In this context, Figure 10 presents a layered architecture of the MARTe2 components in RT system deployment.

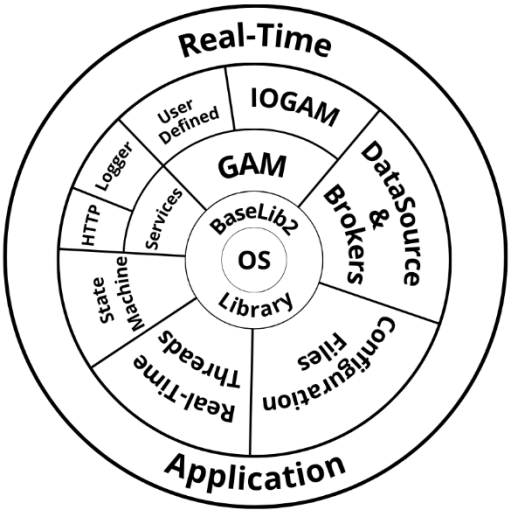


Figure 10: Core MARTe2 components for real-time applications

2.4.2. Generic Application Modules

GAMs are considered "the atomic elements" in MARTe applications, which in turn are entirely structured around them, enabling clear separation between hardware interface, algorithms, and system configuration [13]. In fact, a MARTe application is built by defining GAMs and specifying their order of execution by the RT micro scheduler in a configuration file [17], [25]. These blocks of code are mainly used for interfacing hardware, storing acquired data, executing algorithms, making decisions based on the current state of the system, debugging internal states, and providing RT information about the system [13].

Upon initialization of MARTe, GAMs declare their respective inputs and outputs as configurable named signals with associated data types and, in MARTe2, the corresponding data source for data exchange [13], [31]. Furthermore, GAMs are essentially "unrestricted in functionality" and can be divided into two main categories: user defined computational GAMs and IOGAMs [13]. The latter consists of the component responsible for providing a high-level interface with hardware and exchanging data between both different real-time threads (RTT, plural: RTTs) in a multi-core system as well as across different data sources [13], [31]. Computational GAMs, on the other hand, are responsible for implementing application-specific algorithms, control logic, and data processing, as defined by the user application requirements [13], [17]. As part of the MARTe core abstraction, these are essentially C++ classes that take input data, perform computations based on the scripts, and ultimately produce output data [25]. This logic can be visualized in Figure 11.

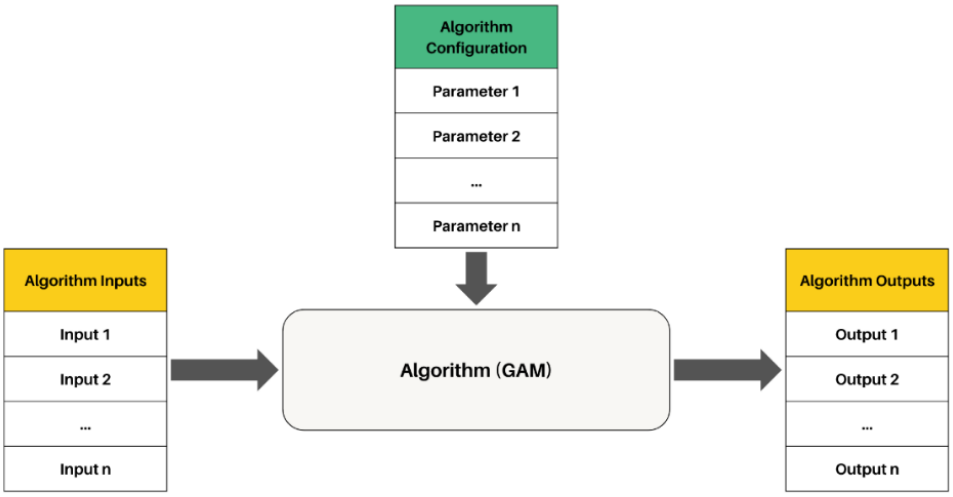


Figure 11: Overview of the internal logic of Generic Application Modules

Examples of computational GAMs that are considered useful for control logic and calculations, and are included in the MARTe2 components repository[11], include the 'PIDGAM' [40] and the 'SimulinkWrapperGAM' [21], [22]. However, novel GAMs can be developed to meet user needs by adhering to the MARTe2 Makefile structure. Additionally, computational GAMs can be chained together in series or parallel to build complex RTAs [13].

2.4.3. DataSources & Brokers

Introduced in MARTe2, DSs are responsible for specific data management tasks and serve as a standardized hardware abstraction layer, facilitating data exchange with system I/O [22], [31]. Built-in instances of DSs include the Timer, Logger, UDP, FileReader, and FileWriter. The Timer DS is responsible for synchronizing component execution [31], while the Logger provides a mechanism for RT streaming, supporting system operation, debugging, and problem analysis without compromising RT performance [13]. Furthermore, the UDP DS enables broadcasting or receiving data via the UDP protocol, while the FileReader and FileWriter DSs enable reading and writing text (CSV) or binary files, respectively. Similarly, the MDSReader and MDSWriter allow for reading reference signals and configuration data from the MDSplus experiment database and streaming RT data into segmented nodes of a target pulse file, respectively [31].

[11] For more information, refer to the MARTe2 Components GitLab repository: https://vcis-gitlab.f4e.europa.eu/aneto/MARTe2-components.

Concurrently, Brokers are instances that oversee and facilitate communication between GAMs and DSs in MARTe2 [31]. Brokers are instantiated by the DS itself, rather than being explicitly declared in a configuration file [31]. Complex data transfer mechanisms, such as interpolation and triggered data transfer, can be provided by reusable Broker classes to simplify the implementation of new data source instances for new actuators and sensors [31]. Figure 12 illustrates the communication logic between DSs, Brokers, and GAMs.

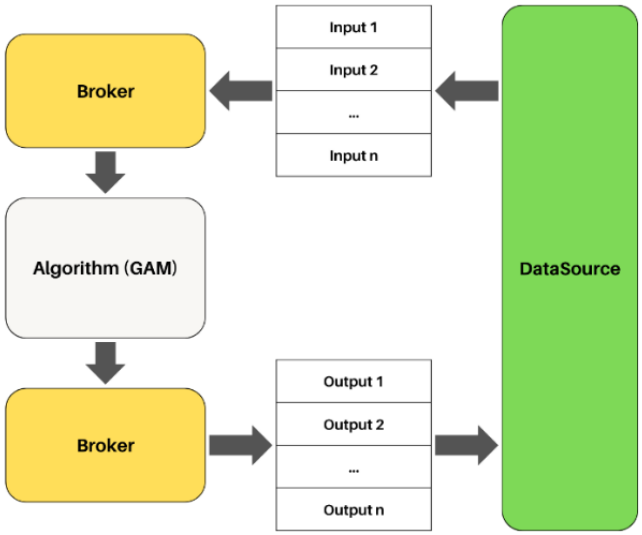


Figure 12: Overview of the communication logic managed by DataSources and supervised by Brokers

2.4.4. Real-Time Threads

Within MARTe's state-machine [22], multiple RTTs can be declared for a given state, with each RTT consisting of a container of GAMs that determines the sequence of their cyclic execution [13], [25], [29]. RTTs can be assigned to specific cores with a designated priority [13], [25]. In fact, to execute MARTe, at least a single RTT is required [13].

Moreover, the RTTs are able to track their individual execution times and store internal timing information for each GAM declared in its configuration [13]. In this sense, the cyclic execution of GAMs within each RTT can be operated at a specific frequency, and, if an execution cycle exceeds the expected completion time, a warning or error is issued [13]. Figure 13 illustrates the simplest RTA that can be created with the aforementioned MARTe features.

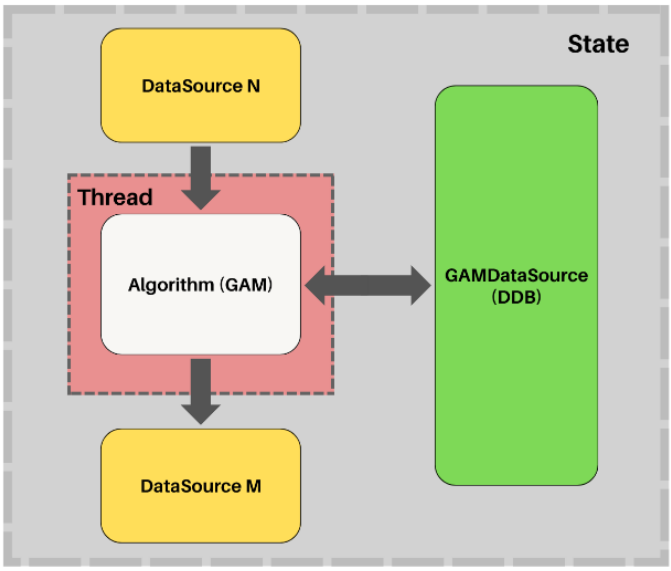


Figure 13: Overview of the structure of the simplest functional MARTe real-time application

2.4.5. Simulink and MARTe for Control Applications

Simulink is featured in various current and envisioned PCSs [21], [22], [39], [57] and is regarded as the "preferred tool for the development of control algorithms" in fusion research, due to its powerful and intuitive block diagram environment, which allows for rapid prototyping and simulation of control algorithms [29]. Indeed, [57] and [22] make a strong case for adopting Simulink to expedite the design of PCSs. Despite its inability to operate in RT [22], the Embedded Coder Simulink add-on enables the automatic translation of the Simulink models into the C code used by RT systems, thus avoiding any translation errors associated with the error-prone and lengthy process of developing code for RTAs [21], [29].

Thanks to this Simulink capability, the critical usability issue associated with the complex MARTe configuration [21], [31], [33] can be overcome when prototyping advanced control strategies, especially if it entails the development of application-specific GAMs. In this context, integrating Simulink with MARTe effectively separates the roles of control and software engineers when designing a control solution using the framework [13]. Indeed, the automatic translation of control solutions created in Simulink into RT system components represents "an important factor in the overall quality of the system" [29] and significantly enhances the user experience with MARTe [22].

A RT control system can be represented at a high-level view as a set of components executing control computation or I/O operations in a block-based fashion, where blocks are interconnected to form a complete system, similar to the organization of both MARTe and

Simulink [21]. In fact, [29] states that "there is a strong conceptual similarity between Simulink blocks and MARTe GAMs", further supporting the case for their integration. Therefore, the path towards enhancing PCSs involves the full integration of Simulink into MARTe2 [21].

The 'SimulinkWrapperGAM' acts as a bridge between the two frameworks by automatically implementing the functions specified in Simulink [22]. The GAM seamlessly wraps the code generated by the Simulink add-on, eliminating the need for C++ code to develop a MARTe2 component for that purpose [21]. It operates by opening the dynamically linked shared object library ('.so' file) produced by the Simulink add-on, parsing parameters and I/O ports information, and finally generating internal dictionaries to represent them [22]. Additionally, errors in setup may occasionally be difficult to interpret, leading to "obscure crashes" upon execution [21]. Therefore, another major contribution of the 'SimulinkWrapperGAM' is providing better clarity in the analysis of logged errors, as the errors launched by the GAM strictly detail the problems that may arise [22].

[29] advocates for a well-round, comprehensive solution for the rapid development of RTAs by integrating MARTe, Simulink and MDSplus. The high-level description in MDSplus enables automatic generation of complex MARTe2 configuration files by specifying components and their corresponding connections within the MDSplus database in a Simulink-like manner [21], [31], [33], significantly reducing manual effort and minimizing the potential for errors in setup. Therefore, adopting this multifaceted approach allows the automated translation of a Simulink-based PCS solution into low-level MARTe2 configuration [21].

3. RESEARCH METHODOLOGY

This chapter outlines the methodology adopted to conduct the research presented in this dissertation. In this sense, it is structured according to the three major phases proposed in the "Introduction" chapter, each corresponding to a specific stage in the development and validation of a RT PCS based on the MARTe2 framework, following an approach similar to the one described in [58]. These phases build sequentially, starting from the foundational steps required to operate MARTe2, progressing through the integration of model-based design tools such as Simulink, and culminating in the development and validation of a RT estimation approach for the plasma current centroid position in ISTTOK.

3.1. MARTe2 Engine Setup and Standalone Validation

3.1.1. Compilation and Configuration of MARTe2

Adopting a well-established framework for the development of RT systems and model-based design provides an "effective approach" to accelerate the commissioning of a new PCS [59]. For this purpose, the MARTe2 framework was selected as the preferred platform for the development of this research, especially because this choice would eventually represent an evolution in the ISTTOK CODAS.

Since, the MARTe2 code is not distributed as an open-source software, its setup mandates compilation of the source code provided in the MARTe2 Gitlab repository[12]. This process enables the design of RTAs within the framework and is detailed in "Appendix A: MARTe2 Compilation Guide". The compilation was carried out on a Linux-based system using standard development tools such as 'make' based on predefined Makefiles provided within the source code repositories [25], allowing for the expedite setup of the RT platform [28].

Once compilation was finished, a series of preliminary tests were conducted to verify the correct configuration of the MARTe2 framework. RT simulation not only plays a crucial role in the development of RT control systems but also ensures their successful operation when applied in real experiments [54], [59], hence this first stage involved simulation of RTAs for simple purposes.

[12] For more information, refer to the MARTe2 GitLab repository: https://vcis-gitlab.f4e.europa.eu/aneto/MARTe2.

The training material made available in the MARTe2-demos-padova repository[13] proved extremely useful. In this context, the provided tutorial configuration files enabled not only a quick verification of a successful MARTe2 engine compilation but also facilitated a faster learning curve, reducing, if not completely eliminating, the need for expert guidance on the framework [31], [33].

3.1.2. Executing a Real-Time Application with MARTe2

As a first experimental MARTe application, the simplest configuration file was executed by running the following initialization hook in a bash terminal inserted into the repository where 'Main.sh' is located: './Main.sh -l RealTimeLoader -f ../Configurations/RTApp-1-1.cfg -s State1'. In this sense, while the tutorial repository provides a default 'Main.sh' file, the one utilized throughout this research consists of a customized instance of this startup script tailored to the specific development environment. Unlike the default version, which assumes a fixed directory structure and hardcoded paths, the use of latter requires manually sourcing the environment setup script 'marte2-exports.sh' with the 'source marte2-exports.sh' command.

This customized startup script is responsible for preparing the runtime environment by exporting system paths, updating the dynamic library path ('LD_LIBRARY_PATH') with all necessary shared libraries for core MARTe2 components and demo modules, and finally invoking the 'MARTeApp.ex' executable with user-specified arguments. This modular and explicit setup is advantageous, as it facilitates the correct execution of MARTe, improves the reproducibility of results, and simplifies the integration of new components, such as additional user-developed GAMs, without requiring extensive reconfiguration of the execution script [23]. As a result, a more maintainable and scalable structure for continued development and experimentation is achieved [13].

After calling the startup script, the command must also include arguments to properly bootstrap the MARTe executor. The accepted arguments and their descriptions are provided in Table 3. It is worth noting that, if neither the '-s' nor '-m' options are specified, the application will simply start and remain idle, waiting for a message or an event to proceed.

[13] For more information, refer to the MARTe2-demos-padova GitLab repository: https://vcis-gitlab.f4e.europa.eu/aneto/MARTe2-demos-padova/.

Table 3: MARTe Startup Arguments

Arguments	Description
-l	Mandatory argument that defines the loader class to be used (e.g. 'RealTimeLoader').
-f	Mandatory argument that defines the path containing the RTA configuration file.
-p	Optional argument that specifies the configuration file format.
-s	Argument that directly defines the first state of the ISS upon loading.
-m	Argument that defines a message to be issued to a target object (e.g. 'StateMachine:START').
-c	Argument that defines the default CPU affinity for objects without specification.
-t	Argument used to replace build tokens defined within the configuration file.
-g	Argument that specifies the scheduler group to be used.

The 'RTApp-1-1.cfg' configuration offers a simple demonstration of MARTe, as it is designed to simply output time signals from 'LinuxTimer'. The RTA structure consists of blocks that define the components shown in Figure 10, where the RTA serves as a receptacle of blocks that define the functions (i.e. GAMs), DSs, ISS states, and the RT scheduler.

Although extremely simple, the resulting configuration consists of 70 lines of code. As is done in the 'RTApp-HTTP-1.cfg', by adding explicit ISS configuration and other MARTe services to this configuration, this number of lines nearly quadruples, highlighting the significant complexity and size of MARTe configuration files [13], [21], [31]. When MARTe is executed with the aforementioned bash script, several lines initially appear in the logger, followed by the output signals handled by the 'LoggerDataSource'. The log for the first 5 seconds of runtime and the subsequent stoppage of MARTe via user input is shown in Figure 14. It is worth noting that analysis of the log revealed that the 'LinuxTimer' issues time values in microseconds, since these were displayed in the order of 10^6 units for a 1 Hz cycle frequency.

```
[Information - RealTimeLoader.cpp:111]: Started application in state State1
[Information - MARTeApp.cpp:135]: Application starting
[Information - LoggerBroker.cpp:152]: Counter [0:0]:1
[Information - LoggerBroker.cpp:152]: Time [0:0]:0
[Information - LoggerBroker.cpp:152]: Counter [0:0]:2
[Information - LoggerBroker.cpp:152]: Time [0:0]:2000000
[Information - LoggerBroker.cpp:152]: Counter [0:0]:3
[Information - LoggerBroker.cpp:152]: Time [0:0]:3000000
[Information - LoggerBroker.cpp:152]: Counter [0:0]:4
[Information - LoggerBroker.cpp:152]: Time [0:0]:4000000
[Information - LoggerBroker.cpp:152]: Counter [0:0]:5
[Information - LoggerBroker.cpp:152]: Time [0:0]:5000000
^C[Information - Bootstrap.cpp:73]: Application recieved SIGINT.

[Information - Bootstrap.cpp:87]: Application successfully stopped.

[NoError - MARTeApp.cpp:137]: Application terminated
```

Figure 14: Log for the first 5 seconds runtime and subsequent MARTe termination of 'RTApp-1-1.cfg'

3.1.3. Implementation of the Water Tank Problem

Once proper setup and understanding of the framework were achieved, a reliable approach based on modeling and simulations to test control algorithms was followed [54], starting with a model derived from the classic water tank problem [36]. The first iteration of the problem was modeled with the framework as an extremely oversimplified static system, meaning it lacked control. Imaginary valve and pump actions were triggered based solely on whether the water level, read from an immutable CSV, was, respectively, either below or above a defined setpoint. This step served as a follow-up to the training previously conducted.

By following this approach, a deeper understanding of working with DSs was gained, as the built configuration employed FileReader and FileReader DSs. Moreover, in addition to the previously applied IOGAM for managing data across different DSs, a GAM for converting data types, as well as another for applying Boolean logic calculations were also employed. This step-by-step procedure was effective in increasing familiarity with MARTe2, thereby facilitating the initiation of increasingly complex configurations.

Besides generating the CSV file containing the time values, water level readings, and valve and pump triggers logical values, this configuration also outputs a log similar to the one shown in Figure 14, with the addition of the 'valve_on' and 'pump_on' Boolean signals. It is worth mentioning that the first line in the output CSV file is a comment with headers consisting of the signal names followed by their data type in parentheses and number of elements in brackets. For the sake of brevity, the log is not displayed herein.

The classic water tank problem is modeled as a first-order differential equation and is of significant interest because it encompasses the fundamental elements of embedded control systems [60]. In fact, this model provides a basis for studying feedback control in RTAs, where the water level is compared to a pre-established reference, and a PID controller

computes the error between the two values to generate control signals with the ultimate goal of eliminating the error [36]. In this context, by simplifying the equation presented in [36] that defines the behavior of the water level in a cylindrical tank with a valve for water insertion and another for removal, the system can be modeled based on Equation (9).

$$\frac{dh(t)}{dt} = (q_i(t) - q_o(t)) \cdot \frac{1}{A}$$
$$q_i(t) = b \cdot V(t) \tag{9}$$
$$q_o(t) = c \cdot \sqrt{h(t)}$$

where h is the water level, q_i and q_o are, respectively, the inlet and outlet flows, A is the area of the water tank cross-section, V is an action signal, understood as a voltage applied to the pump, and b and c are constants related to, respectively, the proportional relationship between the input voltage and the inlet flow rate generated by the pump and the physical characteristics of the outlet. The modeled system is illustrated in Figure 15.

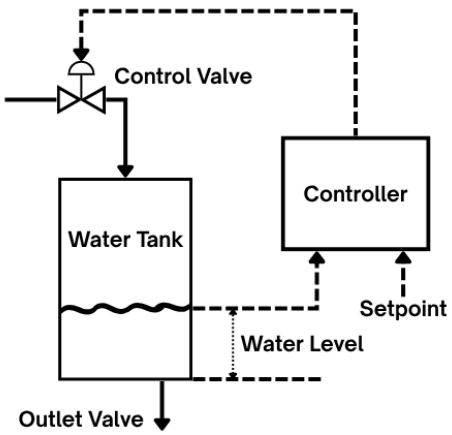


Figure 15: Water tank model used for feedback control analysis

The non-linearity of this model arises from Torricelli's theorem, which is derived from Bernoulli's principle of fluid dynamics. It states that the velocity of a fluid flowing out of an orifice with area A_o, under the influence of gravity (g), is proportional to the square root of the height of the fluid column above the orifice [61]. This velocity v is given by Equation (10).

$$v = \sqrt{\frac{2 \cdot g \cdot h}{1 - \frac{{A_o}^2}{A^2}}} \tag{10}$$

Given that the area of the orifice is considered negligible in comparison to the cross-sectional area of the tank, the velocity of the water flowing out of the orifice simplifies to:

$$v \approx \sqrt{2 \cdot g \cdot h} \tag{11}$$

Consequently, by approximating the discharge coefficient as unity, the outlet flow is expressed as:

$$q_o \approx A_o \cdot v = A_o \cdot \sqrt{2 \cdot g} \cdot \sqrt{h} = c \cdot \sqrt{h} \tag{12}$$

As modern embedded control systems operate using digital computers, this model must be discretized [52]. Using the forward Euler method [52], the model is discretized as follows:

$$h[k+1] = h[k] + \frac{1}{A} \cdot \left(b \cdot V[k] - c \cdot \sqrt{h[k]}\right) \cdot \Delta t \tag{13}$$

Model-based simulations became feasible, leading to the development of both a MARTe configuration and a Simulink model for benchmarking the MARTe solution. In this context, creating a configuration for this purpose involved the laborious task of developing a GAM for the WTM itself [13], as it was initially believed that the 'MathExpressionGAM' class lacked the capability to define the non-linear expression. This process required the help of MARTe2 experts, as proper documentation for the development of new GAMs was not clearly distributed in the training repository.

Creating user-defined GAMs encompasses writing code in two distinct but related C++ files [25]. The primary task in developing a GAM is focused on constructing a source file to define the algorithm and perform signal verification, ultimately ensuring that it operates only when correctly configured. Additionally, a header file must be created to declare the 'WaterTankGAM' class, including its public, protected, and private members, as well as definitions for inline methods that need to be visible to the compiler.

Since the main goal of this dissertation is to leverage Simulink to bypass the GAM development process, this topic is not fully covered here; however, for better understanding,

Equation (13) is defined within the GAM and declared in the 'WaterTankGAM::Execute()' method, which can be visualized in the GitHub repository dedicated for the development of this dissertation[14]. The 'WaterTankGAM' was created by executing a Makefile-based bash command ('make -C WaterTankGAM -f Makefile.gcc') in a dedicated terminal located in a parent repository containing the repository with the GAM source and header files, along with the Makefile instruction files, once variables from 'marte2-exports.sh' were correctly sourced. Finally, the corresponding path to the component built with this command had to be defined in the custom startup script.

On the other hand, to correctly benchmark the MARTe solution, the water tank was effortlessly modeled as a MATLAB function block defining the same algorithm as the 'WaterTankGAM'. With these components in place, simple PID-based control systems were developed in the two frameworks. For the MARTe setup shown in Figure 16(a), a configuration was created that included a 'ConstantGAM' to generate a reference setpoint, a 'PIDGAM' to configure the controller, the 'WaterTankGAM', and a UDP sender for transferring data to an external Python script running in parallel to plot the system response.

In Simulink, a simple block diagram was created for the same purpose, as shown in Figure 16(b), benefiting from built-in blocks in combination with the custom MATLAB function block. As stated in [29], Figure 16 further supports the claim that the frameworks share a strong conceptual similarity, with Simulink offering the added benefit of a more accessible Graphical User Interface. It is worth noting that, due to the way time is handled by the two frameworks, both algorithms for the WTM are identical, with the sole exception that, in MARTe2, Δt from Equation (13) had to be multiplied by 10^{-6}.

[14] For more information, refer to the MARTe2-WaterTank GitHub repository: https://github.com/felipe-aveiro/MARTe2-WaterTank.

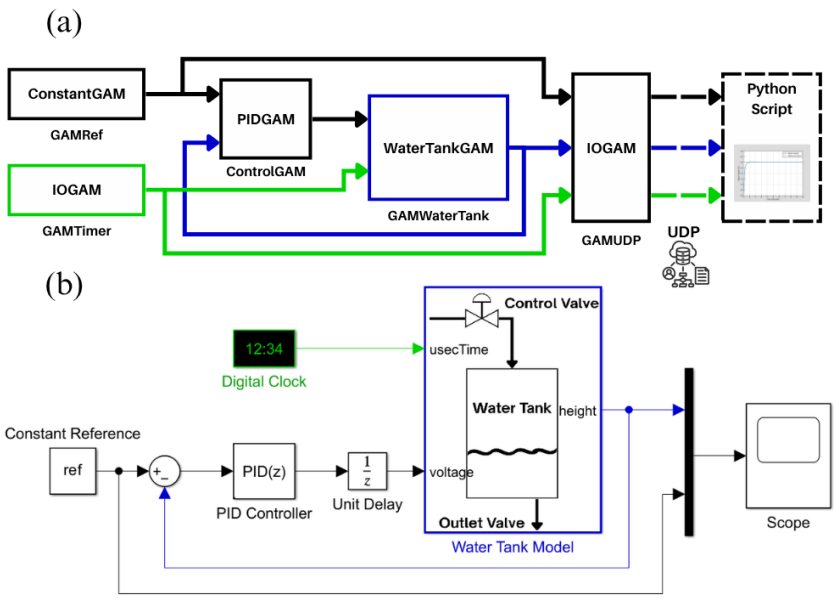


Figure 16: Water tank feedback control simulation models, showing **(a)** the block-based representation of MARTe configuration, and **(b)** the Simulink water tank model used for feedback control analysis

Table 4 presents the parameters set in the WTM configuration in both frameworks. These parameters were specifically chosen to represent a physically realistic water tank, simulating its behavior in 60 seconds of runtime. Concurrently, the PID constants were tuned through trial-and-error, as the dynamics of the model, primarily due to the saturation of the controller, prevented the use of the well-established Ziegler-Nichols method [36].

Table 4: Water tank model and control parameters used in the simulation

Water Tank Parameters	Value
Tank cross-sectional area A [m^2]	2.0
Initial height $h[0]$ [m]	0
Maximum height h_{max} [m]	1.5
Minimum height h_{min} [m]	0
a [$m^{2.5}/s$]	0.001
b [$m^3/s/V$]	0.01
Maximum q_i [m^3/s]	0.05

Control Parameters	Value
Sampling time [s]	0.001
Reference height [m]	0.8
Maximum Voltage [V]	5.0
Minimum Voltage [V]	0
K_P	30.0
K_I	0.015
K_D	0

Figure 17 shows the highly satisfactory result obtained through simulation, with the system exhibiting accurate tracking and no observable overshoot. Most importantly, it

demonstrates the equivalent behavior achieved through simulation in both MARTe2 (a) and Simulink (b), thus validating the use of MARTe2 for RT control [31].

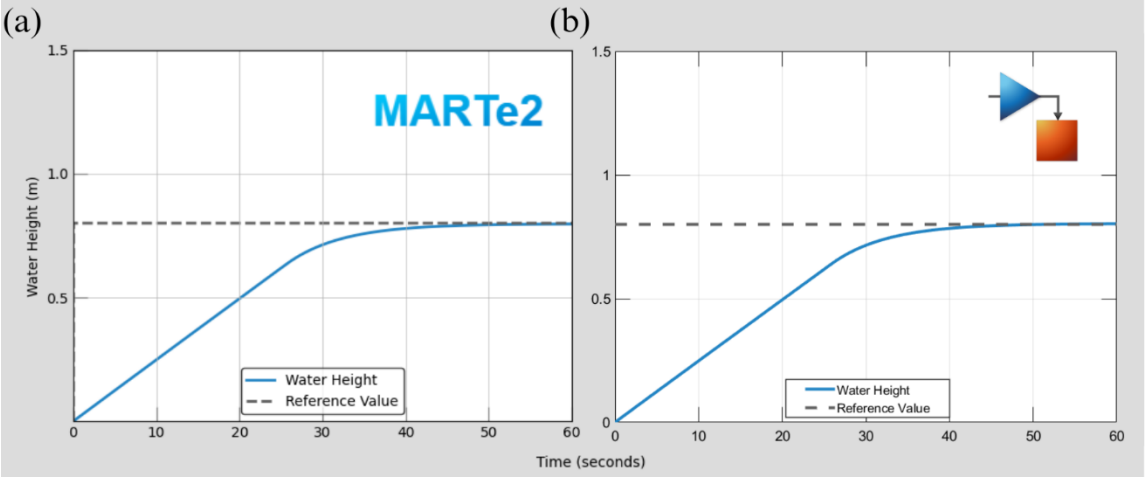


Figure 17: Comparison of the water tank feedback control system responses obtained from **(a)** the MARTe2 simulation and **(b)** the Simulink simulation

3.2. Simulink Integration with MARTe2

3.2.1. C/C++ Code Generation from Simulink Models

To bridge the gap between the graphical model design offered by Simulink and its integration with the MARTe2 framework, the logic implemented in the WTM defined using Simulink block notation had to be translated into portable C++ code for RT deployment [22]. This was effortlessly accomplished with the Embedded Coder add-on [11], [14], [21], which generates the Linux shared libraries interpreted by MARTe2 via the 'SimulinkWrapperGAM'. For the following steps, the MATLAB release 24.2.0.2923080 (R2024b) Update 6 was used.

From the model previously used to benchmark MARTe2 for RT control in standalone mode, a new one was created containing only the WTM block, as the goal was to enable direct comparison to the 'WaterTankGAM'. For the subsequent integration with MARTe2 and correct generation of the low-level code of the WTM, proper setup was essential. For instance, porting input and output signals of the WTM required adding inport and outport blocks. Moreover, a signal naming convention, where the corresponding signal names are assigned to the connections between these additional blocks and others, was mandatory. The porting blocks are required to link GAM signals to the model [22].

Additionally, the tunable parameters declared in the 'WaterTankGAM' were defined as variables in the MATLAB workspace and then invoked within the block's algorithm,

enabling their subsequent variation in the GAM declaration within configuration files. The last step prior to generating the C++ code involved running a small script in the MATLAB terminal after attributing the name of the model to the variable 'model_name'. This script is mandatory, as it determines the appropriate model settings required by the 'SimulinkWrapperGAM'. The newly adapted version of the WTM, ready for conversion to RT code, is shown in Figure 18, and the corresponding MATLAB script is provided in "Annex A: MATLAB Script to Setup Simulink Models for MARTe2 Integration".

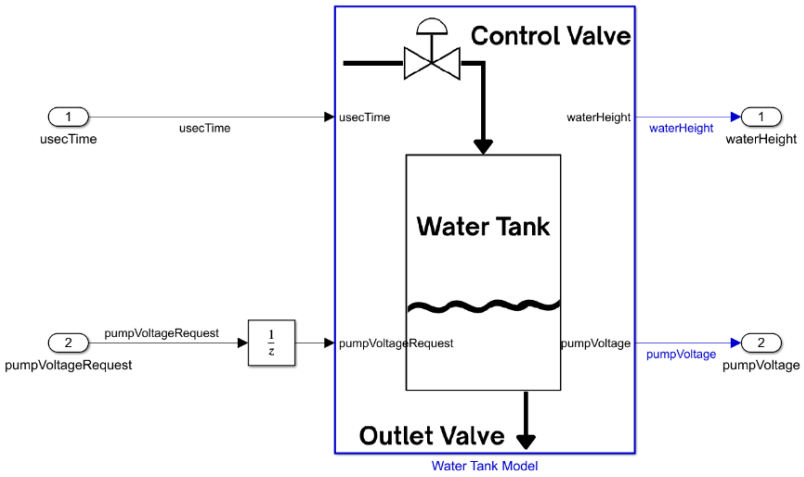


Figure 18: Simulink water tank model for MARTe2 integration

Once the procedure is completed, a number of files are generated, including a Simulink cache archive and a folder corresponding to the embedded RT target. This folder contains source and header files, along with build support components. Additionally, and most importantly, a shared object file for Linux systems is also created, which is the compiled binary file interpreted and executed by external RTAs.

3.2.2. Integration with MARTe2 via 'SimulinkWrapperGAM'

The ability to bypass the lenghty process of manually writing low-level code for RTAs represents the main motivation for leveraging Simulink and the Embedded Coder add-on for quickly designing control solutions in MARTe [29]. In fact, as noted in [22], the "learning curve" for deploying RT algorithms was significantly reduced in comparison to the previous process of creating a GAM for the same purpose. Due to the lack of proficiency in low-level coding and to the user-friendliness of Simulink, the WTM algorithm was not only ready for RT deployment in a much shorter period than the 'WaterTankGAM', but it was also completed without the need for GAM developers.

Assigning the MATLAB installation directory path to the environment variable 'MATLAB_DIR' before compiling MARTe2 was a requirement of the wrapper. Additionally, a directory containing the shared object files generated from the models was created, and its path was appended to the 'LD_LIBRARY_PATH' variable in the custom startup script so that the system locates the required dynamic libraries when executing MARTe2. With the WTM binary file placed in the dedicated directory, the integration of Simulink for modeling alongside MARTe2 for the RTA itself was finally achieved.

3.2.3. Validation and Behavior Analysis

Some nuances related to signal naming and parameters declaration were observed, as the functionality of the 'SimulinkWrapperGAM' is sensitive to the correct porting of information. For instance, unlike other GAMs, it only operates when signals are named exactly as defined in the wrapped model, and parameters must be declared in configuration files with their type defined in parentheses preceding their corresponding value.

Conversely, debugging was found to be straightforward, as error messages declared by the 'SimulinkWrapperGAM' clearly indicate the source of the error, further highlighting its advantage in terms of ease of use. As expected, the simulation of the WTM with Simulink integrated with MARTe2 (a) produced results equivalent to those obtained using the 'WaterTankGAM' (b), as can be seen in Figure 19.

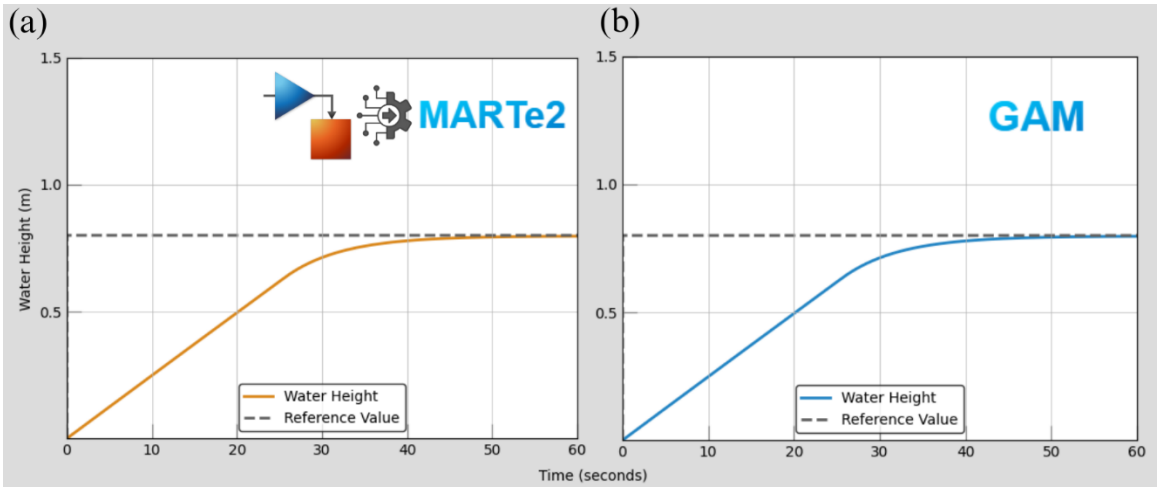


Figure 19: Water tank feedback control system response **(a)** using the Simulink water model integrated into MARTe2 with the 'SimulinkWrapperGAM' and **(b)** using the 'WaterTankGAM'

Moreover, no significant difference was observed between the behavior of the two solutions in terms of jitter, despite the addition of a GAM to convert the time signal to 'float64', the datatype of the time signal ('usecTime') defined in the Simulink model. In fact,

the mean absolute error (MAE) and the root mean squared error (RMSE) for the cycle duration distributions are approximately 0.752% and 1.357%, respectively, indicating a high level of agreement between both solutions.

Figure 20 shows cycle duration histograms from subsequent executions of these two solutions, highlighting their similarity. Therefore, these results not only validate the modeling approach based on using the Simulink integration with MARTe2, but also emphasize the similar jitter behavior in a RT environment when using Simulink models instead of user-developed GAMs. Additionally, this integration between the two frameworks has previously been employed in fusion experiments, such as in TCV [22], supporting its suitability for RTAs and reinforcing its use in the final stage of this research.

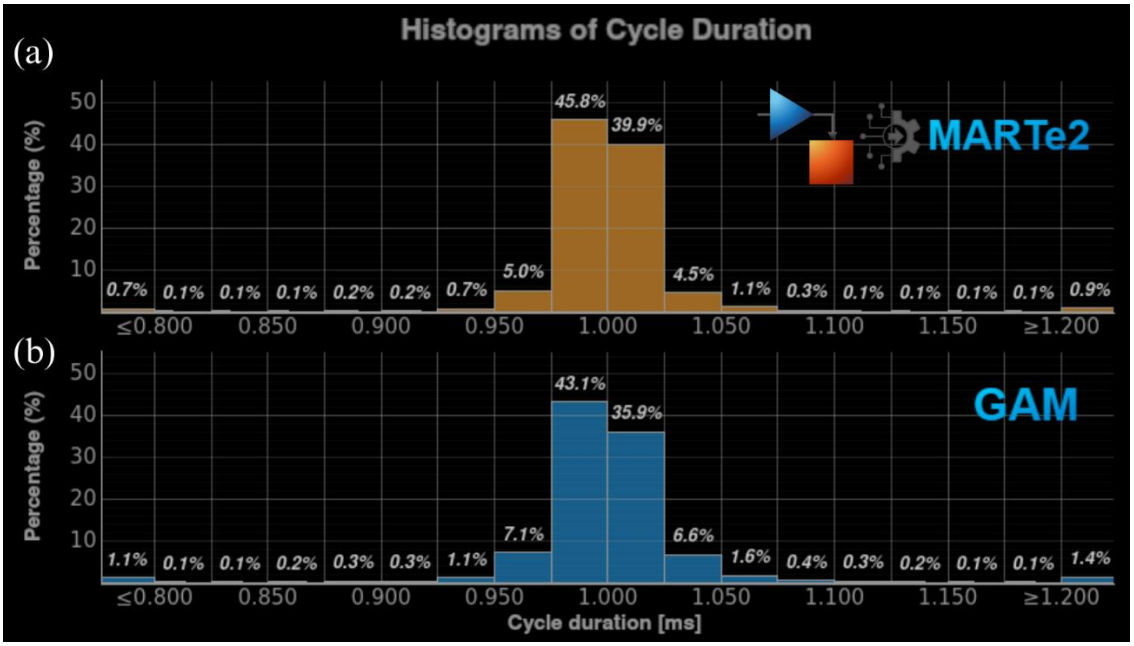


Figure 20: Comparison of cycle duration histograms for different MARTe2 implementations, namely **(a)** the Simulink water tank model integrated into MARTe2 through the 'SimulinkWrapperGAM'and **(b)** the 'WaterTankGAM' implementation

3.3. Plasma Current Centroid Position Estimation in ISTTOK

3.3.1. Acquisition, Processing, and Calibration of Plasma Diagnostics for Position Estimation

The first step in the development of the proposed estimation approach involves the acquisition of signals from the main diagnostic systems of the ISTTOK CODAS. Since plasma position is not a directly measurable parameter [2], [3], [40], [41], these signals are of paramount importance, as they provide measurements for the necessary estimation of the

plasma current centroid position. Since reconstruction of the plasma state from available measurements is essential for feedback control of its parameters [58], the ISTTOK magnetic and electric plasma diagnostics were employed for this purpose.

An oversimplified version of the plasma reconstruction described in [56] was implemented using the poloidal set of 12 Mirnov coils. Due to ISTTOK's large aspect ratio, this approach yields a relatively accurate approximation of the plasma radial and vertical positions, which are computed using similar expressions, except for an added constant term in the radial position calculation. In parallel, reconstruction from measurements obtained with the poloidal array of 4 Langmuir probes was implemented based on the method outlined in [62]. Equations (14) and (15) were used to determine the plasma radial and vertical position from the Mirnov and Langmuir diagnostics, respectively.

$$\begin{aligned} R_p &= \frac{\sum_{i=1}^{N=12} B_{\mathrm{Mirnov}_i} \cdot R_{\mathrm{Mirnov}_i}}{\sum_{i=1}^{N=12} B_{\mathrm{Mirnov}_i}} + \frac{a^2}{4 \cdot R_0} \\ Z_p &= \frac{\sum_{i=1}^{N=12} B_{\mathrm{Mirnov}_i} \cdot Z_{\mathrm{Mirnov}_i}}{\sum_{i=1}^{N=12} B_{\mathrm{Mirnov}_i}} \end{aligned} \tag{14}$$

$$\begin{aligned} R_p &= C_1 \cdot \left(V_{f_{\mathrm{outer}}} - V_{f_{\mathrm{inner}}}\right) + C_2 \\ Z_p &= C_3 \cdot \left(V_{f_{\mathrm{top}}} - V_{f_{\mathrm{bottom}}}\right) + C_4 \end{aligned} \tag{15}$$

R_p and Z_p are, respectively, the plasma radial and vertical position, B_{Mirnov_i} is the magnetic field intensity measured by the i-th Mirnov coil and R_{Mirnov_i} and Z_{Mirnov_i} are the corresponding relative radial and vertical positions of that coil. Regarding the Langmuir reconstruction, $V_{f_{\mathrm{top}}}$, $V_{f_{\mathrm{outer}}}$, $V_{f_{\mathrm{bottom}}}$, $V_{f_{\mathrm{inner}}}$ are the floating potentials measured at the top, outer, bottom, and inner Langmuir probes, respectively. The constants C_1, C_2, C_3, C_4 were derived based on the Mirnov reconstruction, since it is considered the main diagnostic tool for plasma position estimation.

As is common with most sensors, the raw measurements returned by the diagnostics are expressed in voltage values, therefore they typically require processing to extract meaningful physical quantities. In this context, the Langmuir probes directly acquire floating potential values, eliminating the need for conversion of the probes' signals, which can be implemented as-is in the corresponding reconstruction model. On the other hand, the Mirnov coils acquire voltage signals that are proportional to the time derivative of the magnetic flux, requiring an integration and calibration algorithm to retrieve the magnetic field intensity.

In a discrete-time system, this integration is approximated by summing the ADC samples and scaling them according to the sampling frequency. Since the magnetic data is

decimated and sampled in RT, the acquired analog signals are digitized by the ADC, resulting in the discrete values – referred to here as ADC – which are used in computation. Using Equations (2) and (5), this leads to the following conversion algorithm:

$$B_{\mathrm{Mirnov}_i} = (-1) \cdot \frac{K_i \cdot \mathrm{pol}_i \cdot \mathrm{ADC}_i}{S_{\mathrm{eff}_i} \cdot f_{\mathrm{RT}} \cdot d_{\mathrm{RT}}} \tag{16}$$

where pol and K are, respectively, the polarity and gain of a Mirnov coil, and the index i accounts for coil-dependent parameters. Additionally, f_{RT} is the RT system clock frequency, d_{RT} is the RT decimation factor, and the negative sign originates from Faraday's law as represented in Equation (2). "Appendix B: Parameters for Magnetic Diagnostics Signal Processing" presents the coil-dependent parameters according to the coil numbering shown in Figure 8, along with the RT system clock frequency and RT decimation factor. This algorithm was subsequently tested in a simulation environment using shot data from previous shots either retrieved from MDSplus or from the ISTTOK Shared Data Access System (SDAS).

A new ATCA board was installed as part of the MARTe2 upgrade and it was observed that the magnetic data acquired by this new ATCA board (a) – later stored in the MDSplus database – differed from the data stored in the SDAS (b), which originated from the previous one. Figure 21 illustrates this discrepancy using magnetic measurements of shot 53071 acquired by the 12 Mirnov coils. It is worth noting that the difference in time stamps between the two plots solely arises from the way time is stored in each database.

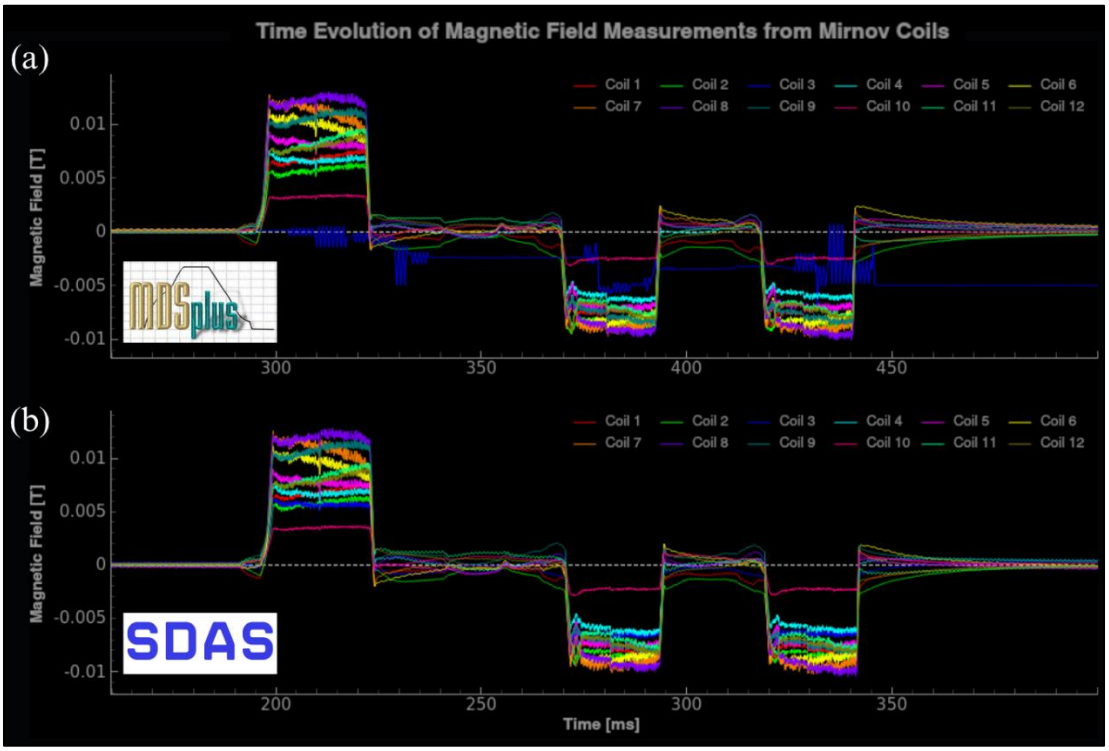


Figure 21: Time evolution of magnetic field measurements from 12 Mirnov coils for shot 53071, obtained through **(a)** MDSplus (new ATCA board) and **(b)** the ISTTOK Shared Data Access System (old ATCA board)

Analysis of Figure 21 reveals that coil 3 on the new ATCA board malfunctioned throughout the discharge, possibly due to cabling issues. Moreover, coil 10 was found to exhibit faulty behavior in both instances, showing a distinct response compared to the other coils, likely due to a direction bias within the tokamak. These findings were critical for the subsequent design of the solution with KFs.

Since none of the available diagnostics could provide an absolutely certain plasma position, it was necessary to conduct an additional indirect validation step. To ensure the reliability of the acquired magnetic data, which is considered the primary diagnostic source and was later used as reference for determining the Langmuir coefficients in Equation (15), a comparison was performed between the plasma current reconstructed from Mirnov coil signals and the measurements obtained with a Rogowski coil. This diagnostic tool is designed to provide direct and integrated measurements of the plasma AC by sensing its rate of change, therefore serving as reference for validating the magnetic data.

$$I_p = \frac{1}{\mu_0} \cdot \oint B \cdot dl \approx \frac{2 \cdot \pi \cdot r_{\text{Mirnov}}}{N \cdot \mu_0} \cdot \sum_{i=1}^{N=12} B_{\text{Mirnov}_i} \tag{17}$$

Equation (17) presents the plasma current magnetic reconstruction algorithm implemented in MARTe2 to validate the magnetic measurements, where N denotes the

number of Mirnov coils and r_{Mirnov} is the radial distance from the VV center to each coil according to ISTTOK's configuration. It is based on a discretization of Ampère's law, shown first in its continuous form, under the assumption the plasma behaves as a single current filament. A similar procedure was performed in [11] to validate the multifilamentary approach adopted for plasma position estimation.

Figure 22 illustrates the comparison between the magnetic reconstruction and plasma current obtained with the Rogowski coil for shot 53071, highlighting the close agreement between both signals, validating the accuracy of the acquired magnetic data. In fact, this is further supported by the computed error metrics, where the MAE and the RMSE were approximately 125.080 A and 230.210 A, respectively, corresponding to approximately 3.2% and 5.8% of the nominal plasma current in AC mode (4 kA). Such discrepancy can be attributed to the approximation of the contour integral to a discretization of 12 points and the faulty behavior of coil 10, as the calculations were performed using data from the SDAS.

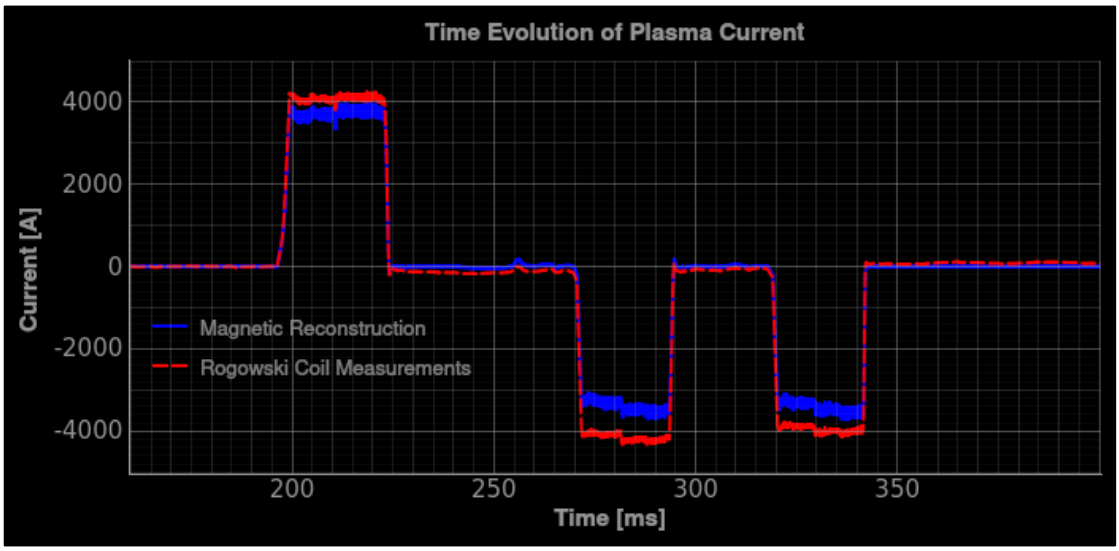


Figure 22: Comparison of plasma current obtained from magnetic reconstruction and Rogowski coil measurements for shot 53071

Once the magnetic measurements were validated, Equation (14) was implemented in MARTe2 to estimate the plasma current centroid position from Mirnov coil signals for shot 53071. Figure 23 shows the estimated plasma radial (a) and vertical (b) position throughout the discharge, where the dashed red lines represent the theoretical limit for the plasma position imposed by the limiter radius (0.085 m). These limits account for the plasma column being highly deformable, as they are based on the distance between the centroid and the limiter rather than accounting for the size of the column. In contrast, the implemented reconstruction model assumes a rigid plasma column, introducing an important conceptual

difference when interpreting these bounds. Therefore, physical contact with the limiter may occur even when the estimated centroid remains within the theoretical boundaries, emphasizing the need for accurate and robust control to keep it away from the internal walls.

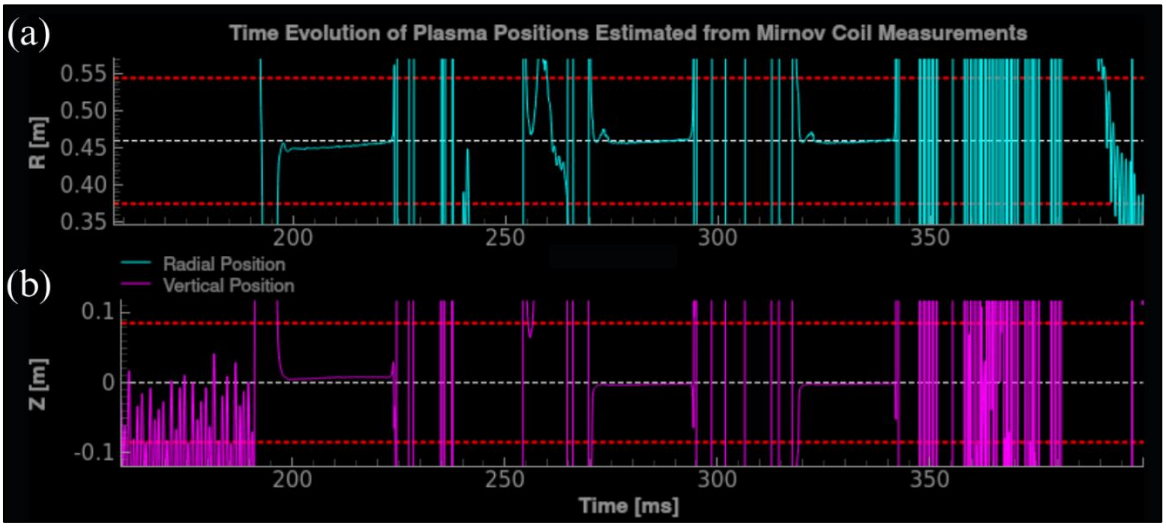


Figure 23: Plasma position reconstruction from magnetic measurements for shot 53071, showing **(a)** the radial and **(b)** the vertical components

As expected, the simplistic model introduced spikes in the estimated vertical and radial positions not only during the AC transitions but also when the sum of the magnetic fields sensed by the coils neared zero, leading to physically inconsistent estimates of the plasma position. This outcome highlights the aforementioned inoperability of the magnetic diagnostics in these scenarios [35]. Nevertheless, during the plasma AC flat top phases, which correspond to the intervals where the estimates exhibit less erratic behavior, the estimated positions remained within the physical bounds.

The next step consisted of estimating the plasma position from electric measurements, which involved solving the forward problem of deriving the Langmuir coefficients based on magnetic reconstructions. Given the strong presence of outliers, mainly associated with the physically inconsistent position estimates, a Huber robust regression was applied instead of a standard linear regression to calculate the coefficients for each selected shot. This method was chosen because it provides robustness against these anomalies, ensuring that the coefficients were not affected by the unrealistic position estimates [63], [64]. Nevertheless, an additional hard filtering step was applied to remove the physically unrealistic estimates, as their presence severely degraded the consistency of the models.

The Langmuir coefficients were then computed as a weighted average of individual shot coefficients obtained for ten shots with distinct characteristics that were selected to

represent a wide range of performance scenarios. The weights were assigned based on a scoring method according to a criteria considering pulse regularity or stability, number of cycles or duration, and current level, with the respective importances of 50%, 30%, and 20%. Each criterion was scored from 1 (substandard) to 3 (excellent), and the weighted sum of these scores determined the resulting weight for each shot.

In this context, higher weights were attributed to shots with highly regular pulses, longer durations, and nominal current levels, whereas those with irregular behavior and shorter durations were assigned lower weights. Table 5 provides an overview of the selected shots, along with their corresponding descriptions and attributed weights.

Table 5: Selected shots for weighted averaging process for determination of Langmuir coefficients

Shot number	Description	Weight
45754	Multiple AC pulses; irregular waveform; premature terminations	1.35
45967	Several positive pulses; extremely well-formed waveform; nominal current	3.0
46241	Multiple AC pulses; excellent regularity; nominal current (ideal case)	3.0
52856	Two AC pulses; irregular waveform; low current	1.0
52857	Six AC pulses; irregular waveform; low current	1.35
53058	One positive + three negative pulses; slight irregularity; nominal current	2.06
53071	One positive + two negative pulses; good regularity; slight current drop	2.41
53099	One positive + three negative pulses; low-quality pulses; low current	1.82
53105	Single AC pulse; regular waveform; slightly low current	2.06
53197	Three AC pulses; irregular waveform; low current	1.35

The Langmuir models obtained for each shot demonstrated overall good agreement with the corresponding magnetic reconstruction. Despite this, the weighted RMSEs for the radial and vertical positions were approximately 1.765 cm and 1.679 cm, while the corresponding median absolute errors were 0.537 cm and 0.654 cm, indicating that typical estimation errors remained well below 1 cm for most of the plasma duration. These values were satisfactory given the influence of outliers introduced by the plasma AC reversals.

The global coefficients derived from this process are presented in Table 6, while Figure 24 shows a comparison between the plasma positions estimated from magnetic measurements and the resulting reconstruction using electric probe signals for shot 53071. As reported in [62], the comparison highlights that the positions estimated from the two reconstructions follow a similar pattern except for the spikes present in the magnetic reconstruction.

Table 6: Resulting global Langmuir coefficients with mantissas rounded to five decimal places

Langmuir Constants	**C1**	**C2**	**C3**	**C4**
	$-1.33105 \cdot 10^{-4}$	$4.58521 \cdot 10^{-1}$	$-1.29928 \cdot 10^{-5}$	$1.81104 \cdot 10^{-3}$

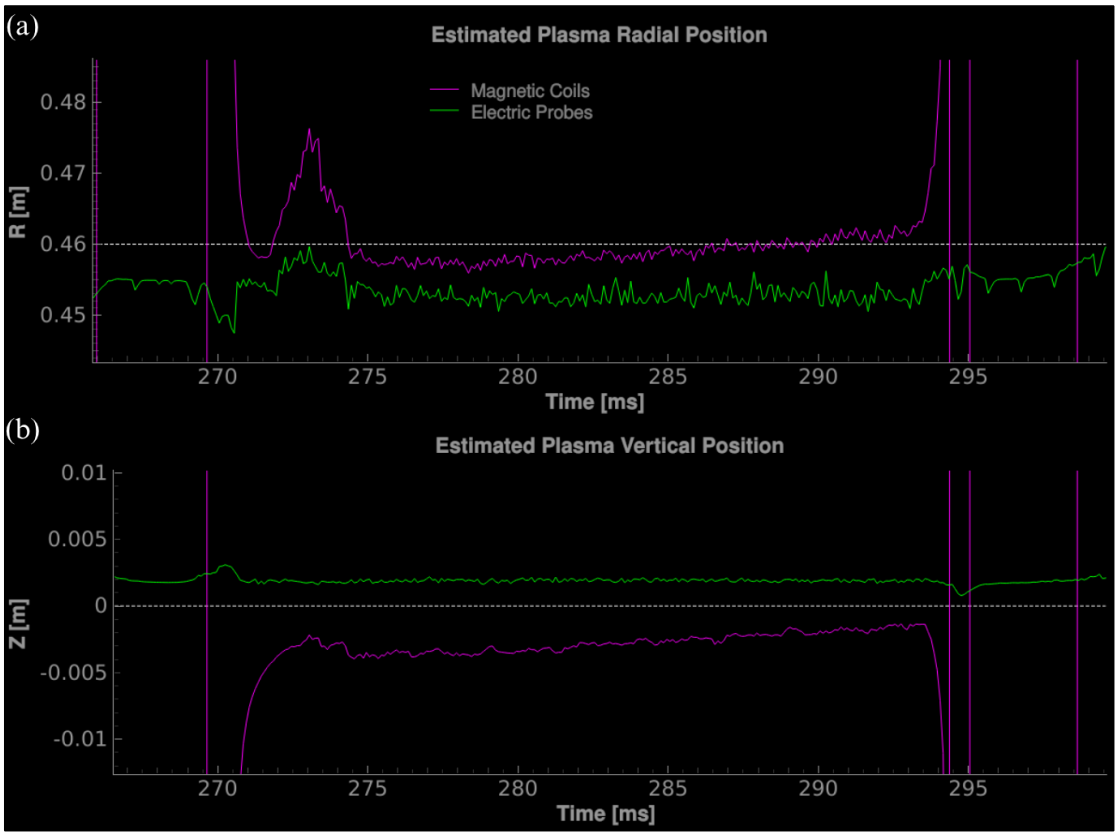


Figure 24: Comparison between **(a)** the radial and **(b)** the vertical plasma positions reconstructed from magnetic and electric measurements for shot 53071

Although the resulting electric reconstruction introduces an inherent bias, it is highly valuable for the sensors fusion approach, as it also estimates the position by using a different electromagnetic property of plasma. In fact, its main advantage lies in its ability to produce consistent plasma position estimations during plasma AC reversals, as reported in [62]. Moreover, Langmuir probes are not subject to the noise from eddy currents that affects magnetic measurements, providing cleaner, albeit potentially less accurate, complementary information for plasma position estimation.

3.3.2. Sensor Fusion for Plasma Position Estimation Using Kalman Filters

Since the envisioned PCS operation depends on uncertain plasma position estimates derived from inherently imprecise measurements, KFs were employed not only for their powerful capability to estimate unknown system states but, more importantly, for their ability to enhance state estimation accuracy through sensor fusion [42]. Rather than relying solely on individual diagnostics, combining magnetic and electric estimations in the fusion process essentially reduces the overall uncertainty associated with the estimated plasma positions relative to following standalone approaches [42].

KFs operate by iteratively enhancing state estimation through two main steps: prediction and update. The prediction step consists of estimating state evolution based on previous known states, accounting for uncertainties through process noise, providing well-grounded estimates before incorporating new measurements. The update step then improves the estimated state by comparing the measurements predicted for the estimated state with the actual measurements, weighting them according to the accumulated uncertainty. For standard KFs, these formulations rely on linear relationships, however extended KFs bypass this requirement by locally linearizing the non-linear functions via their Jacobians for the update step, enabling the implementation of this filtering technique for the plasma position reconstruction.

The chosen methodology consisted of implementing a technique similar to track-to-track or state fusion using two independent extended KFs, as the two diagnostics are used to perform the same task [42]. In this sense, since retrieval of an ISTTOK plasma model has yet to be achieved, these filters were employed not to predict position evolution, but rather to handle the noise in measurements and, most importantly, the inherent uncertainty of both plasma position estimators. Using this configuration, rather than reflecting the physical dynamics of the system, the prediction step of KFs consists of solely propagating previous estimates with added process noise.

For the magnetic reconstruction, the non-linear nature of Equation (14) required the use of an extended KF. In this case there is no exact model to predict magnetic measurements that would be obtained at a given plasma position, therefore an approximate numerical approach was followed for the update step. Similarly, although the Langmuir position estimator is approximately linear, an extended KF was also adopted as there is no direct analytical way to solve the direct problem, that is, to predict floating potential measurements for a given plasma position. The implemented extended KFs estimate plasma positions at a given instant k using the state vector x_k defined as:

$$x_k = \begin{bmatrix} R_{p_k} \\ Z_{p_k} \end{bmatrix} \tag{18}$$

As there was no available model for the plasma dynamics, no state-transition models were formulated, thus the configurations differed from traditional KFs. Therefore, the prediction step was limited to propagating previous estimates, while accounting for uncertainty growth through the process noise covariance matrix Q, as expressed in Equation (19). This is shown in Equation (20), where the process noise was iteratively accumulated in the state covariance matrix P. The notation $k-1|k-1$ refers to the last updated state, while $k|k-1$ denotes the state predicted for instant k using information from the previous update. In this sense, $\hat{x}_{k|k-1}$ represents the *a priori* estimate based on the last updated state, $\hat{x}_{k-1|k-1}$, and the same applies for P.

$$\hat{x}_{k|k-1} = \hat{x}_{k-1|k-1} \tag{19}$$

$$P_{k|k-1} = P_{k-1|k-1} + Q \tag{20}$$

In the update step, new diagnostic measurements, represented by the vector y_k, were compared with their predicted counterparts $\hat{y}_{k|k-1}$, computed through the measurement function $h(\cdot)$, as shown in Equation (21). This function was then locally linearized using its Jacobian ∇h_k, which, together with the predicted covariance matrix $P_{k|k-1}$ and the measurement noise covariance matrix R, was used to compute the Kalman gain K_k, as expressed in Equation (22). In this sense, K_k represents the relative weight attributed to the measurements and current state estimate. Finally, at the end of each iteration, both the state and covariance matrix were updated according to Equations (23) and (24), respectively.

$$\hat{y}_{k|k-1} = h\left(\hat{x}_{k|k-1}\right) \tag{21}$$

$$K_k = P_{k|k-1}\,\nabla h_k{}^T\left(\nabla h_k P_{k|k-1} \nabla h_k{}^T + R\right)^{-1} \tag{22}$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k\left(y_k - \hat{y}_{k|k-1}\right) \tag{23}$$

$$P_{k|k} = (I - K_k \nabla h_k) P_{k|k-1} \tag{24}$$

These formulations were employed in both extended KFs, however, the measurement vectors, state and noise covariance matrices, and, most importantly, the measurement functions differed for each diagnostic. Naturally, the difference in y_k was in terms of its dimension, since, for the electric diagnostic, 4 Langmuir probes were employed, while, for the magnetic, 12 Mirnov coils were used. On the other hand, the covariance matrices differed in the values of their elements (and in dimensions for the case of the R matrix), as these represent distinct sources of uncertainty that were unique to each configuration. Specifically:

- The state covariance matrix (P) elements quantify the uncertainty in the estimated plasma position, where the diagonal terms correspond to the variances of the radial and vertical positions, while the off-diagonal terms represent their correlation.
- The process noise covariance matrix (Q) elements represent the uncertainty growth between updates due to the simplified dynamics or unexpected disturbances, where the off-diagonal terms were set to zero, assuming independence between radial and vertical drift during prediction.
- The measurements noise covariance matrix (R) elements quantify the uncertainty of the diagnostic signals, where the diagonal terms were empirically defined, as specifications of the diagnostics were not available, while the remainder were set to zero under the assumption of uncorrelated sensor noise. For the Mirnov coils, the terms corresponding to coils 3 and 10 were adjusted to reflect the higher noise levels, as previously outlined.

For the sake of brevity, the corresponding matrices along with their elements for the corresponding diagnostics are presented in "Appendix C: Magnetic and Electric Diagnostics Initial Covariance Matrices". As for the corresponding measurement functions, they had to be modelled following case-specific considerations, as both functions reflect underdetermined problems.

For the magnetic diagnostic, the corresponding measurement function derives the expected 12 Mirnov coil signals based on the estimated plasma position at each iteration. This underdetermined problem was bypassed by enforcing the conservation of the total magnetic field and approximating the relative sensitivity of each coil to radial and vertical displacements of the plasma current centroid.

Consequently, this led to the definition of a constraint matrix $A_{\text{Mirnov}_{k|k-1}}$, where the first two rows reflect the sensitivity to the displacements while the third simply sums the signals to conserve the total magnetic field at instant k, $B_{\text{Mirnov}_{\text{total}\,k}}$. The constraint vector c_{Mirnov_k} incorporates these conditions in the system represented in Equation (25).

$$A_{\text{Mirnov}_{k|k-1}}\,\hat{y}_{\text{Mirnov}_{k|k-1}} = c_{\text{Mirnov}_k}$$

$$\hat{y}_{\text{Mirnov}_{k|k-1}} = \begin{bmatrix} \hat{B}_{\text{Mirnov}_1\,k|k-1} \\ \cdots \\ \hat{B}_{\text{Mirnov}_{12}\,k|k-1} \end{bmatrix}$$

$$A_{\text{Mirnov}_{k|k-1}} = \begin{bmatrix} R_{\text{Mirnov}_1} - R_{p_{k|k-1}} + \dfrac{a^2}{4\cdot R_0} & \cdots & R_{\text{Mirnov}_{12}} - R_{p_{k|k-1}} + \dfrac{a^2}{4\cdot R_0} \\ Z_{\text{Mirnov}_1} - Z_{p_{k|k-1}} & \cdots & Z_{\text{Mirnov}_{12}} - Z_{p_{k|k-1}} \\ 1 & \cdots & 1 \end{bmatrix} \tag{25}$$

$$c_{\text{Mirnov}_k} = \begin{bmatrix} 0 \\ 0 \\ B_{\text{Mirnov}_{\text{total}\,k}} \end{bmatrix}$$

where the elements of the first two rows of $A_{\text{Mirnov}_{k|k-1}}$ originate from the manipulation of Equation (14), leading to:

$$\begin{gathered} \sum_{i=1}^{N=12} B_{\text{Mirnov}_i} \cdot \left(R_{\text{Mirnov}_i} - R_p + \frac{a^2}{4\cdot R_0} \right) = 0 \\ \sum_{i=1}^{N=12} B_{\text{Mirnov}_i} \cdot \left(Z_{\text{Mirnov}_i} - Z_p \right) = 0 \end{gathered} \tag{26}$$

Since the system in Equation (25) is underdetermined, the expected measurements were computed as the minimum-norm solution using the pseudo-inverse of $A_{\text{Mirnov}_{k|k-1}}$, as expressed in Equation (27):

$$\hat{y}_{\text{Mirnov}_{k|k-1}} = A^T_{\text{Mirnov}_{k|k-1}} \left(A_{\text{Mirnov}_{k|k-1}}\, A^T_{\text{Mirnov}_{k|k-1}} \right)^{-1} c_{\text{Mirnov}_k} \tag{27}$$

Similarly, for the electric diagnostic, the corresponding measurement function derives the expected 4 Langmuir probe signals based on the estimated plasma position at each iteration. However, to resolve the underdetermined nature of the problem, symmetry between the expected measurements of opposing probes was assumed, such that radial displacements affect only the outer-inner pair, while vertical displacements influence exclusively the top-bottom pair.

This assumption enabled distributing the expected floating potential measurements for each opposing pair proportionally and with opposite signs, which was motivated by manipulating Equation (15) to:

$$\begin{aligned}\left(V_{f_{\text{outer}}} - V_{f_{\text{inner}}}\right) &= \frac{R_p - C_2}{C_1} = \Delta R_{\text{Langmuir}} \\ \left(V_{f_{\text{top}}} - V_{f_{\text{bottom}}}\right) &= \frac{Z_p - C_4}{C_3} = \Delta Z_{\text{Langmuir}}\end{aligned} \tag{28}$$

By assuming a centered average, opposing probes shared equal and opposite contributions, leading to the expected measurements being computed via Equation (29)[15].

$$\begin{aligned}\hat{y}_{\text{Langmuir}_{k|k-1}} = h_{\text{Langmuir}}(\hat{x}_{k|k-1}) &= \begin{bmatrix} 0.5 \cdot \Delta Z_{\text{Langmuir}_{k|k-1}} \\ 0.5 \cdot \Delta R_{\text{Langmuir}_{k|k-1}} \\ -0.5 \cdot \Delta Z_{\text{Langmuir}_{k|k-1}} \\ -0.5 \cdot \Delta R_{\text{Langmuir}_{k|k-1}} \end{bmatrix} \\ \hat{y}_{\text{Langmuir}_{k|k-1}} &= \begin{bmatrix} \hat{V}_{f_{\text{top}}{}_{k|k-1}} \\ \hat{V}_{f_{\text{outer}}{}_{k|k-1}} \\ \hat{V}_{f_{\text{bottom}}{}_{k|k-1}} \\ \hat{V}_{f_{\text{inner}}{}_{k|k-1}} \end{bmatrix}\end{aligned} \tag{29}$$

The final and crucial step in the design of both filters involved the definition of their covariance matrices. Their configuration directly influences the performance and stability of the estimation algorithm, thereby tailoring their design to each diagnostic system was essential. Initially set with arbitrarily defined values, the initial guess of the state covariance matrix (P_0) as well as the R and Q matrices were subsequently fine-tuned within the simulation environment using data from the previously discussed discharges.

[15] Where $\Delta R_{\text{Langmuir}_{k|k-1}} = \left[\left(R_{p_{k|k-1}} - C_2\right)/C_1\right]$ and $\Delta Z_{\text{Langmuir}_{k|k-1}} = \left[\left(Z_{p_{k|k-1}} - C_4\right)/C_3\right]$.

In this context, only the diagonal elements of these matrices were considered for tuning. The R and Q off-diagonal elements were kept at zero in both filters, in accordance with the previously discussed assumptions. It is worth noting that the diagonal elements of P_0 were identical in both filters, as they represent the uncertainty associated with the initial state estimate, which was assumed to be the same for both cases. Conversely, the P_0 off-diagonal elements (i.e. the correlation between the initial estimates of R_p and Z_p) were derived using the same weighted approach previously employed for obtaining the Langmuir coefficients, based on the evolution of estimated plasma position across the analyzed shots. This approach enabled estimating the correlation between radial and vertical directions using both the magnetic and electric position reconstructions.

As for the fusion step, an alternative approach had to be adopted, as estimates started losing consistency when following the traditional state fusion methods described in [42]. In both correlated and uncorrelated fusion formulations, the unprecedented growth of the state covariance matrices led to instability and inconsistency in estimates, particularly when the individual state estimates diverged momentarily. The degraded performance rendered both methods impractical, therefore the adopted approach consisted of dynamically weighting the relative confidence in the states estimated with each diagnostic according to the absolute plasma current magnitude. This choice of criterion stemmed from the observation that the magnetic diagnostic fails exclusively during low-current phases, rendering it unreliable in those regions. In this context, the confidence in the magnetic estimates increased proportionally with the current, while the confidence in the electric decreased accordingly.

This dynamic weighting avoided abrupt transitions and ensured the smooth evolution of the fused state during the plasma AC reversals. This was achieved by utilizing the fusion coefficient $\alpha \in [0, \alpha_{\max}]$, which was determined at each iteration by normalizing the current with respect to a user-defined threshold value $I_{p_{\text{nominal}}}$ that corresponded to its expected nominal value. The coefficient was then scaled by $\alpha_{\max}$, which was also defined by the operator to establish the maximum allowable confidence in the magnetic estimate, as expressed in Equation (30):

$$\alpha = \min\left(1, \frac{|I_p|}{|I_{p_{\text{nominal}}}|}\right) \cdot \alpha_{\max} \tag{30}$$

The final fused estimate was computed as a weighted average of the individual estimates according to Equation (31).

$$\hat{x}_{\text{fused}_{k|k}} = \alpha \cdot \hat{x}_{\text{Mirnov}_{k|k}} + (1-\alpha) \cdot \hat{x}_{\text{Langmuir}_{k|k}} \tag{31}$$

This simple scheme enabled RT implementation with smooth and adaptative state estimation, leveraging two independent estimators based on distinct diagnostics that captured the two main electromagnetic properties of plasma. The final result was a robust and interpretable estimation of the plasma position.

Figure 25 provides a detailed comparison between the fused estimation and those obtained from the standalone diagnostics for shot 53071, highlighting the improved stability and coherence achieved through the proposed state fusion method. The new approach yielded valid results for the entire pulse duration, while preserving higher interpretability and effectively mitigating abrupt deviations.

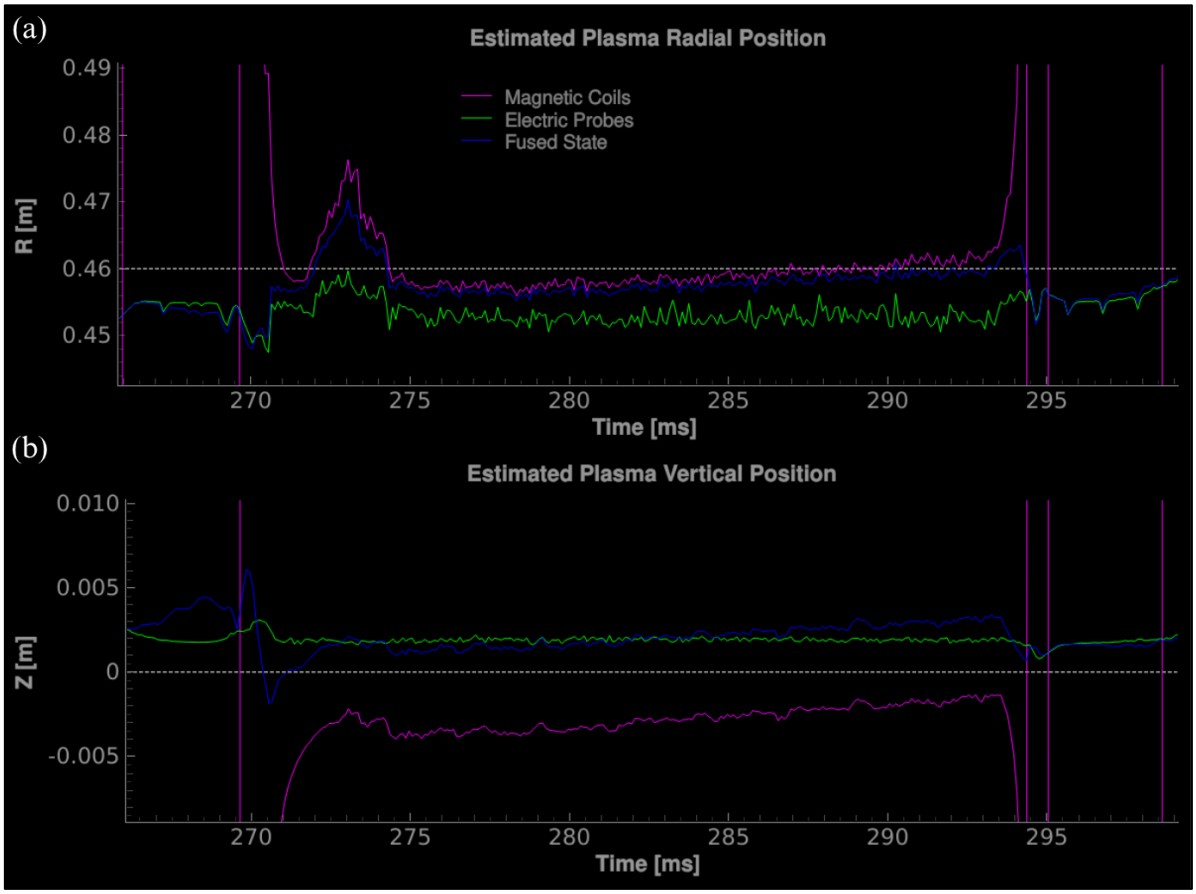


Figure 25: Estimated plasma **(a)** radial and **(b)** vertical positions as obtained by magnetic coils, electric probes, and fused state for shot 53071

The following chapter presents and discusses the experimental results obtained with the implemented framework. The analysis focuses on the comparison of the proposed estimation approach with the reference estimations from past ISTTOK discharges, serving as a benchmark for validation. The evaluation further considers the robustness of the fused

estimation relative to individual diagnostics, as well as the RT performance of the implementation in terms of execution time and jitter. Therefore, these aspects provide a comprehensive assessment of the proposed methodology within a RT environment.

4. RESULTS & DISCUSSIONS

This chapter presents and discusses the results obtained from the application of the proposed RT estimation framework to ISTTOK experimental data. In this sense, it is structured into three main sections, each addressing a key aspect of the evaluation of the developed solution. The analysis begins with the validation of the estimation results against the reference plasma position reconstruction algorithm developed in [11], followed by an assessment of the robustness of the proposed method through sensor fusion analysis. Finally, the RT performance of the estimation algorithm is evaluated in terms of execution time and jitter, ensuring its suitability for deployment in a RT fusion environment.

4.1. Validation of the Estimation Approach

Accurate estimation of the plasma current centroid position is essential for reliable monitoring and subsequent control in tokamak operation [43]. In this context, the proposed estimation approach is evaluated through comparison with reference plasma position reconstructions which served as a basis for the ISTTOK PCS over multiple past successful discharges.

As reference reconstructions were previously used in successful ISTTOK operation, they provide a meaningful benchmark for assessing whether the proposed estimator reproduces physically consistent plasma position trajectories. In this sense, the validation is carried out by comparing three estimation outputs – magnetic, electric, and fused – against the reference reconstruction in order to provide a relevant benchmark. Moreover, this allows the individual behavior of the magnetic- and electric-based estimators to be analyzed before assessing the combined estimation.

The assessment focuses on the temporal agreement with the reference reconstruction, the consistency of the estimated trajectories, and the ability to preserve physically plausible behavior throughout the discharge. The discharges adopted in this evaluation consist of the selection of the 4 best rated shots according to the weighting procedure employed for the Langmuir reconstruction depicted in Table 5 and of the first 4 discharges covered in Appendix A in [11]. The discharge selection is displayed in Table 7.

Table 7: Discharge selection for assessing the developed estimation approach

Shot numbers	
Langmuir reconstruction selection	**Reference reconstruction selection**
45967	48555
46241	48556
53058	48561
53071	48563

The proposed selection allows the estimation approach to be assessed under two complementary perspectives. This provides a broader evaluation scope by combining shots directly related to the calibration of the electric-based reconstruction model with shots directly associated with the established reference data-driven model. Therefore, the comparison is not restricted to a single subset of discharges, enabling the estimators' behaviors to be assessed across different ISTTOK experimental conditions.

The comparison is first performed through a qualitative analysis of the three reconstructed trajectories, followed by a quantitative evaluation based on the error metrics previously introduced in the "Research Methodology" chapter, namely the MAE and RMSE, both computed with respect to the reference reconstruction. In addition, the maximum absolute error (MaxAE) is considered to quantify the largest deviation from the reference, while the Pearson Correlation Coefficient (PCC) [65] is used to assess the degree of temporal agreement between each reconstructed trajectory and the reference reconstruction. While the other metrics quantify the magnitude of the deviations, the correlation coefficient is also of interest since it provides complementary insight into the ability of the estimators to reproduce the overall dynamic behavior of the plasma position. The mathematical formulation adopted for these metrics is presented below.

Let $x_{\mathrm{ref}}(k)$ represent either plasma position component determined by the reference reconstruction at sample k, and $x_{\mathrm{est}}(k)$ the corresponding estimated value obtained from the magnetic, electric, or fused estimator. The estimation error, $e(k)$, is therefore defined as:

$$e(k) = x_{\mathrm{est}}(k) - x_{\mathrm{ref}}(k) \tag{32}$$

Considering the number of samples within the analyzed flat-top interval, denoted by N, these error metrics are defined in Equations (33) to (35):

$$MAE = \frac{1}{N} \cdot \sum_{k=1}^{N} |e(k)| \tag{33}$$

$$RMSE = \sqrt{\frac{1}{N} \cdot \sum_{k=1}^{N} e(k)^2} \tag{34}$$

$$MaxAE = \max_{k} |e(k)| \tag{35}$$

In parallel, the PCC is used to assess the similarity between the temporal evolution of each estimated trajectory and the corresponding reference reconstruction. It is defined as follows:

$$PCC = \frac{\sum_{k=1}^{N} \left((x_{\text{est}}(k) - \bar{x}_{\text{est}}) \right) \cdot \left((x_{\text{ref}}(k) - \bar{x}_{\text{ref}}) \right)}{\sqrt{\sum_{k=1}^{N} (x_{\text{est}}(k) - \bar{x}_{\text{est}})^2} \cdot \sqrt{\sum_{k=1}^{N} (x_{\text{ref}}(k) - \bar{x}_{\text{ref}})^2}} \tag{36}$$

where $\bar{x}_{\text{est}}$ and $\bar{x}_{\text{ref}}$ denote the mean values of the estimated and reference trajectories over the same interval, respectively. The PCC measures the degree of linear dependence between two variables and takes values in the interval [−1,1], where values close to 1 indicate strong positive linear agreement [65].

The analysis begins with shot 53071, which is used throughout this research as a representative discharge of recent ISTTOK operation. Following the time interval analyzed in Figure 25, and considering Figure 22 as a reference for the plasma current behavior during this discharge, Figure 26 and Figure 27 provide a qualitative assessment of the three estimated trajectories for the plasma radial and vertical positions, respectively, against the corresponding reference reconstructions during the same plasma flat top phase.

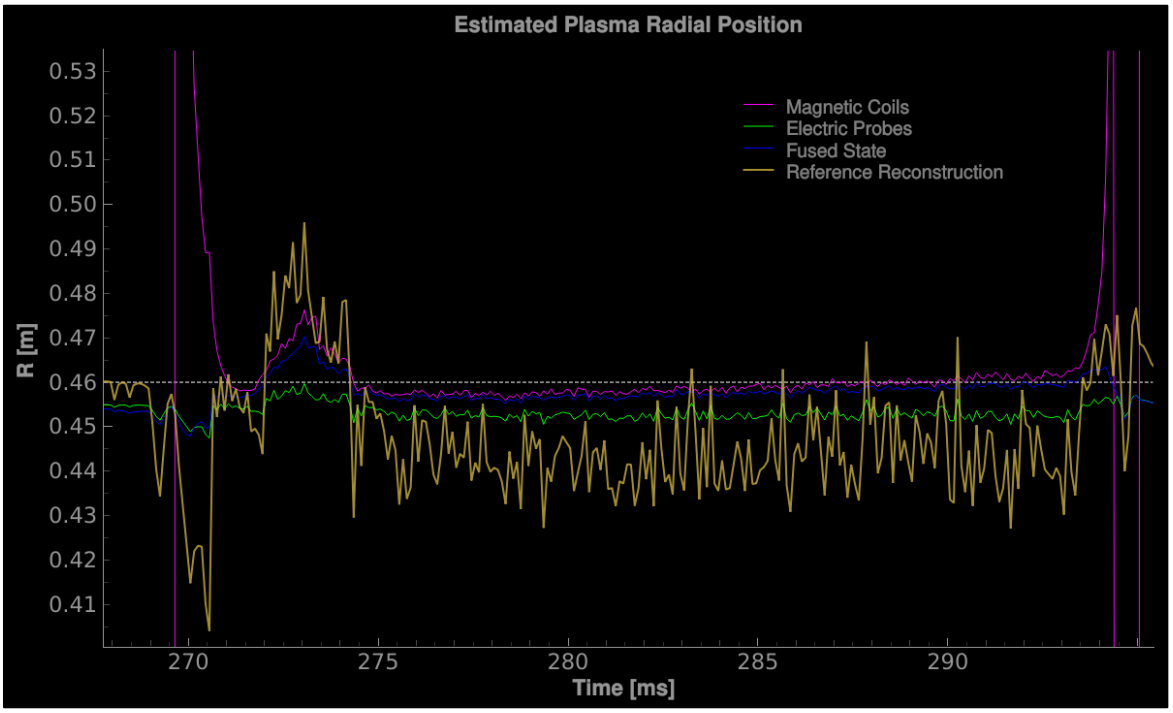


Figure 26: Plasma radial position for shot 53071 estimated by magnetic coils, electric probes, fused state, and reference reconstructions

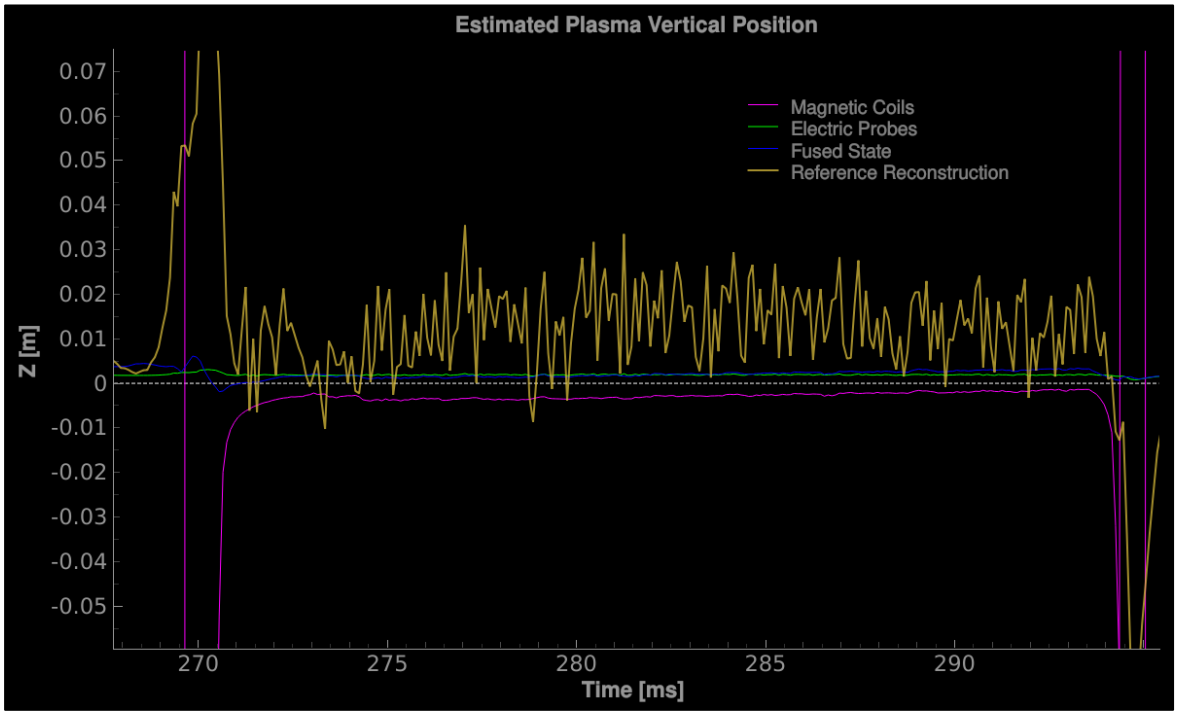


Figure 27: Plasma vertical position for shot 53071 estimated by magnetic coils, electric probes, fused state, and reference reconstructions

A qualitative inspection of the figures above shows that the reference reconstruction presents a more erratic behavior than the three developed estimation methods, with larger amplitude variations throughout the analyzed flat top interval. This behavior is particularly

evident in the vertical position reconstruction, where the reference signal exhibits frequent oscillations of approximately 1 cm around the zero-position reference.

Despite these local discrepancies, the overall trajectory followed by the proposed estimations remains broadly consistent with the reference reconstruction during the analyzed interval. To further support this qualitative analysis and avoid relying solely on visual interpretation, Table 8 presents the mean radial and vertical positions computed over the flat top phase (approximately between 272 and 293 ms) and compares them with the corresponding reference values.

Table 8: Comparison between calculated mean plasma positions during 272-293 ms flat top phase for shot 53071

Estimator	Mean R position [cm]	ΔR to Reference [cm]	Mean Z position [cm]	ΔZ to Reference [cm]
Reference	44.709	N/A	+ 1.272	N/A
Magnetic	45.989	+ 1.280	- 0.277	- 1.549
Electric	45.317	+ 0.609	+ 0.191	- 1.081
Fused	45.838	+ 1.129	+ 0.207	- 1.064

The observed offsets are approximately on the order of one centimeter, which is comparable to the amplitude of the oscillations exhibited by the reference reconstruction during the same interval. This comparison is relevant because it places the differences between the proposed estimators and the reference signal within the same spatial scale as the intrinsic variability observed in the reference itself. Although the proposed estimators do not reproduce the local oscillatory behavior of the reference reconstruction, their trajectories remain confined to a physically plausible range for the analyzed discharge. Therefore, the smoother behavior observed in the developed estimators may not necessarily indicate a loss of accuracy, but rather suggest that the proposed approaches capture the dominant plasma position trend while suppressing higher-amplitude local fluctuations.

The smoother behavior observed in the proposed estimations is most likely associated with the simplified reconstruction model adopted in this work, which does not explicitly account for noise contributions or electromagnetic effects mainly associated with the eddy currents in the VV. In contrast, the reference reconstruction includes additional modeling considerations, which may explain the larger fluctuations observed in its trajectory.

Moreover, since the reference reconstruction is based on a data-driven model calibrated under previous ISTTOK operating conditions, differences between the machine configuration at the time of its development and the conditions of more recent discharges may also contribute to the observed deviations. This does not invalidate the reference reconstruction, but highlights that its comparison with recent shots should be interpreted with caution, since configuration-dependent diagnostics are involved.

To further assess this aspect, the same qualitative and quantitative evaluation is extended to the second flat top phase of discharge 48555, documented in [11]. The second flat top phase was selected to preserve the same analysis criterion adopted for shot 53071, allowing a more consistent comparison between a recent discharge and one closer to the operating conditions under which the reference model was originally developed.

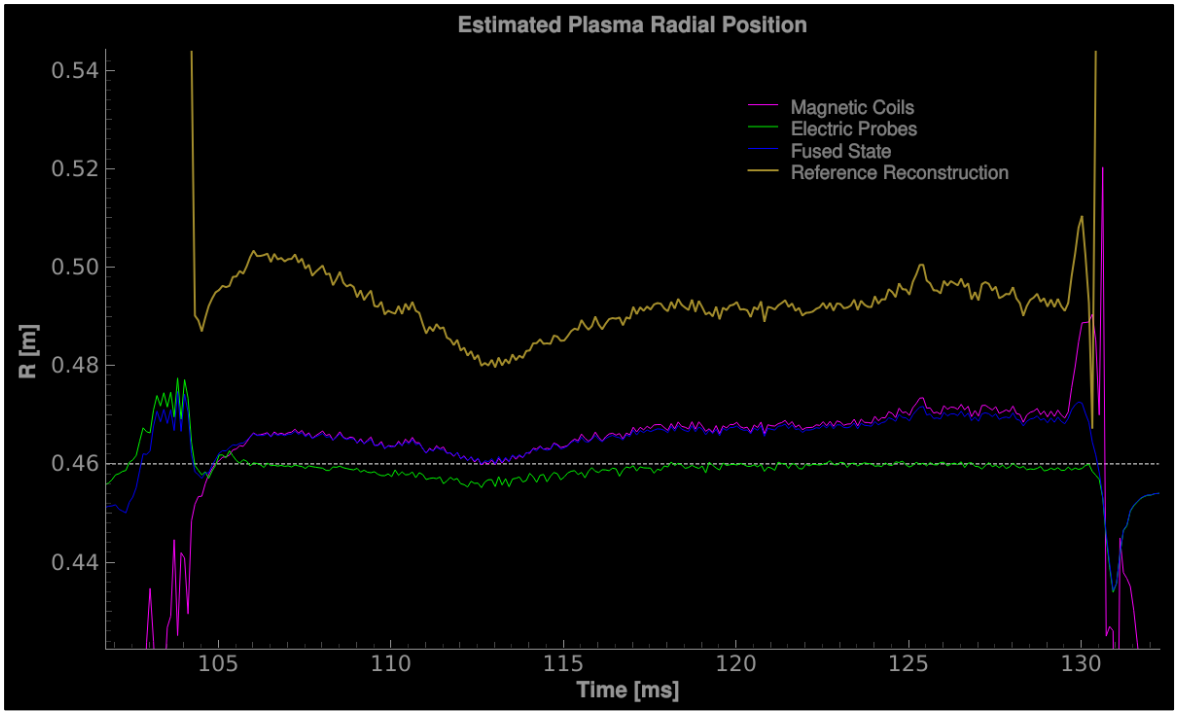


Figure 28: Plasma radial position for shot 48555 estimated by magnetic coils, electric probes, fused state, and reference reconstructions

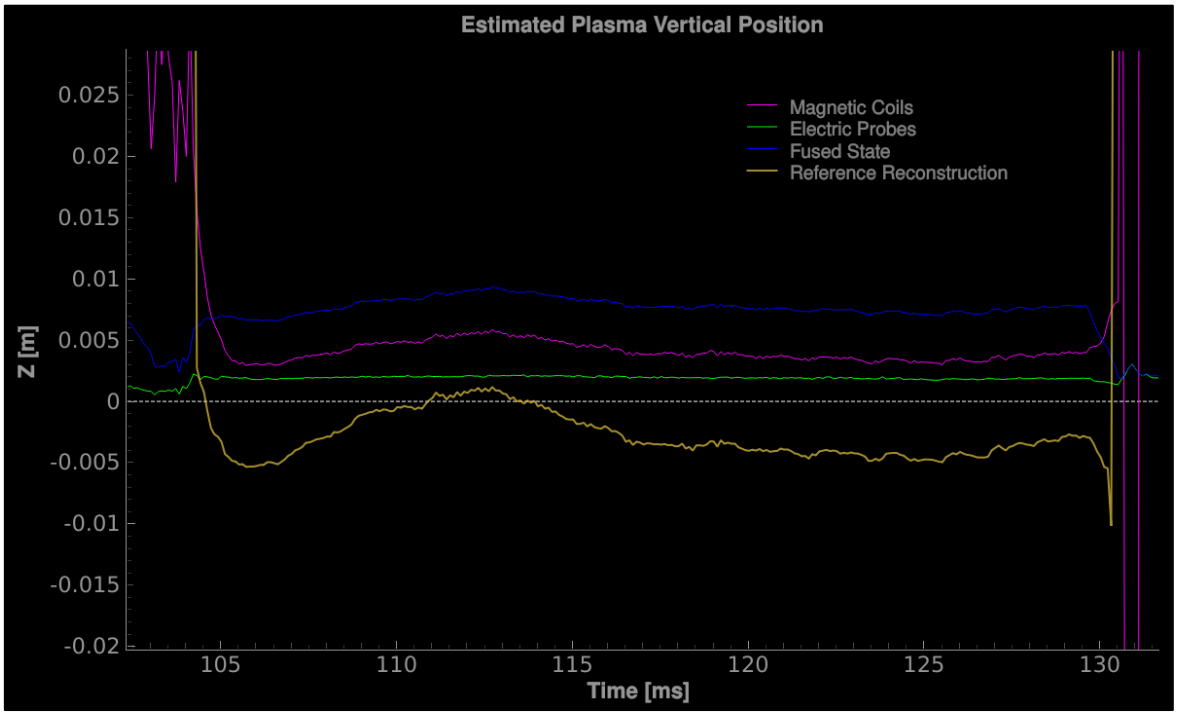


Figure 29: Plasma vertical position for shot 48555 estimated by magnetic coils, electric probes, fused state, and reference reconstructions

The figures above show that, for discharge 48555, the reference reconstruction presents a significantly smoother behavior than that observed for shot 53071, particularly in the vertical position. This supports the hypothesis that part of the more erratic behavior previously observed may be associated with the application of the reference model to more recent operating conditions. Nevertheless, the comparison also reveals a noticeable offset between the proposed estimations and the reference reconstruction, especially in the radial position, where the reference remains consistently above the trajectories obtained with the developed estimators throughout most of the selected flat top phase.

In contrast, the vertical position shows a closer agreement between the proposed estimations and the reference reconstruction. The offset in this case is considerably smaller than that observed for shot 53071, and all trajectories remain approximately within a ±1 cm range relative to the zero-position reference. Additionally, the oscillation amplitudes are reduced to roughly 0.5 cm, indicating more stable behavior during the analyzed flat top interval. To quantify these observations, Table 9 presents the same mean position analysis performed in Table 8 for the second flat top phase of shot 48555 (approximately between 105 and 130 ms).

Table 9: Comparison between calculated mean plasma positions during 105-130 ms flat top phase for shot 48555

Estimator	Mean R position [cm]	ΔR to Reference [cm]	Mean Z position [cm]	ΔZ to Reference [cm]
Reference	49.229	N/A	- 0.288	N/A
Magnetic	46.675	- 2.554	+ 0.404	+ 0.692
Electric	45.896	- 3.333	+ 0.194	+ 0.482
Fused	46.615	- 2.614	+ 0.776	+ 1.064

Comparing Tables 8 and 9, the two discharges reveal distinct agreement patterns. For shot 53071, the main discrepancy is observed in the vertical position, where the reference reconstruction presents larger oscillations and a higher mean value relative to the developed estimators. Conversely, for shot 48555, the vertical agreement improves considerably, while the radial position exhibits a more pronounced systematic offset, with all developed estimators remaining below the reference reconstruction.

These observations indicate that the agreement between the proposed estimators and the reference reconstruction is not uniform across discharges. In this context, although the qualitative comparison indicates clear differences in local signal variability, it also suggests that the proposed estimators reproduce the general plasma position behavior. Therefore, a quantitative evaluation is required to better assess the magnitude of these deviations and compare the performance of the magnetic, electric, and fused estimations against the reference reconstruction.

To quantitatively assess the agreement between the developed estimators and the reference reconstruction, the MAE, RMSE, MaxAE, and PCC were computed for each analyzed discharge. Within each discharge, the metrics were evaluated separately for the radial and vertical plasma position components and, for each component, for the magnetic, electric, and fused estimators. Figure 30 and Figure 31 summarize these metrics for the radial and vertical positions, respectively, considering both representative discharges analyzed in this section.

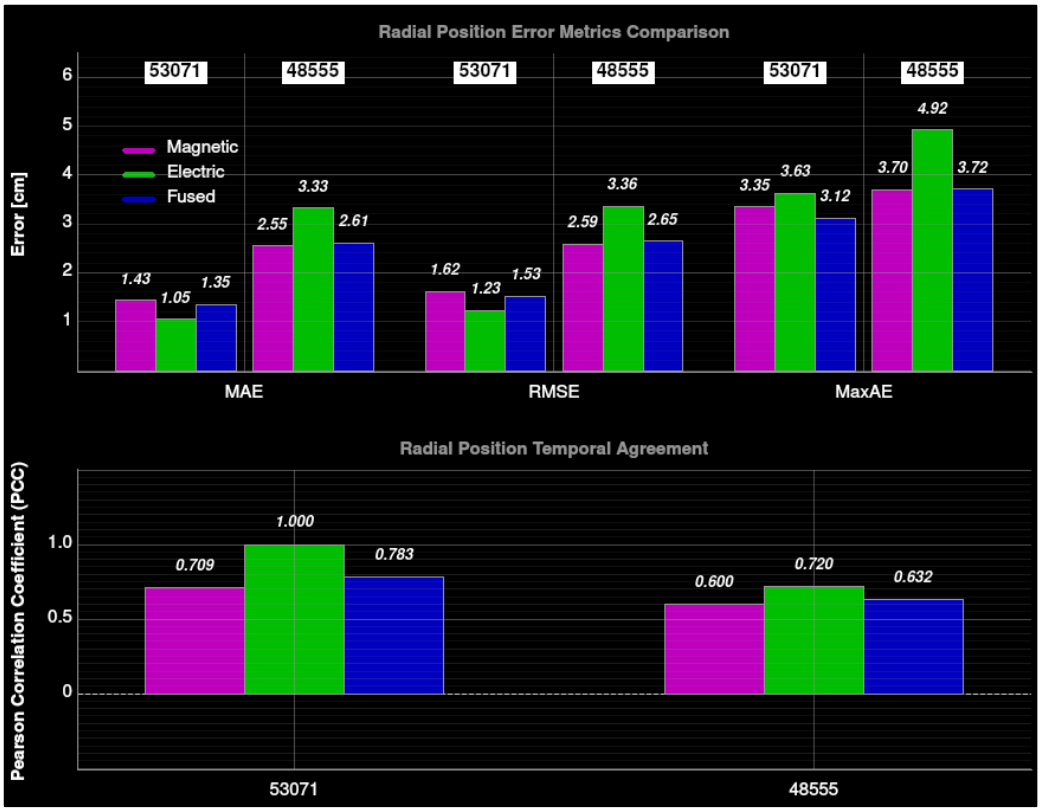


Figure 30: Plasma radial position estimation metrics for magnetic coils, electric probes, and fused state, using the reference reconstruction as benchmark

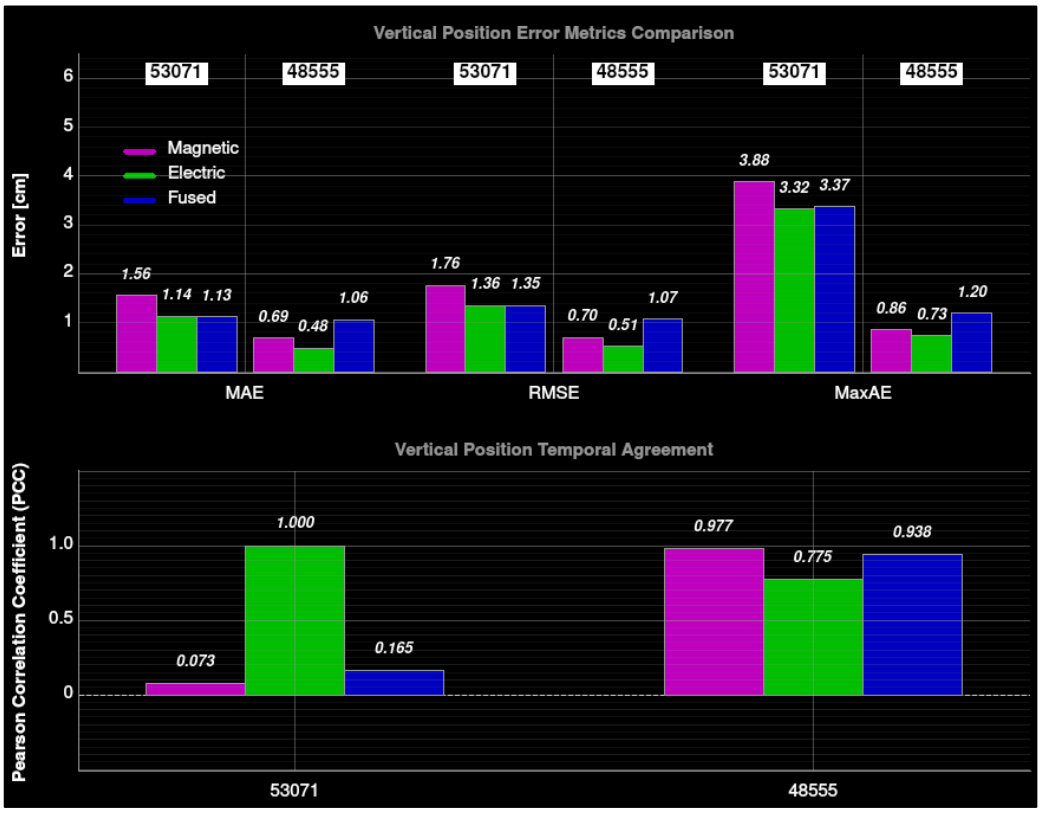


Figure 31: Plasma vertical position estimation metrics for magnetic coils, electric probes, and fused state, using the reference reconstruction as benchmark

Overall, the results indicate that the estimation errors remain within a few centimeters for both analyzed discharges. For the radial position in shot 53071, the electric estimate presents the lowest MAE and RMSE values and an almost perfect temporal agreement with the reference reconstruction. Although it exhibits the highest MaxAE among the three estimators, the differences between the corresponding maximum errors are relatively small. The magnetic and fused estimates show slightly higher average error magnitudes, while their PCC values indicate that the overall temporal evolution is still reasonably captured.

A similar trend is observed for the vertical position in shot 53071. The electric and fused estimates present nearly identical MAE and RMSE values, with the fused estimate achieving the lowest values by a small margin. The electric estimate also exhibits a PCC close to unity, indicating strong agreement with the reference reconstruction in both error magnitude and temporal evolution. By contrast, the magnetic and fused estimates present very low PCC values, despite maintaining comparable error magnitudes. This suggests that their discrepancies cannot be interpreted solely as a systematic positional offset, since their temporal evolution differs substantially from that of the reference reconstruction.

For shot 48555 a markedly different behavior is observed, likely reflecting the weaker applicability of the data-driven reference reconstruction model under the current ISTTOK operating conditions, which differ from those represented in the data used for its original development. For the radial position, the electric estimate presents the highest MAE, RMSE, MaxAE values, although it still achieves the highest PCC among the three estimators. The magnetic and fused estimates exhibit lower error magnitudes than the electric estimate, while maintaining moderate positive correlation with the reference reconstruction. Nevertheless, the radial errors obtained for shot 48555 are generally higher than those observed for shot 53071, indicating a larger systematic discrepancy between the proposed estimators and the reference reconstruction.

On the other hand, for the vertical position in shot 48555, all three estimators exhibit the lowest error magnitudes among the analyzed cases. The electric estimate provides the lowest errors, although it presents the lowest PCC of the three estimators. Even so, it is worth noting that its PCC indicates a good degree of temporal agreement. The magnetic and fused estimates show particularly favorable results, combining relatively low error magnitudes with PCC values close to unity. These results indicate that both estimators reproduce the temporal evolution of the reference vertical position with a high degree of similarity throughout the selected flat top interval.

The vertical position results for shot 53071 must, however, be interpreted with particular care. As shown by the corresponding time-domain evolution in Figure 27, the reference reconstruction presents an atypical and highly irregular behavior when compared with the smoother trajectories provided by the magnetic and fused estimators. This behavior contrasts markedly with the vertical position reconstruction obtained for shot 48555, where the reference trajectory evolves smoothly and exhibits strong agreement with both the magnetic and fused estimates.

The vertical reconstruction of shot 53071 therefore appears as an outlying case within the present analysis and may indicate a reduced applicability of the data-driven reference model under operating conditions that differ from those represented in the data used for its development. Consequently, the very low PCC values obtained for the magnetic and fused estimates in this case do not necessarily indicate a failure to represent the physical plasma motion. Instead, they primarily reflect the disagreement between these estimates and a reference trajectory whose behavior differs substantially from that observed in the other analyzed discharge.

Overall, the observed relationship between MAE, RMSE, MaxAE, and PCC provides a consistent picture of the estimator performance across the analyzed discharges. In most cases, the similarity between MAE and RMSE suggests that the discrepancies with respect to the reference reconstruction are not dominated by isolated extreme deviations. In parallel, the PCC values help distinguish between cases in which the estimation error is primarily associated with systematic offsets and those involving a weaker agreement in temporal evolution.

To determine whether the behavior observed for the two representative discharges extends to the broader dataset captured in in Table 7, the same metrics were computed for all remaining selected shots. A complete shot-by-shot breakdown of the radial and vertical MAE, RMSE, MaxAE, and PCC values, together with the corresponding analyzed flat top intervals, is provided in Appendix D: Detailed Plasma Position Estimation Metrics.

The overall results confirm that the estimators generally preserve the main temporal evolution of the plasma position, although the magnitude of the error varies between discharges and position components. In several cases, relatively high PCC values coexist with non-negligible MAE and RMSE values, supporting the interpretation that a substantial part of the discrepancy arises from a systematic positional offset rather than from an inability to reproduce the plasma dynamics.

This offset may be partly related to physical effects that are not fully represented in the current estimation method, particularly the influence of the VV and the associated eddy currents. Future developments should therefore consider the inclusion of an explicit vessel-response model or an appropriate offset-compensation procedure.

Overall, these results demonstrate that the developed estimators are capable of reproducing the plasma position with errors on the order of a few centimeters while preserving the main dynamical trends of the discharge. Despite the limitations associated with the reference reconstruction, the agreement observed across the analyzed cases provides evidence that the proposed approach constitutes a viable plasma position estimation framework and motivates its further assessment under a broader range of operating conditions.

The comparison also supports the hypothesis that the data-driven reference reconstruction becomes less representative when applied outside the operating conditions for which it was originally established. As previously stated, this limitation is particularly evident in the anomalous vertical reconstruction obtained for shot 53071, highlighting the importance of interpreting the calculated metrics in conjunction with the physical plausibility of the reference signal.

To conclude, among the evaluated estimators, the fused estimate generally provides a balanced representation of the plasma position by combining information from the magnetic coils and electric probes. Although it does not consistently minimize every individual metric, it captures the overall plasma behavior across the analyzed discharges and generally provides an intermediate compromise between the magnetic and electric reconstructions. These findings therefore support sensor fusion as a more robust basis for RT plasma position estimation in ISTTOK, while also identifying vessel-induced offsets and limitations of the reference model as relevant targets for future improvement.

4.2. Sensor Fusion and Robustness Analysis

Following the validation of the proposed estimators against the reference reconstruction, the robustness benefits introduced by the sensor fusion strategy were evaluated. In this context, robustness is not limited to minimizing the estimation error for an individual discharge, but also concerns the ability to preserve the plasma position dynamics while providing a smooth, continuous, and physically plausible signal suitable for subsequent RT control [24].

The magnetic and electric diagnostics provide complementary information about the plasma state, but each is affected by different noise sources and modeling limitations. Kalman filtering is particularly suitable for this problem because it combines model-based predictions with uncertain measurements and provides a recursive state estimate capable of attenuating random measurement fluctuations while accounting for uncertainty in the estimation process [42]. Consequently, this property is especially relevant for the estimation of the plasma position, since the plasma dynamics are represented through simplified physical models and the available diagnostic measurements are inherently imprecise.

Rather than combining the raw magnetic and electric measurements within a single filter, the adopted architecture independently estimates the plasma state through two extended KFs and subsequently combines their outputs. This configuration corresponds to a state-fusion approach, in which different sensor systems perform the same estimation task before their resulting state estimates are combined [42].

The two state estimates are combined through a current-dependent weighting strategy in which the magnetic estimate is assigned a larger contribution during the flat tops, when the plasma current is higher and more stable. Conversely, during current reversals, the plasma current becomes very low or approaches zero, making the magnetic reconstruction more susceptible to unrealistic spikes. In these intervals, the electric estimate assumes a larger role, allowing the estimation process to remain continuous despite the abrupt current variations inherent to the ISTTOK AC operation mode. This strategy proved effective in the analyzed discharges, as the fused trajectories remained within the geometrical limits of the ISTTOK VV while preserving the continuity of the estimation process during AC reversals.

The purpose of the fused estimate is therefore not necessarily to outperform both standalone estimators. Instead, it is intended to reduce the dependence of the final plasma position estimate on the limitations of a single diagnostic, providing a balanced response when either reconstruction is affected by significant levels of noise, poor temporal agreement, or sensitivity to the discharge conditions.

The benefits of the developed fusion-based estimation architecture become particularly apparent in terms of temporal agreement with the reference reconstruction and the attenuation of the sudden spikes observed in the standalone magnetic estimate during plasma-current reversals. Indeed, the analysis presented in the previous section further supports this interpretation, not only through the visual assessment of the fused plasma position trajectories, but also through the consistently acceptable values obtained for the evaluated performance metrics across the analyzed discharges. Although the fused estimate

does not always achieve the lowest error values, its performance generally remains between those of the magnetic and electric estimators or approaches the diagnostic that provides the most favorable response under a given operating condition, further supporting its adoption as a more reliable plasma position estimate.

This aspect is particularly relevant from a control perspective, since abrupt spikes in the estimated plasma position could be interpreted by the controller as rapid physical displacements of the plasma column, potentially leading to excessively large or discontinuous actuator requests. In extreme cases, these requests could exceed the actuator saturation limits, reducing the effectiveness of the control action and introducing additional nonlinear behavior into the closed-loop response. Thus, even when a standalone estimator presents satisfactory agreement with the reference during selected intervals, its direct use as a feedback signal may be less suitable if it exhibits sudden spikes during current reversals.

This is also related to the bumpless transfer problem addressed in the previous ISTTOK control implementation [11], where different controllers were used during distinct phases of the discharge. In that case, switching between control laws required additional care to avoid abrupt changes in the control signal. In the present architecture, the use of a continuous fused position estimate together with a single PID-based control structure would avoid introducing an additional controller-switching problem. As a result, the robustness advantage of the fused estimate is not limited to its validation metrics, but also concerns the continuity and practical suitability of the signal supplied to the RT control system.

An additional robustness advantage of the proposed architecture lies in its model-based formulation. Unlike the data-driven reference reconstruction, the magnetic and electric estimators are not intrinsically tied to a training dataset representing a particular period of ISTTOK operation. Instead, the plasma position is reconstructed from diagnostic measurements through physical and empirical relationships incorporated into the estimation models. Consequently, the proposed approach is not expected to lose applicability solely because the operating conditions evolve beyond those represented in historical data.

This distinction is relevant considering the results presented in the previous section. The reference reconstruction showed a less consistent agreement with the proposed estimators for some of the more recent discharges, particularly in the vertical position, where larger variations are more critical. Rather than being limited to a single shot, this trend was observed across several recent discharges, suggesting a noticeable increase in discrepancies when compared to older shots. Although this behavior does not by itself prove temporal degradation of the reference model, it supports the hypothesis that its predictive fit may

weaken when applied to operating conditions that differ from those represented in the dataset used for its original development.

In contrast, the proposed architecture does not reproduce plasma position through a direct statistical mapping learned from a fixed historical dataset. Instead, its performance depends on the validity of the underlying diagnostic relationships, the adopted physical simplifications, and the calibration of the estimator parameters. This does not make the method independent of operating conditions, since changes in plasma behavior, diagnostic response, or vessel effects may still introduce systematic errors. However, it reduces the dependency on the current discharge being statistically similar to the data used to train the reconstruction model. Future work may focus on refining the modeling assumptions and improving the robustness of the estimator under varying operating conditions.

Within this framework, sensor fusion further improves the practical robustness of the estimation architecture. The magnetic and electric diagnostics respond differently to variations in plasma conditions and measurement quality, meaning that the limitations of one diagnostic are not necessarily reproduced by the other. Therefore, combining both estimates provides a degree of redundancy and reduces the risk that the final position signal is dominated by a single diagnostic whose validity has deteriorated under a particular operating condition.

The resulting architecture should not be interpreted as universally independent of calibration or model validity. Its main advantage is that it combines two physically distinct diagnostic sources without relying on a mapping that is strictly tied to a specific operational regime or dependent on a single data-driven model. This characteristic, together with the smoother response, the attenuation of magnetic spikes, and the positive results obtained for the fused estimate, supports its use as a more adaptable basis for RT plasma position estimation and subsequent RT control in ISTTOK.

4.3. Real-Time Performance Evaluation

After validating the estimation accuracy and robustness of the proposed sensor fusion architecture, it is also necessary to assess its computational performance. Since the developed estimators are intended for integration into the ISTTOK RT control architecture, the execution time of the estimation cycle must remain compatible with the nominal cycle duration of 100 μs required by the PCS [11], [18], [23].

This analysis is particularly important because an accurate plasma position estimate is not sufficient for RT control if the corresponding computation cannot be completed within the specified cycle time [27]. Excessive execution times or irregular timing behavior could compromise the deterministic operation of the complete control system, potentially introducing delays and missed updates in the feedback loop. For this reason, the timing performance of the developed architecture was evaluated as a final validation step before considering its integration into closed-loop control operation.

The evaluation focuses on the execution times of the complete estimation cycle in MARTe2 implementation developed in this work. In this context, the tested execution chain consisted of the minimal configuration chain required to generate the fused plasma position estimates and record the corresponding timing behavior. This sequence was designed to isolate the computational cost of the proposed estimation architecture while keeping the MARTe2 execution chain as simple as possible and can be summarized as follows:

1. acquisition of input diagnostic data;
2. preprocessing and correction of the magnetic signals;
3. execution of the estimation model through the SimulinkWrapperGAM, including the magnetic and electric extended KFs, and the sensor fusion stage;
4. recording of the cycle-duration distribution.

The execution-time measurements were obtained from the measured cycle duration of the RTT and subsequently represented through histograms. Similarly to the procedure previously applied to the WTM RT performance analysis, this representation allows the distribution of the cycle durations to be evaluated, making it possible to identify not only the main concentration of execution times of the proposed architecture, but also the occurrence of occasional timing deviations that could compromise deterministic RT operation.

The main objective of this section is therefore to determine whether the complete estimation architecture can be executed within the expected control-cycle duration for ISTTOK. If the measured execution times remain below the available sampling period, the implementation can be considered compatible with the RT requirements of the system. Conversely, if a significant number of cycles exceed the target duration, further optimization would be required before deployment in closed-loop control operation.

The results of this analysis for shots 48555 and 53071 are presented in Figure 32 and Figure 33. These shots were selected to illustrate the RT performance of the architecture in two distinct operating conditions, for which the reference reconstruction exhibited markedly different behaviors throughout their duration. In both figures, the two shots are shown

together to compare how the execution-time distribution changes between discharges for a given histogram interval.

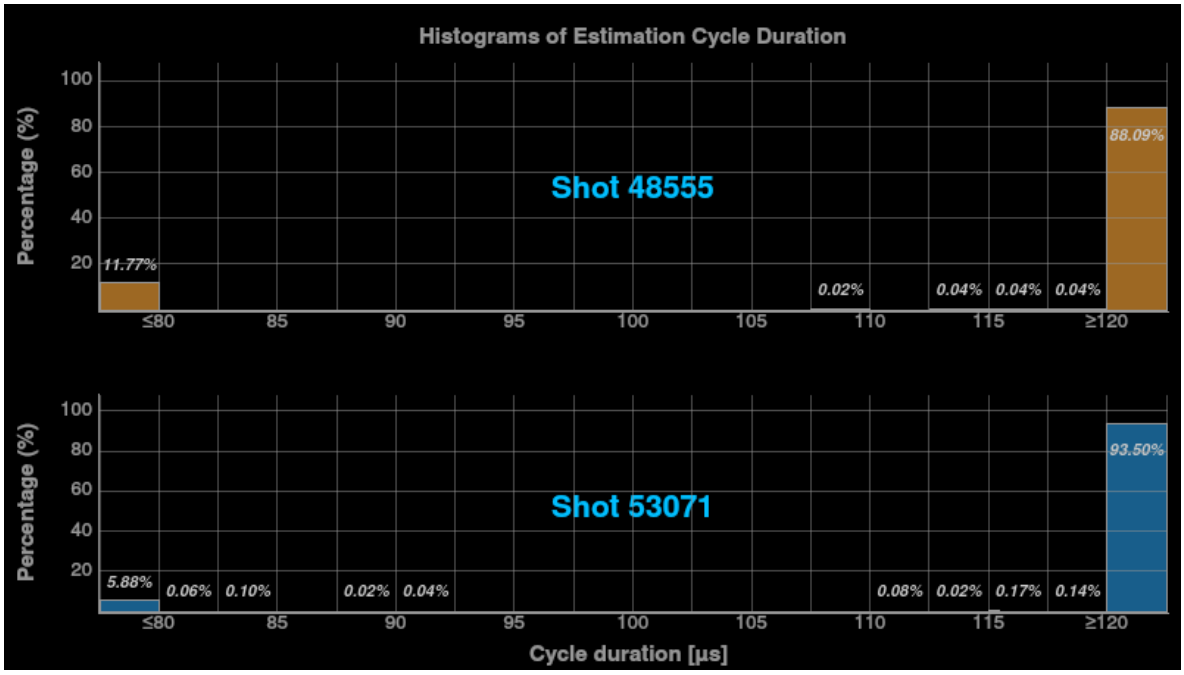


Figure 32: Estimation-cycle duration histograms for shots 48555 and 53071 evaluated around the nominal 100 µs real-time cycle duration

Figure 32 presents the execution-time histograms using an interval centered around the nominal ISTTOK cycle duration of 100 µs, corresponding to the 10 kHz operation frequency. This representation was used as the first reference analysis, since it directly evaluates whether the implemented architecture can operate within the expected RT cycle duration. However, the obtained distributions show that the majority of the measured cycles is not concentrated within this interval, indicating that the current implementation is not able to operate around the nominal 100 µs target.

For this reason, a second representation was considered in Figure 33, using the 160-200 µs interval. This interval does not correspond to the required ISTTOK cycle duration, however, it provides a clearer visualization of the main execution-time distribution observed during the tests. In this representation, the internal bins allow the concentration of samples in the higher execution-time range to be analyzed in more detail, while the leftmost bin groups all cycles completed below 160 µs and the rightmost bin groups all cycles exceeding 200 µs.

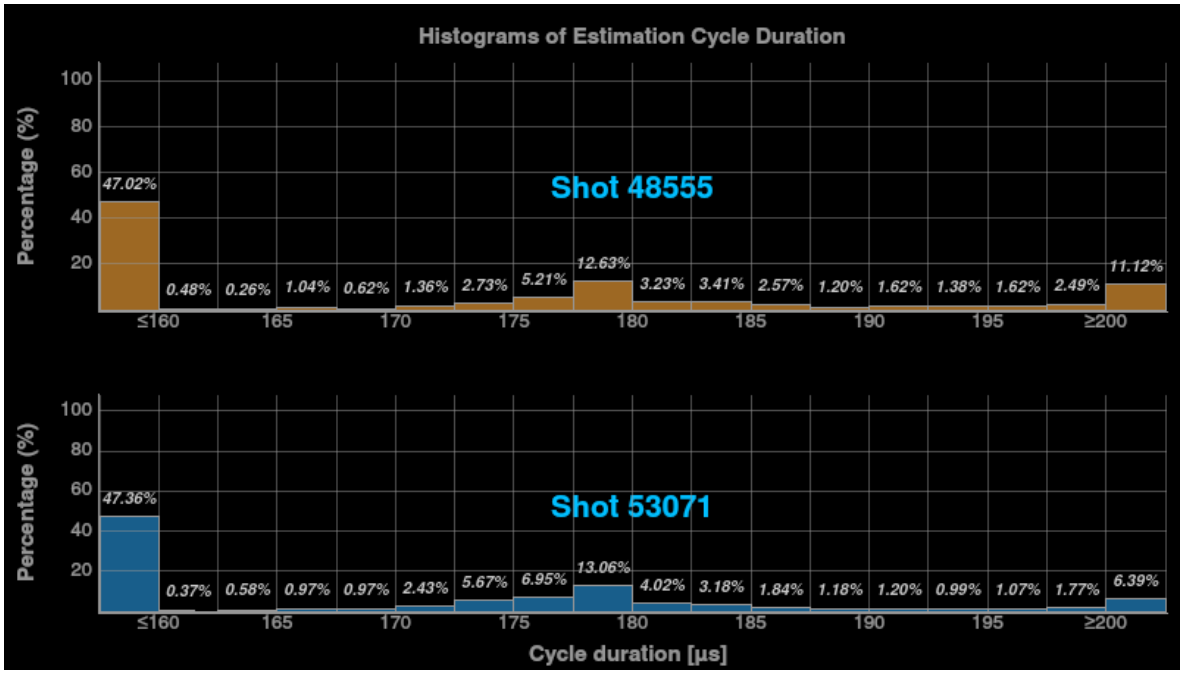


Figure 33: Estimation-cycle duration histograms for shots 48555 and 53071 evaluated in the 160-200 µs interval

The presence of a significant fraction of cycles in the lower bin of Figure 33 must be interpreted carefully. For both shots, this bin does not indicate that the execution time is concentrated exactly at 160 µs, but rather that several cycles were completed faster than this lower limit. This behavior may be associated with intervals in which the input signals are null, small, or less demanding from a computational point of view, reducing the effective execution time of the generated estimation code. Nevertheless, the comparison between both figures shows that the main execution-time behavior of the implemented architecture is better represented above the nominal 100 µs cycle duration.

The obtained results show that the complete estimation cycle is not compatible with the nominal 100 µs cycle duration required for ISTTOK operation. Although part of the measured cycles is completed below the 160 µs lower limit used in Figure 33, the histograms centered around 100 µs show that the execution times are not sufficiently concentrated within the required RT cycle duration. Moreover, the 160-200 µs representation indicates that a relevant portion of the execution-time distribution occurs above the nominal target, with a significant portion also exceeding 200 µs. Therefore, the current implementation cannot be considered suitable for direct deployment in a 10 kHz RT control scenario without further optimization.

It should also be noted that this conclusion must be interpreted in the context of the architecture evaluated in this work. The tested MARTe2 sequence corresponded to a

minimal estimation-oriented configuration instead of including the complete closed-loop control chain, containing other elements such as PID control stages, actuator-output interfaces, or additional supervisory logic. Consequently, the measured execution times already exceed the target cycle duration even before the inclusion of the remaining components that would be required for plasma position control during actual operation.

Despite this limitation, the timing results do not invalidate the proposed estimation architecture as a proof of concept. One of the main objectives of this work was to assess whether a plasma position estimation and sensor fusion strategy could be developed in Simulink and integrated into MARTe2 without requiring a complete manual implementation through native MARTe2 GAMs, which would lead to complex C++ coding. From this perspective, the proposed approach was successfully validated, as the magnetic and electric extended KFs, and fusion logic were executed in MARTe2, and evaluated using experimental ISTTOK data.

The lack of compatibility with the ISTTOK 100 μs timing requirement should therefore be understood as a limitation at the implementation level of the current Simulink-based approach, rather than as a constraint of the estimation concept itself. Future work may focus on optimizing the Simulink model, reducing its computational complexity, simplifying specific operations within the extended KFs, or improving the efficiency of the MARTe2 execution chain. Alternatively, an equivalent sensor fusion architecture could be developed directly using native MARTe2 GAMs. Although this approach would require greater MARTe2 development expertise, it could reduce execution overhead and provide greater control over memory allocation, signal handling, and computational efficiency.

In summary, the timing analysis shows that the current Simulink-based implementation is not yet compatible with the 10 kHz timing requirements of ISTTOK. Nevertheless, it validates the feasibility of using the SimulinkWrapperGAM as a practical development pathway for implementing and testing plasma position estimation and sensor fusion algorithms in MARTe2. The proposed architecture is therefore suitable as a rapid prototyping and validation framework, while further optimization or a native MARTe2 implementation would be required for future closed-loop RT operation.

5. CONCLUSION

This dissertation focused on the development and validation of a RT plasma position estimation architecture for the ISTTOK tokamak, based on diagnostic measurements and implemented through a Simulink-based modeling approach integrated with MARTe2. The motivation for this work stems from the fact that accurate RT knowledge of the plasma position is essential for feedback control, stability preservation, and disruption avoidance in tokamak operation [4], [9], [11]. In the specific case of ISTTOK, this challenge is further reinforced by its AC operation, the need for fast control cycles, and the limitations imposed by the diagnostic and magnetic configuration of the device [18], [34], [43].

The first stage of the work established the basis required for developing and executing RT applications in MARTe2. This included the compilation and configuration of the framework, the execution of simple RT applications, and the implementation of the water tank problem as a validation benchmark. The comparison between the MARTe2 implementation and the corresponding Simulink model confirmed that equivalent dynamic behavior could be obtained in both environments. Furthermore, the integration of the Simulink-generated WTM through the SimulinkWrapperGAM reproduced the expected system response and showed similar cycle-duration behavior to the native MARTe2 implementation. These results support the use of Simulink as a practical tool for rapid prototyping of RT algorithms in MARTe2, while reducing the manual effort associated with the development of native C++ GAMs [21], [22], [29], [31].

The main contribution of the dissertation was the development of a plasma current centroid position estimation architecture using magnetic and electric diagnostics. Since the plasma position is not directly measurable and must be reconstructed from indirect diagnostic signals, the proposed methodology relied on measurements from Mirnov coils and Langmuir probes to estimate the radial and vertical positions of the plasma current centroid [2], [3], [40], [41].

The magnetic reconstruction was based on a simplified formulation [56] using the Mirnov coil array, while the electric reconstruction was obtained from Langmuir probe signals using coefficients derived indirectly from the magnetic-based position estimates, following the formulation detailed in [62]. Before retrieving these coefficients, the plasma current reconstructed from magnetic measurements was compared with Rogowski coil measurements, providing an indirect validation of the consistency of the magnetic diagnostic data used in the estimation process. Once this validation step was completed, the magnetic

position reconstruction was used as the reference basis for determining the Langmuir coefficients, under the assumption that both magnetic and electric diagnostics should provide estimates of the same physical quantity, namely the plasma current centroid position. Therefore, the electric reconstruction was calibrated indirectly from the validated magnetic reconstruction, allowing the Langmuir probe signals to provide a complementary estimate of the plasma position.

To improve the robustness of the diagnostic-based estimates, two independent extended KFs were implemented, one associated with the magnetic diagnostic and another with the electric diagnostic. Since a complete plasma dynamic model was not available for ISTTOK [40], the filters were not used to predict the physical evolution of the plasma position from a state-space plasma model. Instead, they were used to handle diagnostic uncertainty, smooth the individual estimates, and provide a structured basis for sensor fusion. The final fused estimate was obtained through a dynamic weighting strategy based on the absolute plasma current magnitude, reflecting the reduced reliability of magnetic reconstruction during low-current phases and AC reversals [50].

The validation of the proposed approach was carried out by comparing the magnetic, electric, and fused estimates with reference reconstructions obtained from previous successful ISTTOK discharges [11]. The results showed that the developed estimators were able to reproduce the dominant behavior of the plasma position, while remaining within physically plausible ranges during the analyzed intervals. However, the agreement with the reference reconstruction varied depending on the shot, diagnostic, and position component. In particular, the discrepancies observed between the proposed estimators and the reference signals were often dominated by systematic offsets rather than purely random errors. This behavior is consistent with the known influence of eddy currents, passive conducting structures, and magnetic diagnostic contamination in ISTTOK [11], [18], [35], [43].

The sensor fusion strategy did not systematically outperform both individual estimators in every metric or for every discharge. Nevertheless, this was not the sole objective of the fused architecture, as its main contribution was to combine two physically distinct diagnostic sources and reduce the dependence of the final estimate on a single measurement principle [50]. In this context, the magnetic estimator provided useful information during well-established current phases, but was more vulnerable during current reversals and low-current intervals. Conversely, the electric estimator produced smoother estimates and remained useful during regions where magnetic reconstruction degraded, although it introduced its own bias. Therefore, the fused estimate provided a more balanced

and interpretable plasma position signal, mitigating abrupt deviations associated with the standalone magnetic reconstruction while preserving a continuous and stable estimate through the complementary information provided by the electric diagnostic.

The RT execution analysis revealed that the main limitation of the current framework is associated with implementing it in actual ISTTOK operation. In this sense, ISTTOK operation is associated with a nominal cycle duration of 100 µs, corresponding to a 10 kHz RT operation frequency [11], [18], [50]. However, the execution-time histograms obtained for the complete estimation cycle showed that the current Simulink-based architecture was not sufficiently concentrated around this target duration. The analysis centered around 100 µs demonstrated that a significant fraction of cycles exceeded the nominal timing requirement, while the analyzed interval of 160-200 µs provided a clearer representation of the main execution-time distribution observed during the tests. Therefore, the current implementation cannot be considered directly compatible with the closed-loop RT operation in ISTTOK without further optimization.

The timing limitation observed in the RT execution analysis also defines an important boundary for the interpretation of the results of this work. Since the evaluated MARTe2 sequence corresponded to a minimal estimation-oriented implementation, without the additional stages required for closed-loop feedback control, the measured execution times should be regarded as an optimistic representation of the computational cost required by the proposed architecture. The fact that this configuration already exceeds the nominal 100 µs cycle duration of ISTTOK indicates that the current Simulink-based implementation still requires optimization before it can be considered for integration into its complete operational PCS. This is particularly relevant because the actual implementation would also require additional components, such as control computation, actuator-output interfaces, and supervisory logic.

Nevertheless, this limitation does not undermine the main contribution of the dissertation. The proposed work demonstrated that a plasma position estimation and sensor fusion architecture could be developed in Simulink, integrated into MARTe2 through the SimulinkWrapperGAM, and evaluated using experimental ISTTOK data. In this sense, this directly supports the use of Simulink as a practical and user-friendly development pathway for prototyping advanced RT algorithms in MARTe2, reducing the need for immediate manual implementation through native C++ GAMs [21], [22], [29], [31]. Therefore, the dissertation validated the feasibility of the proposed integration workflow, even though the current implementation is not yet suitable for direct operation in the case of ISTTOK.

The main limitations identified throughout the research are related to both the estimation model and its implementation. From the estimation perspective, the simplified reconstruction assumptions, the absence of a complete plasma dynamic model, the dependence on diagnostic calibration, and the systematic offsets observed relative to the reference reconstruction still limit the accuracy of the proposed approach. The influence of eddy currents and passive conducting structures is particularly relevant, since these effects are known to affect magnetic diagnostics in ISTTOK and may explain part of the observed deviations in the reconstructed plasma position [11], [18], [35], [43]. From the implementation perspective, the main limitation is the computational cost of the Simulink-generated code when executed inside the MARTe2 RT chain.

Future work should therefore follow two complementary directions, namely improving the estimation approach itself and enhancing its RT performance. Regarding the estimation framework, the main developments expected to significantly enhance its accuracy are the compensation of systematic offsets, a better representation of VV and eddy-current effects, and the adaptive tuning of the filter covariance matrices. The inclusion of additional diagnostics, such as those discussed in the "Research Methodology" chapter, could also increase redundancy and reduce dependence on magnetic and electric measurements alone, but could further increase the computational cost of the estimation process. For the RT performance, further work should focus on simplifying the Simulink model, reducing the computational complexity of its internal code, and optimizing the MARTe2 execution chain. In parallel, an equivalent sensor fusion architecture could be implemented directly using native MARTe2 GAMs written in C++, in order to assess whether the removal of the Simulink-based interface reduces execution overhead sufficiently for ISTTOK operation.

In sum, this dissertation demonstrated the feasibility of using a Simulink-based modeling approach integrated with MARTe2 for the development and validation of plasma position estimation and sensor fusion algorithms in ISTTOK. Although the current implementation does not yet satisfy the strict 10 kHz timing requirements of the RT control framework, it provides a validated proof of concept for rapid prototyping, diagnostic fusion, and RT execution of advanced estimation algorithms in MARTe2. The work therefore represents a step toward the modernization of ISTTOK's RT estimation and control capabilities, while clearly identifying the improvements required before future closed-loop plasma position control operation.

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ANNEX A: MATLAB SCRIPT TO SETUP SIMULINK MODELS FOR MARTE2 INTEGRATION

The script must be executed in the MATLAB terminal after assigning the Simulink model name to the variable 'model_name'. It is responsible for automating the necessary configuration process, ensuring that the model will be operational and properly configured for integration with MARTe2 via the generated dynamic shared library. This script is part of the MARTe2-components repository[16] and was not developed as part of this research.

```
% Solver
set_param(model_name, 'SolverType', 'Fixed-step');

% Code Generation
set_param(model_name, 'SystemTargetFile', 'ert_shrlib.tlc');
set_param(model_name, 'RTWVerbose', 0);

% Optimization
set_param(model_name, 'DefaultParameterBehavior', 'Tunable');
set_param(model_name, 'OptimizationCustomize', 1);
set_param(model_name, 'GlobalVariableUsage', 'None');

% Report
set_param(model_name, 'GenerateReport', 0);

% Comments
set_param(model_name, 'GenerateComments', 0);

% Custom code (MODEL is a coder keyword for the model name)
set_param(model_name, 'CustomSourceCode', ...
     [ ...
     '#define CONCAT(str1, str2, str3) CONCAT_(str1, str2, str3)'
newline, ...
     '#define CONCAT_(str1, str2, str3) str1 ## str2 ## str3'
newline, ...
     '#define GET_MMI_FUNC    CONCAT(MODEL  , _GetCAPImmi ,  )'
newline, ...
```

[16] This this MATLAB script can be found in: https://vcis-gitlab.f4e.europa.eu/aneto/MARTe2-components/-/blob/master/Source/Components/GAMs/SimulinkWrapperGAM/SimulinkWrapperGAM.h?ref_type=heads.

```
     '#define RT_MODEL_STRUCT CONCAT(RT_MODEL_ , MODEL  , _T )'
newline, ...
     newline, ...
     'void* GET_MMI_FUNC(void* voidPtrToRealTimeStructure)'
newline, ...
     '{'
newline, ...
     'rtwCAPI_ModelMappingInfo* mmiPtr = &(rtmGetDataMapInfo( (
RT_MODEL_STRUCT * )(voidPtrToRealTimeStructure) ).mmi);' newline,
...
     'return (void*) mmiPtr;'
newline, ...
     '}' ...
     ] ...
);
% Interface
set_param(model_name, 'SupportComplex', 0);
set_param(model_name, 'SupportAbsoluteTime', 0);
set_param(model_name, 'SuppressErrorStatus', 1);
set_param(model_name, 'CodeInterfacePackaging', 'Reusable
function');
set_param(model_name, 'RootIOFormat', 'Part of model data
structure');
set_param(model_name, 'RTWCAPIParams', 1);
set_param(model_name, 'RTWCAPIRootIO', 1);
set_param(model_name, 'GenerateAllocFcn', 1);
set_param(model_name, 'IncludeMdlTerminateFcn', 0);
set_param(model_name, 'CombineSignalStateStructs', 1);
set_param(model_name, 'ArrayLayout', 'Column-major');
% Templates
set_param(model_name, 'GenerateSampleERTMain', 0);
```

APPENDIX A: MARTE2 COMPILATION GUIDE

For the following guide on the compilation of the MARTe2 engine, Linux is assumed as the user's operating system (for further information on MARTe2 compilation in different environments, refer to the MARTe2 documentation[17].). It is good practice to store all MARTe components and projects in a designated repository based on the "home" directory. Hence, firstly, the following commands are required to be input into the bash terminal:

```
$ cd
$ mkdir Projects
$ cd ~/Projects
```

Before proceeding, it is necessary to ensure that all required development tools and libraries are installed. On most Debian/Ubuntu-based systems, this can be achieved by running the following commands:

```
$ sudo apt update
$ sudo apt install git build-essential wget cmake libxml2 libxml2-dev bc
$ sudo apt install libncurses5-dev libreadline-dev
```

These commands install Git[18], as it is required to clone the necessary MARTe2 repositories, the standard development toolchain, and additional libraries required to compile and run the framework and its dependencies. Consequently, the following commands are required to clone the repositories into appropriate local directories:

```
$ mkdir MARTe2-dev
$ cd ~/MARTe2-dev
$ git clone http://vcis-gitlab.f4e.europa.eu/aneto/MARTe2.git
$ cd ~/Projects
$ mkdir MARTe2-components
$ cd ~/MARTe2-components
$ git clone http://vcis-gitlab.f4e.europa.eu/aneto/MARTe2-components.git
$ cd ~/Projects
```

[17] MARTe2 documentation, https://vcis.f4e.europa.eu/marte2-docs/master/html/overview.html.
[18] Git documentation, https://git-scm.com/doc.

```
$ mkdir MARTe2-demos-padova
$ cd ~/MARTe2-demos-padova
$ git clone http://vcis-gitlab.f4e.europa.eu/aneto/MARTe2-demos-padova.git
```

After cloning all repositories into their respective local directories, the user must define a number of environment variables to identify the paths to the MARTe2 engine and its components.

```
$ export MARTe2_DIR=~/Projects/MARTe2-dev
$ export MARTe2_Components_DIR=~/Projects/MARTe2-components
$ export MARTe2_Demos_DIR=~/Projects/MARTe2-demos-padova
$ export LD_LIBRARY_PATH=$LD_LIBRARY_PATH:$MARTe2_DIR/Build/x86-linux/Core/
$ export TARGET=x86-linux
$ export MDSPLUS_DIR=
$ source ~/Projects/MARTe2-demos-padova/marte2-exports.sh
```

Finally, to compile MARTe2, its components, and the training repository, navigate to the appropriate directories and run the following commands:

```
$ cd ~/MARTe2-dev
$ make -f Makefile.gcc
$ cd ~/MARTe2-components
$ make -f Makefile.gcc
$ cd ~/MARTe2-demos-padova
$ make -f Makefile.gcc
```

Once these commands have been executed, the system will be ready for use. The user is encouraged to test the configuration files contained in the training repository to verify that the compilation was successful. If any issues arise, the user is advised to consult the MARTe2 documentation for troubleshooting.

APPENDIX B: PARAMETERS FOR MAGNETIC DIAGNOSTICS SIGNAL PROCESSING

Table 10: Relevant Parameters for the Mirnov Coil Signal Conversion Algorithm

Mirnov Coil nº	Parameter	Value
1	K_1	$1.73887 \cdot 10^{-5}$
	S_{eff_1} [m^2]	$2.793 \cdot 10^{-3}$
	pol_1	-1
2	K_2	$1.73938 \cdot 10^{-5}$
	S_{eff_2} [m^2]	$2.814 \cdot 10^{-3}$
	pol_2	-1
3	K_3	$1.74276 \cdot 10^{-5}$
	S_{eff_3} [m^2]	$2.839 \cdot 10^{-3}$
	pol_3	1
4	K_4	$1.74994 \cdot 10^{-5}$
	S_{eff_4} [m^2]	$2.635 \cdot 10^{-3}$
	pol_4	-1
5	K_5	$1.73827 \cdot 10^{-5}$
	S_{eff_5} [m^2]	$2.579 \cdot 10^{-3}$
	pol_5	1
6	K_6	$1.74440 \cdot 10^{-5}$
	S_{eff_6} [m^2]	$2.202 \cdot 10^{-3}$
	pol_6	1
7	K_7	$1.74532 \cdot 10^{-5}$
	S_{eff_7} [m^2]	$2.183 \cdot 10^{-3}$
	pol_7	1
8	K_8	$1.73724 \cdot 10^{-5}$
	S_{eff_8} [m^2]	$2.218 \cdot 10^{-3}$
	pol_8	1
9	K_9	$1.71925 \cdot 10^{-5}$
	S_{eff_9} [m^2]	$2.305 \cdot 10^{-3}$
	pol_9	1
10	K_{10}	$1.72739 \cdot 10^{-5}$
	$S_{\text{eff}_{10}}$ [m^2]	$1.686 \cdot 10^{-3}$
	pol_{10}	1

Real-Time Decimation Factor	Real-Time System Clock Frequency
200	10000

11	K_{11}	$1.73810 \cdot 10^{-5}$
	$S_{\mathrm{eff}_{11}}$ [m^2]	$2.705 \cdot 10^{-3}$
	pol_{11}	-1
12	K_{12}	$1.74413 \cdot 10^{-5}$
	$S_{\mathrm{eff}_{12}}$ [m^2]	$2.442 \cdot 10^{-3}$
	pol_{12}	1

APPENDIX C: MAGNETIC AND ELECTRIC DIAGNOSTICS INITIAL COVARIANCE MATRICES

The initialization of the covariance matrix was carried out independently for the two diagnostics considered. For the initial state covariance matrices, the diagonal entries represent the uncertainty attributed to the initial plasma position, set to ±2.5 cm in both radial and vertical directions. This corresponds to a variance of $(0.025\ \text{m})^2 = 6.25 \cdot 10^{-4}\ \text{m}^2$, ensuring a balanced confidence interval at the beginning of the estimation process, regardless of the diagnostic employed.

The off-diagonal terms represent the coupling between the radial and vertical position estimates. Rather than being arbitrarily fixed, these values were obtained from a filtered and weighted statistical analysis across a representative set of discharges. In practice, the correlations observed between R_p and Z_p for each diagnostic provide a direct measure of how deviations in one coordinate tend to affect the other. To ensure consistency of units with diagonal terms, the correlation coefficients were scaled by the predefined position variance thus yielding the off-diagonal covariance values in the matrices.

Lastly, for the process and the measurement covariance matrices (Q and R), only the diagonal elements were considered, due to independence between the corresponding process and measurement noises. In the case of the Mirnov-based filter, the diagonal elements of Q were set to $2.25 \cdot 10^{-4}\ \text{m}^2$, corresponding to an assumed uncertainty of ±1.5 cm in the radial and vertical directions. For the Langmuir-based filter, the first diagonal entry was set to $4.00 \cdot 10^{-4}\ \text{m}^2$ and the second to $1.00 \cdot 10^{-4}\ \text{m}^2$, corresponding to assumed uncertainties of ±2.0 cm in the radial direction and ±1.0 cm in the vertical direction. The values of all matrices were tuned empirically according to the relative confidence placed on the model dynamics and the diagnostic measurements, guided by the behavior of the fused estimates across the analyzed discharges to ensure consistency with both diagnostics and to avoid incoherent results.

The resulting matrices are presented below, with the terms rounded to six decimal places for clarity. P_0 and Q are expressed in m^2, while the measurement covariance matrices are expressed in the units of the respective diagnostics: T^2 for the Mirnov coils and V^2 for the Langmuir probes. For the measurement covariance matrices, only the diagonal elements are reported, as the off-diagonal terms were set to zero.

- Mirnov

$$P_{0_{\text{Mirnov}}} = \begin{bmatrix} 6.250000 \cdot 10^{-4} & -4.820681 \cdot 10^{-5} \\ -4.820681 \cdot 10^{-5} & 6.250000 \cdot 10^{-4} \end{bmatrix}$$

$$Q_{\text{Mirnov}} = \begin{bmatrix} 2.250000 \cdot 10^{-4} & 0 \\ 0 & 2.250000 \cdot 10^{-4} \end{bmatrix}$$

$$R_{11_{\text{Mirnov}}} = 1.000000 \cdot 10^{-6}$$

$$R_{22_{\text{Mirnov}}} = 1.000000 \cdot 10^{-6}$$

$$R_{33_{\text{Mirnov}}} = 2.250000 \cdot 10^{-4}$$

$$R_{44_{\text{Mirnov}}} = 1.000000 \cdot 10^{-6}$$

$$R_{55_{\text{Mirnov}}} = 1.000000 \cdot 10^{-6}$$

$$R_{66_{\text{Mirnov}}} = 1.000000 \cdot 10^{-6}$$

$$R_{77_{\text{Mirnov}}} = 1.000000 \cdot 10^{-6}$$

$$R_{88_{\text{Mirnov}}} = 1.000000 \cdot 10^{-6}$$

$$R_{99_{\text{Mirnov}}} = 1.000000 \cdot 10^{-6}$$

$$R_{10,10_{\text{Mirnov}}} = 5.625000 \cdot 10^{-5}$$

$$R_{11,11_{\text{Mirnov}}} = 1.000000 \cdot 10^{-6}$$

$$R_{12,12_{\text{Mirnov}}} = 1.000000 \cdot 10^{-6}$$

- Langmuir

$$P_{0_{\text{Langmuir}}} = \begin{bmatrix} 6.250000 \cdot 10^{-4} & 3.147274 \cdot 10^{-5} \\ 3.147274 \cdot 10^{-5} & 6.250000 \cdot 10^{-4} \end{bmatrix}$$

$$Q_{\text{Langmuir}} = \begin{bmatrix} 4.000000 \cdot 10^{-4} & 0 \\ 0 & 1.000000 \cdot 10^{-4} \end{bmatrix}$$

$$R_{11_{\text{Langmuir}}} = 2.250000$$

$$R_{22_{\text{Langmuir}}} = 2.250000$$

$$R_{33_{\text{Langmuir}}} = 2.250000$$

$$R_{44_{\text{Langmuir}}} = 2.250000$$

APPENDIX D: DETAILED PLASMA POSITION ESTIMATION METRICS

Table 11: Detailed plasma position estimation metrics computed over the selected flat top intervals for shots presented in Table 7

Shot number	Analyzed interval [ms]	Position	Estimator	MAE [cm]	RMSE [cm]	MaxAE [cm]	PCC
45967	53 - 77	Radial	Magnetic	1.538	1.587	2.635	0.932
			Electric	1.666	1.751	3.087	0.750
			Fused	1.547	1.604	2.742	0.926
		Vertical	Magnetic	0.747	0.749	0.921	0.557
			Electric	0.190	0.197	0.434	0.094
			Fused	0.584	0.586	0.719	0.481
46241	35 - 59	Radial	Magnetic	1.688	2.051	4.020	0.817
			Electric	2.538	2.944	6.002	0.601
			Fused	1.898	2.257	4.473	0.822
		Vertical	Magnetic	0.070	0.086	0.232	0.747
			Electric	0.176	0.204	0.670	0.212
			Fused	0.068	0.094	0.303	0.714
48555	105 - 130	Radial	Magnetic	2.554	2.590	3.703	0.600
			Electric	3.333	3.363	4.918	0.720
			Fused	2.614	2.647	3.717	0.632
		Vertical	Magnetic	0.692	0.700	0.862	0.977
			Electric	0.482	0.511	0.729	0.775
			Fused	1.064	1.070	1.200	0.938

Shot number	Analyzed interval [ms]	Position	Estimator	MAE [cm]	RMSE [cm]	MaxAE [cm]	PCC
48556	105 - 130	Radial	Magnetic	2.743	2.863	4.654	0.549
			Electric	2.971	3.092	5.113	0.633
			Fused	2.682	2.804	4.432	0.572
		Vertical	Magnetic	0.855	0.881	1.186	0.950
			Electric	0.691	0.765	1.174	0.053
			Fused	1.238	1.262	1.610	0.885
48561	104 - 128	Radial	Magnetic	2.714	2.841	4.659	0.378
			Electric	2.881	3.002	4.438	0.478
			Fused	2.635	2.764	4.500	0.394
		Vertical	Magnetic	0.835	0.857	1.177	0.984
			Electric	0.692	0.759	1.234	0.334
			Fused	1.223	1.242	1.544	0.947
48563	103 - 127	Radial	Magnetic	2.117	2.350	4.534	0.359
			Electric	2.185	2.407	3.918	0.677
			Fused	2.018	2.255	4.096	0.437
		Vertical	Magnetic	0.806	0.855	1.240	0.958
			Electric	0.562	0.696	1.177	0.130
			Fused	1.182	1.223	1.664	0.897
53058	39 - 64	Radial	Magnetic	1.079	1.310	3.168	0.484
			Electric	1.123	1.397	4.074	1.000
			Fused	1.091	1.315	3.406	0.653
		Vertical	Magnetic	1.136	1.306	3.116	0.159
			Electric	0.941	1.207	3.998	1.000
			Fused	0.968	1.206	3.844	0.256

Shot number	Analyzed interval [ms]	Position	Estimator	MAE [cm]	RMSE [cm]	MaxAE [cm]	PCC
53071	272 - 293	Radial	Magnetic	1.472	1.648	3.355	0.727
			Electric	1.085	1.255	3.630	1.000
			Fused	1.392	1.553	3.121	0.809
		Vertical	Magnetic	1.564	1.767	3.882	0.002
			Electric	1.145	1.367	3.321	1.000
			Fused	1.132	1.357	3.370	0.126